Paediatric Urology 2

Condensed rapid revision · facts already answered in Papers 1–4 point to that answer

highlighted number to memorise · **KEY POINT** and **ALGORITHM** boxes carry the reasoning · MCQ exam trivia · † added beyond the source notes · P2 topic = covered in full in that Paper's answer book

Chapter	Topic
Ch 32	Embryology of the Genitourinary Tract
Ch 33	Urologic Aspects of Paediatric Nephrology
Ch 34	Perinatal Urology
Ch 35	Urologic Evaluation of the Child
Ch 36	Paediatric Urogenital Imaging
Ch 37	Infection & Inflammation of the Paediatric GU Tract
Ch 38	Perioperative Management in Children
Ch 39	Laparoscopic & Robotic Surgery in Children
Ch 40	Clinical & Urodynamic Evaluation of LUTD
Ch 42	Bladder Anomalies in Children
Ch 43	Exstrophy–Epispadias Complex
Ch 44	Prune-Belly Syndrome
Ch 45	Posterior Urethral Valves
Ch 46	Neuromuscular Dysfunction of the LUT (Neurogenic Bladder)
Ch 47	Functional Bladder & Bowel Disorders, Enuresis
Ch 48	Lower Urinary Tract Reconstruction in Children
★	Putting it together: common paediatric pathways

32 Embryology of the Human Genitourinary Tract

1 · Overview

Every congenital anomaly is a normal step gone wrong.

- Development starts in the first trimester and is mostly complete by the second.
- The **indifferent stage** lasts to about **8–9 wk**; then androgens drive the sexes apart.

2 · Kidney and ureter

Bud position decides kidney quality and orifice site.

Three successive kidneys

Kidney	Features
†Pronephros	Cervical, week 4; non-functional and regresses
Mesonephros	†Thoracolumbar and transient, but its **Wolffian duct** persists. Tubules become efferent ductules and paradidymis (male), or epoöphoron and paroöphoron (female)
Metanephros (permanent)	†Week 5; ureteric bud and metanephric blastema form the nephrons

Reciprocal induction †

- The **ureteric bud** grows from the Wolffian duct into the metanephric mesenchyme. Its stalk becomes the ureter; its branches form pelvis, calyces and collecting tubules.
- The mesenchyme secretes **glial cell line-derived neurotrophic factor (GDNF)**, which binds **RET** on the bud to drive branching.
- **Wilms tumour 1 (WT1)** makes mesenchyme competent; **paired box 2 (PAX2)** maintains bud and duct, and its loss causes agenesis or hypoplasia (renal-coloboma syndrome).
- Bud tips induce epithelial change: renal vesicle, comma, S-shape, nephron. Nephrogenesis ends at about 34–36 wk; none after birth.

> **KEY POINT**
> One defect gives both dysplasia and an abnormal orifice: bud position on the Wolffian duct sets where it meets the blastema (kidney quality) and where its orifice lands in the bladder.

Ascent and rotation †

- The kidney rises from pelvis to lumbar region (weeks 6–9), taking successively higher arteries; the hilum rotates 90° from anterior to medial.
- Failure gives a pelvic or ectopic kidney, malrotation or multiple arteries. Fused lower poles make a **horseshoe kidney**, trapped under the inferior mesenteric artery.

3 · Cloaca, bladder, trigone and urethra

This explains reflux, ectopia, exstrophy and urachal remnants.

- P2 Embryogenesis of urinary bladder — *urorectal septum, vesicourethral canal vs UGS, wall layers, trigone (apoptosis, mostly bladder muscle), urachus, Weigert–Meyer*

Not in the paper answer:

- Anorectal malformations arise from the **cloacal membrane**, not the urorectal septum (weeks 5–6).
- Bladder and urethra are alike in both sexes except the larger male prostatic urethra.

Weigert–Meyer rule (duplex kidneys)

- P3 Embryology and management of ectopic ureter. — *bud too low gives lateral refluxing lower pole; too high gives caudal/medial ectopic upper pole; male vs female ectopic sites and sphincter*

Not in the paper answer:

- The upper-pole orifice rotates dorsally relative to the lower-pole one.
- **Ectopic vas**: the vas joins the ureter instead of the verumontanum, and both drain into the prostatic urethra.
- Male ectopia stays above the external sphincter, so no continuous wetting; †female ectopia below it causes continuous dribbling with normal voids.

4 · Gonads

Two testicular hormones explain every disorder of sex development (DSD).

- P1 Testicular development, gonadal descent, abnormalities of descent and bilateral undescended testis — *indifferent gonad, PGC migration, SRY then Sertoli cords, rete/efferent ductules, AMH/testosterone/INSL3 table, remnants*

Not in the paper answer:

- Primary sex cords appear in week 6; human seminiferous cords appear to derive from mesonephric tubules.
- Müllerian ducts respond to MIS only during 9–13 wk.
- Ovarian development (below).
- **Sex-determining region Y (SRY)** makes Sertoli cells enclose primordial germ cells (PGCs) in cords at about **6 wk**. Leydig cells appear from 8 wk.
- **Müllerian-inhibiting substance (MIS)**, also called anti-Müllerian hormone (AMH), from Sertoli cells, makes the Müllerian ducts regress at **8–10 wk**. Without it, the uterus and tubes persist.
- **Testosterone** from Leydig cells turns the Wolffian duct into epididymis, vas and seminal vesicle; without it, the ducts regress.

Ovary — the default

- No SRY means no Sertoli cells, MIS or androgen, so female. Cortex and medulla separate at about 8 wk; germ-cell **nests** form at 10–11 wk.
- FOXL2-positive cells make the primordial follicle, which is primary at about 20 wk, secondary at term and antral after birth. Oocytes arrest in meiosis I until puberty.

> **KEY POINT**
> Two independent testicular signals act on two independent duct systems: MIS decides Müllerian, testosterone decides Wolffian and external genitalia. Every DSD is a combination of the two.

5 · Gonadal descent

The failed phase predicts where the testis lies.

- P1 Testicular development, gonadal descent, abnormalities of descent and bilateral undescended testis — *CSL + gubernaculum; transabdominal (10–15 wk: androgen, INSL3) and inguinoscrotal (7th month: genitofemoral nerve CGRP) phases; female ligaments*

Not in the paper answer:

- *Insl3*-null mice have isolated intra-abdominal cryptorchidism. Gubernacular swelling is androgen-independent; the inguinoscrotal phase is androgen-dependent.

6 · Genital ducts

Explains 5α-reductase deficiency and combined Müllerian–renal anomalies.

Male accessory glands

- The **prostate** buds at **10 wk** from FOXA1-positive urogenital sinus (UGS) epithelium below the ureteric insertion. Most lie lateral to the **verumontanum**, where Wolffian ducts and utricle open.
- Androgen acts via the **androgen receptor (AR) in mesenchyme**; stromal type 2 5α-reductase (5αR2) makes **dihydrotestosterone (DHT), which has about 10× the AR affinity** of testosterone.
- So 5αR2 deficiency gives a rudimentary prostate and severe hypospadias. Mesenchyme can induce prostate even from bladder epithelium.
- Oestrogen causes squamous metaplasia via oestrogen receptor 1 (ESR1) in stroma **and** epithelium.
- The **seminal vesicle** is a Wolffian bud at 10–12 wk; its main anomaly is a cyst.

Female tract

- In week 5 the Müllerian ducts use the **Wolffian duct as a guidewire** to reach the UGS, hence combined anomalies.
- In week 8 they fuse into the uterovaginal canal, whose septum is gone by week 9. Unfused cranial ducts become tubes; the canal becomes uterine body and cervix.
- The PAX2-positive Müllerian **vaginal** plate is replaced (14–21 wk) by FOXA1-positive UGS epithelium, so the **entire vaginal epithelium and hymen are UGS-derived**, supporting Bulmer over Koff. Canalisation occurs at 15–19 wk.
- Female Wolffian remnants: epoöphoron, paroöphoron, **Gartner duct**.

7 · External genitalia

Explains hypospadias and the safe plication site.

- P3 Embryology of the penis and management of proximal hypospadias (including Koyanagi repair) — *anogenital distance first, germ layers, double zipper (opening androgen-independent, closing dependent), glanular canalisation, timetable, 12 o'clock nerve-free zone*

Not in the paper answer:

- The meatus marks the junction of ectoderm and endoderm.
- The tunica is thickest at the 12 o'clock nerve-free zone, the dorsal plication site for mild to moderate curvature.
- The clitoris is a penis without a urethra, with the same 12 o'clock zone.
- The shaft is fused by **16 wk**; the prepuce depends on the glanular urethra.
- In **females** the vestibular plate canalises but folds do not fuse, leaving an open vestibule.

8 · Developmental step and resulting anomaly

Step that fails	Anomaly
†No bud or no induction (GDNF–RET, PAX2)	Renal agenesis (absent hemitrigone)
†Abnormal bud–blastema contact	Dysplasia, hypoplasia, multicystic dysplastic kidney (MCDK)
†Early bifurcation or two buds	Bifid ureter or complete duplication

Step that fails	Anomaly
Bud too low / too high	Vesicoureteral reflux (VUR) / ectopic ureter
†Ascent or rotation fails	Pelvic, malrotated or horseshoe kidney
Cloacal membrane or septation	Anorectal malformation; †cloacal anomalies, exstrophy
Allantois persists	Urachal remnants
Fold fusion arrests (shaft–glans junction commonest)	Hypospadias
Urethral plate splits (Sonic hedgehog)	Urethral duplication (complete from bladder; Y-shaped along shaft)
Androgen excess in 46,XX	Vestibular fusion, giving virilisation (congenital adrenal hyperplasia, CAH)
Cranial suspensory ligament (CSL) retained	Intra-abdominal testis
Gubernacular attachment abnormal	Canalicular or ectopic testis (perineal, femoral, prepenile)
Gubernacular migration fails	Congenital cryptorchidism
Processus fails to close / to involute	Hydrocele, hernia / acquired (ascending) testis
One Müllerian duct absent	Unicornuate uterus
Ducts fail to fuse	Uterus didelphys
Septum persists	Uterine obstruction or duplication
Vaginal plate fails to canalise	Transverse vaginal septum (middle–upper third)
Inferior end fails to open	Imperforate hymen — commonest vaginal obstruction
Müllerian and Wolffian both abnormal	Obstructed hemivagina and ipsilateral renal anomaly (OHVIRA)

9 · Clinical correlations

Where embryology changes work-up or surgery.

- **Hypospadias** (about 1 in 250): ectopic meatus, dorsal hood, ventral skin deficiency, with or without curvature. Abortive spongiosal tissue alongside makes it an **arrest, not a deformation**. Cause mostly unknown (genetics, endocrine disruptors).
- **CAH** (autosomal recessive) virilises 46,XX infants, who may look fully male. Clues: **impalpable gonads and hyperpigmentation**; it mimics hypospadias with bilateral impalpable testes.
- A **streak gonad** (45,X; 46,XX or 46,XY dysgenesis) makes no MIS or testosterone, so the child is female. Give hormones and **remove the streak if 46,XY**, because of gonadoblastoma risk.
- In **ovotesticular DSD** each duct follows its own-side gonad; female fertility is possible, spermatogenesis not reported.
- **Persistent Müllerian duct syndrome (PMDS)**: 46,XY with uterus and tubes (hernia uteri inguinale), usually an MIS-receptor defect, found at exploration for bilateral abdominal testes.
- **OHVIRA** presents after menarche with cyclic pain, discharge or pyocolpos. The kidney is often **dysplastic with an ectopic ureter into the obstructed hemivagina**, not absent; if missed, leakage persists after septum resection.

KEY POINT

Homologues: Littre and Skene; Cowper and Bartholin; spongiosum and vestibular bulb; gubernaculum and the ovarian plus round ligaments; labioscrotal swellings and labia majora. Traps: the vagina is UGS-derived; the appendix testis is Müllerian and the appendix epididymis Wolffian.

33 Urologic Aspects of Pediatric Nephrology

1 · Renal function in the child

Neonatal values are not adult values.

- Congenital anomalies of kidney and urinary tract (CAKUT) are the **commonest cause of paediatric end-stage kidney disease (ESKD)**.
- Glomerular filtration rate (GFR) doubles in 2 weeks, reaching adult level by **2 years**; at term it is about 40 mL/min/1.73 m².
- Creatinine in the first 48 h is **maternal**; by day 7 it is the neonate's own (**<0.5 mg/dL** at term).
- †Schwartz: estimated GFR = 0.413 × height (cm) / creatinine (mg/dL).
- Neonates dilute to 50 mOsm/L but concentrate poorly; bicarbonate matures over 3–4 weeks.

> **KEY POINT**
> The neonatal kidney is "hormonally deaf": maternal creatinine, week-1 salt wasting, hyperkalaemia (aldosterone-unresponsive tubule), low bicarbonate and poor concentration (antidiuretic hormone-unresponsive) reflect immaturity, not disease.

2 · Haematuria

Microscopy decides the work-up.

- Microscopic haematuria is **>5 red blood cells (RBCs)/high-power field** (fresh spun urine); about 1% of school children, mostly resolving.
- **Fresh microscopy is critical**: red urine is non-glomerular; brown "tea-coloured" urine with RBC casts is glomerular. A positive dipstick without RBCs means haemoglobin or myoglobin.
- **Urinary tract infection (UTI) is the commonest cause of gross haematuria**; culture it. Trauma, pain or clots need urological work-up, often computed tomography (CT). Stones: *see Ch 54*.
- Suspected glomerulonephritis (GN): check creatinine, albumin, C3/C4, antistreptolysin O (ASO)/anti-DNase B; refer to nephrology.
- In isolated microscopic haematuria, ultrasound (US) is low-yield; **voiding cystourethrogram (VCUG) and cystoscopy are not indicated**.
- Gross-only: urethrorrhagia, Wilms or bladder tumour, renal vein thrombosis, papillary necrosis, nutcracker. Microscopic-only: thin basement membrane, interstitial nephritis, sickle cell.

3 · Proteinuria

Most isolated proteinuria is benign.

- Normal urine protein-to-creatinine ratio (UPCR): **<0.2 mg/mg** (<0.7 in newborns); timed <4 mg/m²/h; adolescents <150 mg/24 h.
- **Transient** proteinuria (fever, exercise) is >75% of isolated cases. **Orthostatic** proteinuria (normal first-morning UPCR) is benign.
- **Fixed** proteinuria (glomerular, tubular, reflux, obstruction, dysplasia) needs full evaluation with or without biopsy.

4 · Glomerular disease

Age and complement narrow the diagnosis.

Under 1 year: genetic (Denys-Drash, Frasier). 1–10 years: minimal change, post-streptococcal GN, Henoch-Schönlein purpura (HSP), haemolytic uraemic syndrome (HUS). ≥7 years: lupus, IgA, membranoproliferative GN (MPGN), Alport.

Nephrotic syndrome

- Proteinuria **>40 mg/m²/h or UPCR >2.0** with hypoalbuminaemia and hyperlipidaemia.
- Congenital disease (<3 months) ignores immunosuppression. In primary disease give **steroids first, without biopsy**,: minimal change is 90% under 10 years and about 90% steroid-sensitive.

Category	Definition	Action
Steroid-sensitive	Remission within 8 weeks	~50% relapse
Frequently relapsing	≥2 relapses in 6 months or ≥4 in 12	Steroid-sparing: cyclophosphamide, mycophenolate, rituximab, calcineurin inhibitors
Steroid-dependent	2 consecutive relapses on steroids	
Steroid-resistant	No remission after 6–8 weeks	Biopsy; usually focal segmental glomerulosclerosis (FSGS), which recurs after transplant

Nephritic syndrome

Haematuria, proteinuria, hypertension and acute kidney injury (AKI); complement sorts causes:

C3	C4	Diagnoses
Low	Normal	Post-infectious GN, C3 glomerulopathy
Low	Low	Lupus nephritis, MPGN
Normal	Normal	IgA, HSP, Alport, antineutrophil cytoplasmic antibody (ANCA) GN, anti-glomerular basement membrane (anti-GBM)

- **Post-streptococcal GN** follows pharyngitis or pyoderma by weeks. C3 normalises by **8–12 weeks**; if it stays low, biopsy.
- **IgA nephropathy** is the commonest primary GN: painless gross haematuria **during** a respiratory infection.
- **Alport** (type IV collagen, mostly X-linked COL4A5) gives haematuria and high-frequency sensorineural deafness; about 50% of males reach ESKD by 25.
- **HUS** is the commonest cause of severe AKI in a healthy young child. Shiga-toxin E. coli HUS gets supportive care (about half dialysed); atypical (complement) HUS gets eculizumab.

5 · Tubulopathies

Obstruction can cause reversible renal tubular acidosis (RTA).

- P2 Renal tubular acidosis — *types 1/2/4 compared: defect, urine pH, K^+, stones, causes (obstruction gives type 4), treatment*

Not in the paper answer:

- Fanconi syndrome adds glucosuria, aminoaciduria and phosphaturia (cystinosis, drugs).
- Distal RTA also causes rickets in children.

- Distal RTA causes hypercalciuria and **nephrocalcinosis or stones**. Type 4 (**hyperkalaemia**) from obstruction improves after urological correction.

6 · Cystic disease

MCDK needs reflux screening and follow-up.

- **Multicystic dysplastic kidney (MCDK)**: non-communicating cysts, no function; bilateral is lethal. Vesicoureteral reflux (VUR) or pelviureteric junction obstruction occurs in up to 25%, so do a VCUG, especially with hydroureteronephrosis.
- Follow with US and blood pressure (BP) 3-monthly in year 1, 6-monthly until involution or ≥5 years, then yearly into adulthood.
- **Autosomal recessive polycystic kidney disease (ARPKD**, *PKHD1*): enlarged kidneys, hepatic fibrosis; >75% ESKD by 5 years. The dominant form (ADPKD, *PKD1/PKD2*) is usually adult.
- Nephronophthisis causes polyuria and anaemia, with ESKD by adolescence (Senior-Løken, Joubert, Bardet-Biedl).

7 · Hypertension

Age predicts the cause.

- Hypertension: repeated BP **>95th percentile** for age, sex and height; adolescents normal below 120/80. Up to a quarter are white-coat.
- Cause by age: infants, coarctation or renovascular; early childhood, Wilms or neuroblastoma; school age, **renal parenchymal disease including reflux nephropathy**; obese children >6 years and adolescents, essential.
- Over 6 years, no screening US unless urinalysis or function is abnormal. Echocardiogram when drugs start.
- Lifestyle for stage 1 without end-organ damage; drugs for stage 2, symptoms, end-organ damage or a secondary cause. Target <90th percentile, or **<50th** in chronic kidney disease (CKD).

8 · Acute kidney injury

Creatinine lags behind injury.

KDIGO stage	Creatinine	Urine output
I	Rise ≥0.3 mg/dL in 48 h or 1.5–1.9× baseline	<0.5 mL/kg/h for 6–12 h
II	2.0–2.9×	<0.5 for ≥12 h
III	≥3× or ≥4.0 mg/dL or clearance <35 (<18 years)	<0.5 for ≥24 h or <0.3 for 12 h

- Creatinine lags GFR by 24–72 h, rising only after about 50% loss.
- Fractional excretion of sodium (FeNa) **<1%** means prerenal, >3% intrinsic; on diuretics use fractional urea <35%.
- Prerenal AKI: isotonic fluid, no potassium until urine flows. Dose drugs as GFR <15.
- Renal replacement therapy (RRT) is for diuretic-resistant overload, refractory hyperkalaemia or acidosis, uraemia or nutrition. Two-thirds later develop CKD.

9 · Chronic kidney disease

Growth, bone and BP matter most.

- CKD is abnormality lasting >3 months; stages (age >2 years) by GFR: I ≥90, II 60–89, III 30–59, IV 15–29, V **<15** or dialysis.
- Causes: CAKUT or cystic disease when young; glomerular in adolescents.
- Complications: growth failure (growth hormone **resistance**), bone disease, anaemia (erythropoietin and iron), acidosis, cardiovascular death.
- **Do not restrict protein**; strict BP control (50th percentile) is the only proven way to slow progression.

10 · Dialysis and transplantation

The urologist prepares the lower tract.

- P1 Urological evaluation and management of pediatric kidney transplant patient. — *lower-tract evaluation and reconstruction before transplant, native nephrectomy indications, ureteric techniques, complications*

Not in the paper answer:

- Peritoneal dialysis (PD) is first choice if **<9 years / <20 kg**. Vesicostomy is acceptable; omphalocele, gastroschisis, diaphragmatic hernia and exstrophy contraindicate it.
- Transplant is often pre-emptive living-related; many await **≥10 kg and 65 cm** because of graft thrombosis.
- Living vs deceased donor: delayed graft function 5% vs 15%; graft half-life 20 vs 12.5 years.
- Risks: cytomegalovirus, Epstein-Barr and BK virus; malignancy about 7-fold.

34 Perinatal Urology

1 · Fetal renal development and prenatal imaging

Amniotic fluid is fetal urine, so obstruction threatens lungs as well as kidneys.

- P3 Antenatal hydronephrosis (UTD) and fetal uropathy (LUTO) — *UTD 1–3% of pregnancies, second after cardiac; 18–20 wk scan; APD, SFU and UTD grading; A1/A2–3 and P1–P3 tables; causes with incidence; postnatal plan*

Not in the paper answer:

- AFI cut-offs; which anomalies prenatal ultrasound diagnoses well (below).
- Pyramidal thickness; function by SFU grade (§2).
- Fetal intervention: prognosis, window, complications, shunt versus cystoscopy, outcome (§4).
- Voiding cystourethrogram (VCUG) yield, renogram pitfalls, prophylaxis trials, megaureter predictors (§5–7).
- Great Ormond Street pyeloplasty trial (§9).
- The metanephric kidney appears at **day 28**; urine production starts at **14 weeks**. After **~16 weeks** amniotic fluid is mainly urine, so obstruction causes oligohydramnios and pulmonary hypoplasia.
- Amniotic fluid index (AFI), the sum of four largest pockets: normal 8–18 cm; oligohydramnios **<5–6 cm**.
- Ultrasound (US) severity tracks pelviureteric junction (PUJ) obstruction and posterior urethral valves (PUV), not vesicoureteral reflux (VUR) or duplex. Exstrophy and neural tube defects are usually diagnosed correctly; genital ambiguity is not.
- Fetal magnetic resonance imaging (MRI) is only for anatomy: cloaca, urogenital sinus, exstrophy, duplex with outlet obstruction.

> **KEY POINT**
> Prenatal detection is for triage and counselling (differential, postnatal plan, natural history); there is no proof it improves outcome.

2 · Grading systems

Grade sets follow-up but does not predict reflux.

- **Anteroposterior pelvic diameter (APD)** is measured transversely at the hilum (error: sagittal). It rises with gestational age (GA), so it is **meaningless without its GA**. Severe: >10 mm (2nd trimester), >15 mm (3rd).
- Postnatal pathology rises from about 12% (mild) to 88% (severe), but **VUR shows no trend**.
- **Society for Fetal Urology (SFU)** grading (postnatal): 1, sinus barely split; 2, major calyces; 3, minor calyces, parenchyma spared; 4, parenchymal thinning.
- All SFU 3 kidneys keep differential renal function (DRF) >40%; about 70% of SFU 4 do. It is subjective and ignores ureter and bladder.
- Pyramidal thickness **≤3 mm** predicts pyeloplasty and reduced function.
- **Urinary tract dilation (UTD)** classification uses six parameters plus oligohydramnios, with a normal APD ceiling that rises with age (box in §3). It is objective and subjective, pre- and postnatal, and covers the whole tract.

3 · Prenatal follow-up

The prenatal grade sets rescan frequency.

ALGORITHM

1. Normal APD ceiling, all else normal: **<4 mm (16–27 weeks) → <7 mm (≥28 weeks) → <10 mm (postnatal >48 h)**.
2. If A1 (APD 4 to <7 mm at 16–27 wk, or 7 to <10 mm at ≥28 wk, all else normal), repeat US once after 32 weeks.
3. If A2–3 (any abnormal parameter), scan every 4 weeks until delivery, because the lesion may progress and a second anomaly is likelier.
4. In both, do a postnatal US after 48 h.

- Multidisciplinary prenatal consult: exstrophy, cloaca or urogenital sinus, autosomal recessive polycystic kidney disease (ARPKD), outlet obstruction (PUV, prune-belly syndrome [PBS], urethral atresia), myelomeningocele.

4 · Fetal LUTO and oligohydramnios

Fetal intervention for lower urinary tract obstruction (LUTO) saves lungs, not kidneys.

- Picture: male fetus, bilateral hydroureteronephrosis, non-cycling bladder, **"keyhole" sign** and oligohydramnios. Only fetal cystoscopy can separate PUV, urethral atresia and PBS.
- "Pop-offs" (urinary ascites, urinoma, VUR, unilateral dysplasia) show the system is under pressure.
- Poor prognosis: oligohydramnios **before 27 weeks**, cortical cysts, echogenicity. Onset after 27 weeks favours the lungs.
- Fetal urine biomarkers are **not validated** (European consensus 2022). †Classic favourable serial values: Na <100 mEq/L, Cl <90 mEq/L, osmolality <200 mOsm/L, β2-microglobulin <4 mg/L.

Ruano stage	Fluid / kidneys	Urine biochemistry	Intervention
I — mild	Normal fluid and echogenicity	Favourable	Not indicated
II — severe, function preserved	Oligo/anhydramnios; hyperechoic; no cysts	Favourable within 3 samples	**Indicated** — for lungs and kidneys
III — severe, function abnormal	Anhydramnios; hyperechoic, with or without cysts and dysplasia	Unfavourable after 3 samples	Maybe — for lungs only

Options

- Options: early delivery, vesicoamniotic shunt (VAS), fetal cystoscopy, serial amnioinfusion. Window **18–24 weeks**; about 1 in 60,000 need it.
- Complications about **44%**: shunt failure (commonest), preterm labour, membrane rupture, ascites, chorioamnionitis, bowel herniation.

Feature	VAS	Fetal cystoscopy
Diagnoses cause	No	Yes — views posterior urethra
Mechanism	Bypass from bladder to amniotic cavity	Laser valve ablation lets the bladder cycle
Specific complications	Blockage, migration, bowel herniation	Urethrocutaneous or rectourethral fistula (≤10%)

Feature	VAS	Fetal cystoscopy
Survival / renal function (PUV)	Improved / not improved	Improved / short-term only

- One series found VAS before 16 weeks gave better lung and renal outcome; newer shunts seem safe before 14 weeks.
- Serial amnioinfusion supports lungs in bilateral renal agenesis (no urine to shunt); RAFT trial ongoing.
- After VAS, over half have renal insufficiency, a third need renal replacement therapy (RRT) and a third need clean intermittent catheterisation (CIC).

> **KEY POINT**
> Counsel: intervention improves perinatal survival (lungs); complications exceed 40%; it does not reliably prevent renal replacement, because dysplasia is often established first.

5 · Postnatal imaging

Right timing and test avoid missed obstruction and needless radiation.

- **First US after 48 h**, because neonatal dehydration under-reads dilation; delay up to 2 weeks is harmless.
- Postnatal P1 (APD 10 to <15 mm, central calyces), P2 (≥15 mm, peripheral calyces or abnormal ureter) and P3 (abnormal parenchyma or bladder) roughly equal SFU 1–2, 3 and 4. **Classify by the highest-risk feature**: parenchymal thinning is P3 at any APD.
- **VCUG** finds VUR in up to 31%, and up to 25% with VUR have a normal first US. Order by **urinary tract infection (UTI) risk**: females, uncircumcised males, high-grade dilation, dilated ureter, abnormal bladder.
- Use ALARA dosing, a non-balloon catheter and cycling; outpatient unless outlet obstruction is suspected (bladder wall ≥0.5 cm, distended bladder, bilateral hydroureteronephrosis).
- Contrast-enhanced voiding urosonography (CeVUS): no radiation, possibly more sensitive for VUR, poor urethral view. Surveillance of known VUR is safe; finding asymptomatic VUR has unproven benefit.
- **Mercaptoacetyltriglycine (MAG3)** is preferred over DTPA ("well-tempered" protocol). Consider surgery if **DRF <40%**, half-time (T½) >20 min, or retention persists upright.
- Pitfalls: dehydration, furosemide before peak uptake into a capacious pelvis, region-of-interest placement.
- **Dimercaptosuccinic acid (DMSA)**: dysplasia or scars in VUR, non-function in multicystic dysplastic kidney (MCDK), duplex upper-pole function. **Magnetic resonance urography (MRU)**: ectopic ureter, megaureter, crossing vessels; needs sedation.

6 · Postnatal management, UTI and prophylaxis

Most dilation is harmless; protect the few at real risk.

- **P1**: US at 3–6 months; no continuous antibiotic prophylaxis (CAP), VCUG or MAG3; it resolves by 2–4 years.
- **P2**: CAP at discretion, but **yes if the ureter is >7 mm**; VCUG and MAG3 at discretion.
- **P3**: US at about 1 month, CAP, VCUG, and MAG3 if dilation persists.

- Dissent: reflux is raised at every grade, so some base VCUG on ureteral dilation or LUTO features.
- In P3 without VUR, give CAP until pyeloplasty or **1 year** (about 90% of UTIs occur in year 1).
- UTI occurs in 8–22%. Agreed risks: **female sex, high-grade dilation, intact foreskin**; debated: ureteral dilation, VUR, no CAP.
- Circumcision has a number needed to treat (NNT) of **10**; odds ratio about 0.3.
- CAP evidence is weak; target high-grade UTD, females, uncircumcised boys, dilated ureter or VUR. SFU registry (low-grade, UTI 8%) showed no benefit and McMaster (high-grade, 19%) benefit, reflecting case mix.
- Agents: amoxicillin or a first-generation cephalosporin from birth; nitrofurantoin or trimethoprim only from **≥8 weeks**, because hepatic metabolism is immature.

ALGORITHM

1. US after 48 h: if normal (<10 mm, all else normal), the dilation was transient.
2. If P1, repeat US at 3–6 months.
3. If P2, give CAP if the ureter is >7 mm; VCUG and MAG3 at discretion.
4. If P3, give CAP, do a VCUG and repeat US at about 1 month; if dilation persists, do a MAG3.
5. If bilateral severe dilation (with or without oligohydramnios), treat as outlet obstruction: decompress, CAP, urgent evaluation.

7 · Specific antenatal diagnoses

Each cause has its own natural history and surgical trigger.

- **Transient** dilation (50–70%) is commonest. Mild dilation resolves in 90–98%: SFU 1 by about 2.5 years, about 95% of SFU 2 by 4 years. Warn that VUR is possible.

PUJ obstruction pattern

- Commonest **surgical** cause. Third-trimester APD **15 mm** has about 80% sensitivity and specificity for pyeloplasty; postnatal APD predicts better. Even with severe UTD, about a fifth need surgery.
- VCUG excludes VUR and checks the posterior urethra in boys. MAG3 at 1 month for persistent P3; serial US suffices if P2, APD <20 mm and pyramidal thickness >3 mm (DRF >40% likely).
- Operate for rising dilation, DRF <40% with poor drainage, falling DRF, upright retention or symptoms (*see Ch 53*).

Vesicoureteral reflux

- Prenatally detected VUR: male, high-grade, bilateral, congenital dysplasia (up to 40%), often early resolution. After febrile UTI: female, lower grade, acquired scarring.
- Screen girls with high-grade UTD and uncircumcised boys; do not screen circumcised boys with low-grade UTD.

Megaureter / ureterovesical junction obstruction

- Types: obstructed, non-obstructed, refluxing, refluxing-obstructed. Exclude single-system ectopic ureter.
- Most resolve. Surgery is predicted by a ureter **>13 mm** and DRF <30%.

- BAPU: prenatal ureter **>7 mm after 30 weeks** needs postnatal US, CAP and VCUG for all; consider MAG3.
- Operate for DRF <40%, febrile UTI or pain. **Under 1 year, temporise** (cutaneous ureterostomy, pyelostomy, refluxing reimplant); reimplant after 1 year, †because a tapered ureter in a small bladder obstructs. Follow into adulthood.

> **KEY POINT**
> In PUJ, VUR and megaureter, dilation alone is not an indication: operate for function loss, infection or progression; grade sets surveillance.

LUTO conditions

- **PUV** is the commonest LUTO (1.4–2.2/10,000) (*see Ch 45*). **PBS**: megacystis, deficient abdominal wall, impalpable testes, with or without urethral atresia or megalourethra (*see Ch 44*).
- **Urethral atresia**: a catheter will not pass; vesicostomy, then progressive augmented dilation of the anterior urethra (PADUA).
- **Megacystis from massive reflux**: lateral orifices; urine cycles between bladder and ureters; no oligohydramnios.
- **Megacystis-microcolon-intestinal hypoperistalsis syndrome (MMIHS)**: distended non-obstructed bladder; vesicostomy or CIC; CAP only for recurrent UTI.

> **KEY POINT**
> Bilateral dilation with normal fluid means urine still leaves the fetus (reflux, megacystis); with low fluid it means true outlet obstruction.

Neonatal management of LUTO

> **ALGORITHM**
> 1. Have the team at delivery and resuscitate the lungs first.
> 2. Pass a 5 Fr feeding tube, with a finger in the rectum to guide it over the high bladder neck.
> 3. Confirm position on a plain abdominal film (KUB).
> 4. If it fails, place a US-guided suprapubic catheter.

- **Do not use a balloon catheter**: the balloon obstructs the posterior urethra and compresses the ureteric orifices.
- Do not do an immediate VCUG with renal insufficiency; judge decompression by the **creatinine nadir**. Warn of post-obstructive diuresis.
- Oliguria after decompression suggests a thick bladder obstructing the ureters; treat with antimuscarinics.

Multicystic dysplastic kidney

- MCDK: non-communicating cysts, no central cyst. VUR in about 20%; contralateral PUJ obstruction can occur.
- There is **no Wilms association**, so no routine nephrectomy. No CAP; VCUG and DMSA optional. Serial US, blood pressure and proteinuria through puberty; remove if a solid component appears.

Duplex, ureterocele, ectopic ureter

- **Meyer–Weigert**: the upper-pole orifice is inferomedial and obstructs or is ectopic; the lower-pole orifice is superolateral and refluxes ("drooping lily").
- Suspect a duplex when the ureter is more dilated than the pelvis. A ureterocele is a thin-walled cyst in the bladder; in girls it can cause LUTO.
- Examine the introitus; US with bladder; CAP from birth; VCUG; scintigraphy.
- Endoscopic puncture is increasingly first-line, and mandatory for outlet obstruction or sepsis. Without VUR use an upper-tract approach; high-grade VUR usually needs ureterocele excision and common-sheath reimplant.

Exstrophy

- Both forms show a non-visualised bladder, low umbilicus and pubic diastasis.
- **Cloacal exstrophy (OEIS)** adds omphalocele, imperforate anus and spinal, renal and limb anomalies, so get KUB, renal and spinal US, with or without MRI.
- Bladder exstrophy: renal US, not an emergency; about 50% detected prenatally.

Cloaca, urogenital sinus, CAH

- Distension of both bladder and vagina means outlet obstruction; reflux through the tubes causes urinary ascites.
- **Cloaca** (~1/50,000 girls): single channel, pelvic cyst, calcified meconium. Polyhydramnios suggests tracheo-oesophageal fistula (TOF). **Urogenital sinus**: no bowel obstruction and normal external genitalia.
- **Congenital adrenal hyperplasia (CAH)**: 21-hydroxylase (6p), autosomal recessive; 75% are salt-wasting. Only **1 in 8** at-risk pregnancies is an affected girl, so prenatal treatment is controversial. Check electrolytes first, then 17-hydroxyprogesterone (17-OHP) and genetics.

Myelomeningocele

Fetal closure before 26 weeks reduces ventriculoperitoneal shunting and improves motor and cognitive outcome. It risks preterm labour and uterine dehiscence, and gives **no bladder benefit**.

8 · Neonatal urological emergencies

These need a quick, correct first step.

Urinary ascites and retention

- Urine causes up to 30% of neonatal ascites (respiratory distress, low sodium, high potassium, raised creatinine). **PUV is the commonest cause**.
- Bladder rupture also causes it: spontaneous rupture heals with a catheter, but injury from an umbilical catheter needs repair.
- Retention >24 h suggests PUV or atresia in boys, and urogenital sinus or ureterocele in girls.
- If a catheter fails, use an angiocatheter (16 G takes a 3.5 Fr; 14 G a 5 Fr). Persistent distension needs vesicostomy, with or without vaginostomy.

Abdominal masses

- **Congenital mesoblastic nephroma** is the commonest solid renal mass under 6 months (hypertension, polyhydramnios); Wilms peaks at 1–5 years. Ovarian cysts mostly resolve by 6 months.

- **Adrenal haemorrhage versus neuroblastoma**: haemorrhage has no Doppler flow, shrinks on 1–2-weekly US and calcifies like an eggshell; neuroblastoma is vascular, persists and has stippled calcification.
- Haemorrhage is usually right-sided and bilateral in 10–15%. It may track down the processus and mimic an acute scrotum.

Anorectal malformation and ambiguous genitalia

- Anorectal malformation has genitourinary or spinal anomalies in up to **75%**: renal and spinal US; CAP until videourodynamics at 6 weeks.
- Ambiguous genitalia needs the disorder of sex development (DSD) team; defer gender assignment. The commonest cause is CAH (46,XX): impalpable gonads and a uterus.
- **Exclude salt-wasting first**. Then send karyotype, electrolytes, 17-OHP, testosterone and dihydrotestosterone (*see Ch 59*).

Interlabial masses

Mass	Position	Management
Imperforate hymen	Midline bulging membrane	Hymenal incision
Urethral prolapse	Midline collar at meatus	Conservative
Prolapsed ureterocele	Off-midline from meatus; hymen intact	US and VCUG
Paraurethral cyst	Off-midline, displaces meatus	Most resolve

Gross haematuria

First exclude an oestrogen-withdrawal bleed and urate crystals, then get a Doppler US. Haematuria with hypertension means severe disease.

Feature	Renal vein thrombosis	Renal artery thrombosis	Cortical necrosis
Risk	Maternal diabetes, prematurity, umbilical venous catheter, polycythaemia; prothrombotic state in ≤50%	Umbilical artery catheter	Hypoperfusion
Presentation	**Triad**: haematuria, flank mass, thrombocytopenia	Hypertension, heart failure	Renal failure
Treatment	Hydration, with or without anticoagulation	Remove catheter, with or without thrombolysis	Supportive

Scrotal masses and torsion

- Differential: extravaginal torsion, hernia, hydrocele, meconium peritonitis (stippled calcification), adrenal haematoma and tumour (yolk sac, teratoma, juvenile granulosa).
- **Prenatal torsion** gives a hard, non-tender, discoloured testis at birth. Salvage is nearly nil, so do urgent contralateral orchidopexy, with or without exploration.
- **Postnatal torsion** (first 30 days) has salvage up to 44%: explore immediately and fix the other side, to prevent asynchronous contralateral torsion.

9 · Landmark trials

The evidence behind counselling.

Trial	Design	Result	Take-home
PLUTO	Randomised controlled trial (RCT), VAS vs conservative in fetal LUTO; 31 randomised, stopped early	Fewer perinatal deaths; no renal benefit; 44% complications	Shunting saves lungs, not kidneys
Great Ormond Street pyeloplasty RCT	APD ≥15 mm, DRF >40%: pyeloplasty vs observation	Most observed kidneys stable or improved; a minority (APD 20–40 mm) lost function	Observation safe with monitoring
MOMS†	RCT, prenatal vs postnatal myelomeningocele closure	Less shunting, better motor outcome; no urodynamic benefit	Fetal closure is not for the bladder

35 Urologic Evaluation of the Child

1 · Triage by acuity

First decide how fast to act.

Triage asks: **what is lost by waiting** — an organ, a life, or nothing?

Tier	Conditions
Emergent	Acute scrotum; acute abdomen; acute kidney injury (AKI); obstruction with urinary tract infection (UTI) and/or AKI; retention; incarcerated hernia; paraphimosis; priapism; trauma; newborn ambiguous genitalia, exstrophy, renal vein thrombosis
Urgent (24 h)	Gross haematuria **without infection**; newborn bilateral or solitary-kidney hydronephrosis; renal or testicular tumour; spina bifida
Semiurgent (1–3 wk)	After treated febrile UTI; postnatal dilation (no outlet obstruction); reducible hernia; interlabial mass; amenorrhoea
Routine	Hydrocele; circumcision; incontinence or enuresis; hypospadias; labial adhesions; meatal stenosis; microscopic haematuria; recurrent UTI; undescended testis; varicocele

KEY POINT

Context shifts the tier: a hernia is emergent if incarcerated but semiurgent if reducible; hydronephrosis is urgent if bilateral or in a solitary kidney.

2 · History

History often holds the diagnosis.

- **Prenatal** history (key for hydronephrosis): in vitro fertilisation (IVF), growth restriction, maternal diabetes, anomalies, karyotype discordant with ultrasound (US) genitalia.
- Ask gestation at detection, progression and **amniotic fluid**, a proxy for fetal renal function.
- **Consent from guardian, assent from child**; know who legally holds consent (foster care). Bullying or abuse may surface. Adolescents get a private history.
- In neurogenic bladder, mobility, hand dexterity and sensory level decide self-catheterisation.

3 · Examination

Children hide shock until late.

- Minimum systolic blood pressure (BP): **≥60** neonate; ≥70 at 1 month–1 year; **70 + 2×age** at 1–10 years; ≥90 above 10 years.
- **Shock shows first as tachycardia and tachypnoea; hypotension means decompensation.** Weigh every visit (dosing).
- Examine boys with a parent or chaperone. Prepubertal: supine frog-leg; adolescent: standing with Valsalva (hernia, varicocele).
- P1 Testicular development, gonadal descent, abnormalities of descent and bilateral undescended testis — *examining for undescended testis: warm room, sweep from ASIS, cross-legged/squat, ectopic sites; retractile; no imaging*

Not in the paper answer:

- One hand sweeps from the anterior superior iliac spine (ASIS) while the other delivers the testis; soap or lubricant helps.
- Palpate the mons, perineum and upper thigh for an ectopic testis.

- **Tanner staging**: the first male sign is testicular and scrotal enlargement (II), not hair; hair reaching the medial thighs is V.
- **Micropenis**: stretched length below mean − 2.5 standard deviations (SD), **1.9 cm** at term and 9.3 cm in adults. It barely changes at 5–11 years; preterms need gestation-specific values.
- Girls: frog-leg; **caudal traction on the labia majora** opens the vestibule.
- A **simple sacral dimple** (inside the gluteal cleft, <5 mm, base visible, <25 mm above the anus) needs no imaging. Image for lipoma, tag >5 mm, deep or atypical dimple, or a cleft deviated by a mass.

4 · Urine and blood tests

Poor samples give false UTI diagnoses.

- Febrile under 24 months: **catheter or suprapubic aspirate** for urinalysis and culture. **Bag specimens have high false-positive rates** (useful only if negative). †Toilet-trained: clean-catch midstream.
- American Academy of Pediatrics (AAP) UTI: pyuria and/or bacteriuria with **≥50,000 colony-forming units (CFU)/mL** (catheter or aspirate) in a symptomatic child.

Test	Threshold
Microscopic haematuria	**>5 red cells (RBC)/high-power field** (adult ≥3), on >2 occasions
Calcium/creatinine ratio (hypercalciuria)	>0.4 at age 2–5 years; >0.2 above 5 years
Normal protein/creatinine ratio	<0.7 in the first year; <0.4 up to 4 years; <0.2 after 5 years; >3.0 is nephrotic
Specific gravity	<1.010 dilute (water excess, diabetes insipidus); >1.020 dehydration

- Always **confirm dipstick blood by microscopy**. Test proteinuria on a **first-morning** sample (orthostatic is common); if persistent on two, refer to nephrology.

5 · Imaging

Each test answers one question.

Only US and magnetic resonance imaging (MRI) avoid radiation; keep doses as low as reasonably achievable (ALARA).

Modality	Answers
US	Kidney, bladder, scrotum, torsion, spine
Voiding cystourethrogram (VCUG)	Reflux, urethra, genitogram
Computed tomography (CT)	Severe trauma, stones
MRI	Tumours, ectopia, duplication, Müllerian or sinus anatomy; MR urogram also gives function
Mercaptoacetyltriglycine (MAG3) renogram	**Obstruction** and differential function
Dimercaptosuccinic acid (DMSA) scan	**Scarring** or pyelonephritis, and differential function

6 · Emergent presentations

Delay here costs an organ or a life.

Acute scrotum

- Commonest is **torsion of the appendix testis** (prepubertal, point tenderness); its US hyperaemia mimics epididymitis. Treat supportively.
- **Cord torsion** (mostly over 10 years): severe pain, vomiting, high-riding testis, absent cremasteric reflex (a present reflex does not exclude it).
- It is a **clinical diagnosis**: explore without imaging. Manual detorsion does not replace exploration; fix the other side.

TWIST item	Points	Score and action
Testicular swelling	2	0: no US
Hard testis	2	1–5: scrotal US
Absent cremasteric reflex; nausea or vomiting; high-riding testis	1 each	**≥6**: explore without US

- **Neonatal (extravaginal) torsion**: firm, painless testis, often prenatal; authors prefer urgent exploration.

Other emergencies

- **Ambiguous genitalia**: exclude **salt-wasting congenital adrenal hyperplasia (CAH)** (21-hydroxylase in about 95%). Crisis comes on day **4–15**, so no early discharge. Check **17-hydroxyprogesterone (17-OHP) and electrolytes**.
- **Exstrophy**: silk cord tie (no clamp), cling film over the plate, transfer.
- **Infected obstruction**: drain promptly (stent or nephrostomy; in neonates pyelostomy, ureterostomy, ureterocele puncture, vesicostomy).
- **Renal vein thrombosis**: haematuria, flank mass, thrombocytopenia; Doppler.
- **Paraphimosis**: exclude a **hair tourniquet**. Compress the glans with traction under analgesia; if it fails, dorsal slit.
- **Priapism** (≥4 h) is ischaemic, usually from **sickle cell disease**: oxygen, fluids, analgesia; cold packs only if sickle cell is not suspected. Neonatal priapism is benign, resolving in about 4–5 days.
- **Trauma**: haematuria and vital signs are unreliable, so **mechanism** decides CT. After straddle injury consider **sexual abuse**; meatal blood in a boy needs a retrograde urethrogram, not a blind catheter.
- **Retention**: expected capacity (mL) = **(age + 2) × 30**. Try analgesia, sitz bath and disimpaction before a catheter; if no cause, evaluate neurologically.

7 · Common referral presentations

Most clinic work, with clear investigation rules.

Antenatal hydronephrosis

- Scan **before discharge** if bilateral, solitary kidney or suspected outlet obstruction (third-trimester anteroposterior diameter >7 mm or urinary tract dilation A2–3); otherwise after **48 h**. Creatinine is maternal for 48 h, so follow the trend.
- A boy with bilateral dilation, a thick bladder and a dilated posterior urethra needs a **VCUG before discharge**, catheterised meanwhile. Voiding does not exclude obstruction.

Haematuria

- Glomerular: tea-coloured, dysmorphic RBCs, casts, protein; check C3, C4, antinuclear antibody, antistreptolysin O.
- Urological: bright red, clots, terminal; US, then cystoscopy and retrogrades if persistent.
- All gross haematuria: urinalysis, culture, US. Haemorrhagic cystitis follows marrow transplant (BK virus) or cyclophosphamide. Wilms is the commonest renal tumour under 15 (peak 1–5 years).
- Isolated microhaematuria is usually benign (thin membrane, IgA nephropathy, hypercalciuria): repeat over 2–3 weeks; if persistent, US and calcium/creatinine.

Bladder bowel dysfunction and enuresis

- Use validated questionnaires; **always examine the back**, gait, toe-walking. Treat stepwise: behavioural, then medical, biofeedback, physiotherapy, psychology; surgery is rare.

Scrotal, inguinal and genital

- **Hernia or hydrocele**: a communicating hydrocele is smaller in the morning; a parent's history or photo suffices; high ligation at the internal ring.
- An **undescended testis** with hypospadias or micropenis suggests a disorder of sex development (DSD).
- **Varicocele** (about 15% of adolescents): Lambert volume L×W×H×**0.71**. If non-decompressing supine or right-sided, do a retroperitoneal US.
- Prepubertal **testicular tumours** are mostly benign (teratoma, epidermoid; malignant is yolk sac). α-fetoprotein, β-human chorionic gonadotrophin (β-hCG) and lactate dehydrogenase guide testis-sparing surgery.
- **Phimosis**: never forcibly retract; treat lichen sclerosus; steroid if UTI risk.
- **Meatal stenosis**: circumcised boys, deflected thin stream, meatus <5 Fr; meatotomy under topical anaesthetic cream (EMLA).
- **Labial adhesions** (peak 13–23 months): observe; if symptomatic, oestrogen or steroid cream.
- **Interlabial mass**: US for upstream obstruction; consider rhabdomyosarcoma. **Amenorrhoea** with cyclic pain suggests outflow obstruction.

> **KEY POINT**
> Red flags: absent cremasteric reflex with acute pain; ambiguous genitalia with impalpable gonads; bilateral hydronephrosis in a boy; fever with obstruction; hypotension; abdominal mass; meatal blood; lumbosacral lipoma or deviated cleft.

8 · Child comfort

A calm child may avoid anaesthesia.

- Distraction, midazolam, intranasal ketamine and EMLA cream can avoid general anaesthesia.

36 Pediatric Urogenital Imaging

1 · Principles — ALARA and the three harms

Why this matters: children live long, so each test is weighed against radiation, sedation and contrast harm.

- **ALARA** ("as low as reasonably achievable", ICRP) is the standard. Radiosensitivity cannot be predicted, so **every child is treated as radiation sensitive**: developing tissue and a long lifetime raise malignancy risk.
- **Sedation risk usually exceeds contrast risk** (gadolinium nephrogenic systemic fibrosis in immature neonatal kidneys; iodinated nephropathy).
- Magnetic resonance imaging (MRI) needs sedation in young children; avoid fetal MRI in the first trimester. Ultrasonography (US) is **safe at all ages and gestations**.

> **KEY POINT**
> Every choice trades **radiation, sedation and contrast** against yield; the long lifespan tips it towards US, and the question is which test fills US's functional gap (obstruction, reflux, pyelonephritis) most cheaply.

2 · Ultrasonography

Why this matters: US comes first, but shows anatomy, not function.

- P1 USG in neonates — *technique; neonatal pyramids mimic dilation; under-read in first 48 h; SFU grades 0–4; PUV keyhole, MCDK, ARPKD, masses; normal US does not exclude VUR*

Not in the paper answer:

- Dehydration also hides hydronephrosis.
- Reports vary, so review the images yourself.
- Normal US does not risk-stratify pyelonephritis.
- Fetal urine is made from **10–12 weeks**; by **20 weeks** US detects most anomalies. Check cord insertion and abdominal wall (exstrophy).
- Society for Fetal Urology (SFU) grade 3 is heterogeneous; grade 4 is defined by parenchymal thinning, not more dilatation.
- For obstructive vs non-obstructive megaureter or ureterocele, use diuretic renography or MR urography (MRU). **Bilateral hydroureteronephrosis with a thick, poorly emptying bladder needs a voiding cystourethrogram (VCUG)** for posterior urethral valves (PUV).

Cysts, masses, stones

Lesion	Side	US hallmark
MCDK (multicystic dysplastic)	Unilateral	Non-communicating random cysts, scant echogenic parenchyma
ARPKD (recessive polycystic)	Bilateral	Large homogeneously **bright** kidneys, not discrete cysts
ADPKD (dominant)	Bilateral	Parenchyma almost replaced by cysts
Multilocular cystic nephroma	Unilateral	Like MCDK, more parenchyma; infancy
Cystic Wilms	Unilateral	Overlaps; 2–4 years

- Cysts do not communicate; dilated calyces join the pelvis.

- A neonatal solid mass is usually **mesoblastic nephroma**; a toddler's, Wilms. Neonatal enlargement with acute kidney injury and hypertension means **renal vein thrombosis**. Contrast computed tomography (CT) follows US.
- **Stones**: US is used first and for follow-up. A stone is echogenic **with** shadowing (fat: without); Doppler **twinkling artefact** confirms it.

Infection and trauma

- In urinary tract infection (UTI), US is not routine at the start. **If there is failure to improve on antibiotics, do US** (abscess, obstruction). It misses low-grade vesicoureteral reflux (VUR), duplication, acute pyelonephritis (APN) and scars, where DMSA is better. **Normal US neither risk-stratifies APN nor predicts VUR.**
- In trauma, US compares favourably with contrast CT for follow-up.

Contrast-enhanced voiding urosonography (CeVUS)

- Sulfur hexafluoride microbubbles grade VUR on five grades with **no radiation**. Area under curve **0.97**, sensitivity ~92%, specificity ~91–94%, so it is recommended for VUR.
- It detects **intrarenal reflux** (during voiding, lower pole, rising with grade), which mirrors where DMSA scars occur.

Bladder/urethra

- The **keyhole sign** (thick bladder and dilated posterior urethra) means PUV. Measure pre- and post-void volumes.
- Clot vs tumour: scan supine then prone; a **clot moves**. Viral cystitis gives a thick wall mimicking a mass, after an upper respiratory prodrome, with gross haematuria.

Scrotum and genitalia

- **Never delay exploration for suspected torsion**.
- The **whirlpool sign** gives torsion sensitivity ~96% vs ~76% for colour Doppler, specificity ~99%. No flow with heterogeneous parenchyma means non-viable.
- Epididymo-orchitis shows hyperaemia (also after spontaneous detorsion). Appendage torsion is an avascular nodule with surrounding hyperaemia. Idiopathic scrotal oedema shows the **fountain sign**.
- Prepubertal tumours: **teratoma** is commonest (solid, cystic, calcified); yolk sac is homogeneous and vascular; epidermoid shows **onion rings** and no flow.
- **Do not image a routine undescended testis** (AUA Choosing Wisely) except in an obese child. US is primary for internal structures in differences of sex development (DSD).

3 · Voiding cystourethrography

Why this matters: VCUG is the reference test for reflux and urethral anatomy.

- No sedation is needed. The scout may show spinal anomalies. A non-midline catheter suggests utricle or ectopic ureter.
- **After UTI**, do it once improving on antibiotics; a 4–6-week wait is unnecessary. Very early VCUG may **overgrade** VUR because endotoxin dilates the ureter.
- Indications: hydronephrosis with **dilated ureter(s)**; febrile or recurrent UTI; suspected PUV; boys with epididymitis (ectopic ureter/utricle); UTI in a circumcised boy.
- **Occult VUR**: VCUG finds VUR in only ~40% of DMSA-proven APN. **PIC cystography** (positional instillation of contrast: a cystoscopic stream at the orifice) is used when VCUG is

negative but febrile UTI recurs.

- **Trauma**: with meatal blood, do a **retrograde urethrogram (RUG) before any catheter**, then gravity fill. Capacity (mL) = **(age + 2) × 30**, max 400. A "pie-in-the-sky" bladder means pelvic fracture with extraperitoneal injury.
- **Genitography** (DSD, cloaca, sinus) defines the urethrovaginal confluence (Shopfner classification).

4 · Nuclear medicine

Why this matters: nuclear scans show scars, split function and drainage.

DMSA

- Dimercaptosuccinic acid (DMSA) is taken up by proximal tubules, so APN shows photon-deficient areas.

Feature	Acute pyelonephritis	Scar
Contour	Preserved	Volume loss, concavity
Timing	Within days; resolves over ~5 months	Lesion at **≥6 months** = scar

- A third to a half of acute lesions become scars.

MAG3 diuretic renography

- P1 Radionuclide studies and diuretic renography in PUJ obstruction — *MAG3 vs DTPA vs DMSA; well-tempered protocol (hydration, catheter, diuretic timing, ROI); DRF, T½ from maximal filling; biphasic curve; pitfalls*

Not in the paper answer:

- Region-of-interest placement localises the level (ureteropelvic or ureterovesical).
- The catheter also lowers gonadal dose.
- Give furosemide **1 mg/kg IV** at maximal filling; if too early, drainage is measured before the system is full. †DTPA is noisier in the immature neonatal kidney.
- There is **no validated paediatric half-time (T½) cut-off**, so decide on age, function, T½, curve, upright drainage and serial studies. A **biphasic curve** and poor upright drainage predict progression.
- In Dietl's crisis, cortical retention lasts up to 48 h after an attack; a scan between attacks may be normal.

Radionuclide cystography and testicular scan

- **Direct** cystography needs a catheter; it is more sensitive and lower-dose than VCUG, but anatomy is poor and grading coarse, so it suits follow-up. **Indirect** is insensitive and not routine.
- Testicular scan: cold means torsion, hot means inflammation. Use only if equivocal, without delay.

5 · MR urography

Why this matters: MRU gives anatomy and function without radiation, but needs time and sedation.

- It suits complex anomalies (ectopic ureter, duplication, calyceal diverticulum).
- Protocol: hydration and catheter, then **furosemide 1 mg/kg (max 20 mg)**, T2, gadolinium and dynamic T1; function by **Patlak** analysis.

- Limits: up to 1 h; sedation under 6 years; equivalence to renography unproven. Poor for stones, infection, trauma and scrotum.
- Best for **bladder rhabdomyosarcoma** (non-enhanced sequences first) and **local staging of renal tumours** and contralateral lesions.

6 · CT and radiography

Why this matters: CT is chosen for speed, not superiority.

- CT serves trauma and staging, not hydronephrosis.
- **In trauma, arterial-phase images miss collecting-system injury, so delayed excretory images are mandatory.** CT cystography separates intra- from extraperitoneal (and augment) rupture.
- CT quantifies stones best but rarely changes care. A kidney–ureter–bladder (KUB) film shows stones and constipation. Intravenous pyelography (IVP) is obsolete. In retrograde pyelography, free retrograde flow does not prove antegrade patency.

7 · Choice of test by clinical question

Why this matters: this is the one-page answer to "which test next?"

Question	First line	Next	Avoid
Hydronephrosis — obstruction?	US	MAG3; MRU	CT, IVP
Hydroureteronephrosis, with or without thick bladder	US	VCUG	—
VUR	VCUG	CeVUS; direct cystogram; PIC if occult	Indirect cystogram
APN / scar	DMSA	—	Grey-scale US; MRI
Stones	US	Low-dose CT; KUB	Reflex CT
Renal mass	US	Contrast CT; MRI	—
Bladder mass	US (full bladder)	MRI	—
Trauma	CT with delayed phase; US	RUG; cystography	Catheter before RUG if meatal blood
Acute scrotum	Clinical, with or without Doppler/whirlpool	Scintigraphy if equivocal	Imaging delaying exploration
Undescended testis	Examination	US only if obese	Routine US, MRI
DSD	US and genitography	MRI	CT

37 Infection and Inflammation of the Paediatric Genitourinary Tract

1 · Overview and the paradigm shift

Why this matters: UTI is common, and practice now investigates far less.

- Urinary tract infection (UTI) affects ~**1% of boys, 3–5% of girls**, and 30–50% recur. It is the commonest bacterial infection in febrile children (~7% of febrile infants).
- Old view: febrile UTI with vesicoureteral reflux (VUR) caused scarring, hypertension and chronic kidney disease (CKD), so work-up was aggressive. **Now** congenital dysplasia matters more, so the approach is **selective imaging, prompt treatment and prophylaxis only for the high-risk child**.

2 · Pathogenesis and virulence

Why this matters: how bacteria stick and hide explains recurrence despite correct antibiotics.

- Infection is **ascending**: gut bacteria colonise the perineum, enter the bladder, adhere to umbrella cells, invade, then ascend and may cause bacteraemia. On catheters, fibrinogen deposits bind bacteria, forming biofilm.
- Uropathogenic *E. coli* (UPEC) causes most UTIs; its factors let it **adhere, feed, hide and damage**.

Factor	Mechanism	Association
Type 1 fimbriae (mannose-sensitive)	FimH binds uroplakins, allowing entry	Cystitis, asymptomatic bacteriuria (ASB)
P fimbriae (mannose-resistant)	Bind Galα1–4Galβ (P blood group)	**Pyelonephritis**
Aerobactin	Siderophore	Growth in iron-poor urine
α-Haemolysin, cytotoxic necrotising factor 1 (CNF1)	Cell lysis, cytoskeletal damage	Inflammation, invasion
Capsule (K antigen)	Antiphagocytic, anticomplementary	Pyelonephritis
Biofilm, flagella	Shield; motility	Recurrence; ascent

- Host toll-like receptor 4 (TLR4) triggers cytokines that recruit neutrophils. Without attachment, carriage is asymptomatic; if clearance fails, inflammation is destructive.
- **Intracellular bacterial communities** (IBCs) form a **quiescent intracellular reservoir** that lasts months and reseeds the lumen, causing same-strain recurrence despite adequate antibiotics.

> **KEY POINT**
> IBCs and biofilms shield bacteria from neutrophils and antibiotics; a biofilm survives 1000–1500× normal killing concentrations (poor penetration, slow growth, altered targets).

3 · Host risk factors

Why this matters: many factors are modifiable.

- In the first year, UTI affects boys **3.7%** vs girls 2%; by 6 years, boys 2% and girls 7%. A febrile girl aged 2 months–2 years has ≈ 2.3× a febrile boy's risk. White > Black.
- P1 phenotype occurs in girls with recurrent pyelonephritis; Lewis Le(a−b−) and non-secretors are linked to scarring.
- **Circumcision** lowers UTI in the first 6 months **~10-fold**. US Army cohort: 0.11%, uncircumcised 1.12% and girls 0.57% MCQ. Number needed to circumcise: **111** normal boys, 11 with recurrent UTI and **4** with high-grade VUR. It protects in antenatal hydronephrosis.
- **Faecal reservoir**: the same strain cycles. Antibiotics, especially trimethoprim–sulfamethoxazole (TMP-SMX), destroy protective flora.
- **VUR** is found in 1–2% of newborns and **25–40%** after a first UTI. Most pyelonephritis occurs without VUR, but grade III–IV doubles pyelonephritic change *(see Ch 41)*.
- **Bladder and bowel dysfunction (BBD)** (infrequent voiding, incomplete emptying, constipation, overactivity): treating it cuts recurrence and aids VUR resolution.
- **Neurogenic bladder** causes ~2.5 UTIs/year, and 40–80% of children on clean intermittent catheterisation (CIC) are bacteriuric. **ASB without VUR needs no antibiotics**, because prophylaxis breeds resistance. Catheter type is irrelevant; frequent catheterisation prevents UTI.
- **Catheter-associated UTI (CAUTI)** is the commonest nosocomial infection, at 3–10% per catheter-day, so remove catheters early.
- An IgG nadir at 1–3 months and little urothelial IgA make early infancy highest risk. Obesity raises risk.

> **KEY POINT**
> Host factors either raise inoculum/dwell time (prepuce, faecal flora, stasis, constipation, catheter) or lower defence (immature immunity, receptor-bearing blood groups); prevention reverses whichever applies.

4 · Classification

Why this matters: the recurrence type decides the remedy.

- UTI is **complicated** (anomaly, dysfunction, foreign body; all neonatal/infant UTI) or uncomplicated; upper or lower; first or recurrent.
- Febrile UTI is presumed pyelonephritis; afebrile UTI is cystitis. Localisation is poor: only **50–66%** of febrile children show acute changes on dimercaptosuccinic acid (DMSA) scan, the gold standard.

Recurrence	Between episodes	Cause	Remedy
Unresolved bacteriuria	Never sterile	Non-compliance, malabsorption, resistance	Culture-directed retreatment
Bacterial persistence	Sterile; same organism	Nidus: stone, anomaly, papilla, catheter, stent	**Remove nidus**
Reinfection	Sterile; new organism	New ascent	Prevention

Asymptomatic bacteriuria

- Kass definition: two cultures of **≥10^5 colony-forming units (CFU)/mL** of the same organism, without symptoms.

- Infants (<1% term, 3% preterm): treat and do renal–bladder ultrasound (RBUS). School-age girls without VUR or anomaly: **do not screen or treat**, because this selects resistance and replaces a harmless strain.

5 · Diagnosis

Why this matters: contaminated samples cause overtreatment and needless imaging.

Clinical

- Fever is ≥38 °C. A toxic look, age <30 days and ≥39.4 °C predict bacteraemia. Infants present nonspecifically (irritability, poor feeding, jaundice, vomiting).
- Under 2 years, risk factors are fever >40 °C, previous UTI, suprapubic tenderness and an **uncircumcised penis**. **Another fever source, such as otitis, only halves UTI risk.**
- Examine genitalia (phimosis, labial adhesions, ectopic ureter), spine and for constipation. In adolescents, consider gonorrhoea and Chlamydia.

Urine collection

Method	Contamination	Role
Bag	50–60% (false-positive culture up to 75%)	**Reliable only when negative**; never culture
Clean catch	~26%	Toilet-trained; preferred by NICE (UK National Institute for Health and Care Excellence)
Catheter	10–15%	Sensitivity 95%, specificity 99%; **standard <2 years non-toilet-trained**
Suprapubic aspiration (SPA)	<1%	Reference; US guidance gives 80–90% success

- Collect **before antibiotics** unless delay is dangerous (NICE). For 2–24 months, the American Academy of Pediatrics (AAP) accepts bag urinalysis; if positive, culture a catheter or SPA sample. Examine urine within 1 h, or 4 h if refrigerated.

Test	Sensitivity	Specificity	Use
Leukocyte esterase (LE)	~79%	~87%	Negative helps **rule out**
Nitrite	~50%	**~98%**	Positive **rules in**
LE or nitrite or microscopy	99.8%	70%	Best rule-out
Pyuria ≥5 white cells per high-power field (WBC/HPF)	~74%	~86%	No advantage over LE
Gram stain, fresh urine	~91%	92–96%	Most reliable rapid test

- **Nitrite is falsely negative** with short bladder dwell (infants), dilute urine and gram-positives (Enterococcus).
- Pyuria is less with Klebsiella and Enterococcus, so culture anyway. UTI without pyuria is rare, which separates it from ASB. A threshold of **≥10 WBC/HPF** works better under 2 years.
- **Culture is essential** unless urinalysis is entirely normal.

Guideline	Specimen	Significant count (CFU/mL)
Kass	Voided	$\geq 10^5$
AAP 2011 (2–24 months)	Catheter/SPA	**≥50,000 + pyuria**

Guideline	Specimen	Significant count (CFU/mL)
European Association of Urology (EAU)	Midstream/catheter	10^3–10^4 monoculture
Royal Children's Hospital (RCH) Melbourne	SPA / catheter	Any gram-negative / $>10^3$
ISPN (Italy)	Catheter / midstream	$>10^4$ / $>10^5$

- **The cleaner the specimen, the lower the accepted count.** Contamination: low counts, mixed growth, lactobacilli, coagulase-negative staphylococci, diphtheroids.
- C-reactive protein (CRP) is sensitive, not specific. **Procalcitonin ≥0.5 ng/mL** best predicts acute pyelonephritis (APN) and later scarring.

6 · Imaging after UTI

Why this matters: guidelines differ on whom to image.

- No evidence shows routine imaging prevents scarring, hypertension or renal failure, so **order a test only if it changes management**.
- Ultrasound (US) is abnormal in ~15% after a first febrile UTI, but only 1–2% need action. It is **urgent if the child is unwell or not responding at 48 h**. Pitfalls: oedema, lobar nephronia mimicking tumour, and endotoxin dilating the pelvis. US is poor for VUR and scars.
- Voiding cystourethrogram (VCUG) is the gold standard for VUR. Do it with sterile urine, in a well child. Cyclic filling increases sensitivity, and the volume at reflux onset predicts resolution. Contrast-enhanced voiding urosonography (CeVUS) is an alternative.
- DMSA (~1 mSv) shows APN best within **1 week**; a defect at **≥6 months** is a scar. About 15% of acute lesions scar, and a scar cannot be told from dysplasia. Computed tomography (CT, 10–15 mSv) is only for occult stones or mass vs inflammation.

Guideline	Ultrasound	VCUG	Late DMSA
AAP (2–24 months)	Yes, after first febrile UTI	If US abnormal (or recurrence)	No
NICE <6 months	Yes	Atypical/recurrent	Atypical/recurrent
NICE >6 months	Atypical/recurrent	No	Atypical/recurrent
Top-down	No	If acute DMSA abnormal	If acute DMSA abnormal
RCH Melbourne	If <3 months or seriously ill	No	No

- **NICE "atypical" UTI** means seriously ill, poor urine flow, abdominal or bladder mass, raised creatinine, septicaemia, no response within **48 h**, or a non-*E. coli* organism. Late DMSA is at 4–6 months.
- NICE and AAP miss ~half of grade III–V VUR but spare most children a VCUG, because high-grade VUR declares itself at recurrence. The AAP Urology Section still accepts VCUG (RIVUR).

7 · Treatment

Why this matters: delay increases scarring.

- Urine is sterile within 24 h of effective therapy. **Delay matters**: new-scar odds double if antibiotics start after 72 h vs within 48 h. Start empirically after urine collection.
- **Oral therapy equals intravenous (IV)** for non-toxic children over 2 months who tolerate oral drugs.

- **Admit** if under 1 month (some say 2–6 months), toxic, dehydrated, vomiting or of doubtful compliance. Neonates need a septic work-up (bacteraemia ~20%, hyponatraemia, hyperkalaemia) and IV therapy, switching to oral after ≥24 h afebrile; ≤2 days IV is safe in young infants.
- 90% are afebrile by 48 h. **If not improving at 48 h, do RBUS and broaden cover**.

Infection	Duration
Febrile UTI	**7–14 days** (shorter is inferior)
Afebrile cystitis	**2–4 days** (beats single dose; equals 7–14 days)
Lobar nephronia	≥3 weeks
Renal abscess	Antibiotics; drain if no response
Pyonephrosis	Antibiotics and drainage

- Choose by the **local antibiogram**. TMP-SMX and amoxicillin are poor empirical agents. Prefer **cephalexin**, and nitrofurantoin for cystitis. Urinary levels exceed serum breakpoints >100×, so do not escalate needlessly.
- *E. coli* causes 83% in outpatient girls, ~37% in inpatient boys. **Proteus is a boys' organism**.
- **Neonates** need Enterococcus cover, so give **ampicillin with a third-generation cephalosporin or an aminoglycoside**.
- Use broad cover after previous UTI, recent antibiotics or admission, or with an anomaly; aminoglycoside for Pseudomonas. Fluoroquinolones are reserved (Food and Drug Administration: complicated UTI/pyelonephritis).

Agent	Dose (mg/kg/day)	Restriction
Cephalexin	50–100 in 4	—
Cefixime	8 once daily	—
Amoxicillin–clavulanate	20–40 in 3	—
TMP-SMX	6–12 (TMP) in 2	**<6 weeks/premature**: kernicterus
Nitrofurantoin	3–5 in 2	**<3 months**, glomerular filtration rate (GFR) <50%, G6PD deficiency; **never for febrile UTI**
Ceftriaxone (IV)	75 once daily	—
Gentamicin (IV)	7.5 (once daily acceptable)	Nephro-/ototoxicity †

No repeat culture is needed if the organism was sensitive.

8 · Renal scarring and sequelae

Why this matters: scar risk climbs with each pyelonephritis.

- Scars favour the **poles**, where compound papillae allow intrarenal reflux; risk rises with VUR grade.

Pyelonephritic episode	1st	2nd	3rd	4th	5th
Scar risk	5%	10%	20%	40%	60%

- A small kidney with diffusely low uptake is usually **dysplasia**. Steroids are not routine.

- Childhood UTI causes **<1% of CKD**, and end-stage kidney disease (ESKD) follows in ~1 in 10,000. Scars lead to hypertension and pre-eclampsia.
- Follow height, weight and blood pressure (BP). Bilateral scars need lifelong BP, function and proteinuria checks (renin–angiotensin blockade).

9 · Prevention and prophylaxis

Why this matters: prophylaxis breeds resistance, so it is targeted.

- Recurrence is **10–30%**, highest in the first 3–6 months; in girls up to 40%.
- **Continuous antibiotic prophylaxis (CAP)** is not given after a first UTI (AAP, NICE). It is for the **high-risk** child: girls with dilating VUR (≥III), BBD or recurrent febrile UTI. Low risk: circumcised boy, no BBD, normal imaging.
- Give **~¼ therapeutic** dose at bedtime, using a different agent from treatment, because prior exposure triples resistance. Options are TMP-SMX (not under 6 weeks), nitrofurantoin (spares faecal flora) and a cephalosporin (young infants).
- **BBD**: with VUR and BBD, recurrence is ~45% vs 15%. Treatment steps up: timed voiding and laxatives, anticholinergics, biofeedback, then CIC *(see Ch 47)*.
- Non-antibiotic options: **circumcision** for high-risk boys (recurrent febrile UTI, high-grade VUR, hydroureteronephrosis, obstructive uropathy). Topical steroid for phimosis is reasonable; cranberry is mixed, probiotics adjunct, D-mannose unproven.

10 · Special infections

Why this matters: these need drainage, longer courses or surgery.

- P1 Xanthogranulomatous pyelonephritis. / Fungal UTI — *XGP: Proteus, stone–obstruction triad, mimics tumour, nephrectomy (diffuse) vs partial (focal), hard dissection; Candida risks, C. glabrata resistance, treat >10,000–15,000 CFU/mL, fungus ball drainage + amphotericin*

Not in the paper answer:

- In children, xanthogranulomatous pyelonephritis (XGP) has calculi in only 38–70% and presents with failure to thrive and anaemia.
- Paediatric Candida risks include prematurity and central or umbilical lines.
- With catheter funguria, first change the catheter and stop antibiotics.
- **Lobar nephronia** shows wedge-shaped hypoenhancement on CT that mimics a mass; treat with antibiotics for ≥3 weeks. **Pyonephrosis** must be **drained**. **Renal abscess**: urine culture is negative in up to 20%; US detects lesions ≥1 cm and CT is most accurate.
- **Viral haemorrhagic cystitis** is due to adenovirus 11 (and 21). After marrow transplant, severe disease carries 50–80% mortality. **BK virus** is treated with **cidofovir**.
- †**Prepubertal epididymitis** is usually sterile. If culture-positive or recurrent, do US and VCUG for ectopic ureter, utricle or obstruction. Exclude torsion.
- †**Balanoposthitis** is treated with soaks, hygiene and a topical agent. Recurrent episodes or pathological phimosis need circumcision.

11 · Landmark trials

Why this matters: these define who benefits from prophylaxis.

Trial	Design	Result	Take-home
RIVUR	607 children, VUR I–IV, TMP-SMX vs placebo, 2 years	Recurrent UTI ~halved (number needed to treat 8); **no difference in scarring**	Prevents recurrence, not scars; most benefit with BBD, febrile index UTI
Swedish Reflux Trial	Randomised, dilating VUR, 1–2 years	Prophylaxis reduced recurrence in girls	CAP for girls with dilating VUR

38 Core Principles of Perioperative Management in Children

1 · Perioperative surgical home and enhanced recovery

Why this matters: standard pathways shorten stay without adding complications.

- The **perioperative surgical home** (American Society of Anesthesiologists, ASA) uses surgeon–anaesthesiologist-led standardised pathways through to after discharge.
- **Enhanced recovery after surgery (ERAS)** combines counselling, shorter fasting, judicious intravenous (IV) fluid, no routine tubes or drains, opioid-sparing analgesia and early feeding and mobilisation. It shortens stay with no more complications.

2 · Preoperative assessment

Why this matters: missed airway or infection problems cause most avoidable events.

- Risk is raised by upper respiratory tract infection (URTI), asthma exacerbation, dehydration, uncontrolled diabetes, prematurity and fever.
- **Preoperative anxiety** (up to half) causes more pain, emergence delirium and new behaviour problems (enuresis); child life specialists, explanation and play help.
- **Jehovah's Witness**: a child is not allowed to die for lack of blood (court order); correct anaemia and coagulopathy beforehand.
- **Investigations are indication-driven**: no routine bloods, chest X-ray or ECG, and a murmur alone is not an ECG indication. Check urine human chorionic gonadotrophin (hCG) in females of menstrual age; MCQ it can be falsely positive after augmentation cystoplasty.
- Send a **urine culture** before any procedure opening the urinary tract; reschedule until sterile. Always culture in neurogenic bladder. Treat asymptomatic bacteriuria before endourology, because mucosal injury can cause sepsis.
- **URTI** leaves the airway hyperreactive for **6–8 weeks**, which risks laryngospasm, bronchospasm and desaturation. A laryngeal mask is safer than an endotracheal tube. Postpone elective surgery ≥ 4, ideally 6 weeks.
- **Sepsis**: give antibiotics within **1 h** of septic shock and avoid neuraxial blocks. **Trauma**: assume a full stomach and use rapid-sequence induction.

3 · Fasting (NPO, nil by mouth)

Why this matters: over-fasting dehydrates small children.

Minimum fast	Material
2 h	Clear liquids, carbohydrate drinks (European and Canadian 2022: 1 h)
4 h	Breast milk
6 h	Infant formula, non-human milk, light meal, tube feeds
8 h	Regular meal

- Trap: non-human milk counts as a light meal. List the youngest first (they dehydrate faster). Inpatients get IV maintenance; medicines go with clear fluid up to 2 h.

- **Metabolic disease**: ketogenic diet, no carbohydrate (including suspensions); glycogen storage and urea cycle disorders, glucose fluid throughout; mitochondrial disease, avoid lactated Ringer and propofol.
- **Diabetes**: list the child first and stop metformin 24 h before. Adjust insulin rather than stop it (half the long-acting dose the night before). A long case or emergency needs IV insulin.
- Omit angiotensin-converting enzyme (ACE) inhibitors that morning. Stop warfarin and aspirin 7–10 days before. Never stop clonidine abruptly (rebound hypertension).

4 · Anaesthesia considerations

Why this matters: neonates and comorbid children need precautions.

Neonate and preterm infant

- Little reserve: surfactant deficiency, bronchopulmonary dysplasia (BPD), small blood volume, retinopathy of prematurity (accept saturation in the low 90s) and hypothermia.
- **Postoperative apnoea** peaks at 12–23 h, so give 23-h cardiorespiratory monitoring if postconceptional age is **< 50–60 weeks** (ex-preterm) or **< 44 weeks** (term). Defer elective surgery beyond these.

Anaesthetic neurotoxicity

- The Food and Drug Administration (FDA) warning (2016) says repeated or lengthy (**> 3 h**) anaesthesia at **< 3 years** or in third-trimester pregnancy may affect brain development. This rests on animal neuroapoptosis.
- Human trials of a single short exposure are negative (§11). Counsel that a single exposure < 1 h has not been shown harmful, and do not delay necessary surgery.

Comorbidities

- **Asthma**: give a β_2-agonist before induction, and stress steroids if on systemic or daily inhaled steroids. **Congenital heart disease**: highest risk is with single ventricle, pulmonary hypertension, left ventricular (LV) outflow obstruction and cardiomyopathy.
- **Chronic kidney disease (CKD)** brings anaemia, platelet dysfunction and hyperkalaemia. Avoid surgery within 4–6 h of haemodialysis, and give no non-steroidal anti-inflammatory drugs (NSAIDs) in CKD 3–4.
- **Sickle cell disease**: avoid dehydration (most important: first case, IV fluid during fast), hypoxia, acidosis, cold and stasis (no tourniquets). Acute chest syndrome is the main cause of death.
- **Phaeochromocytoma**: give α-blockade *before* β-blockade, with volume loading. Expect hypoglycaemia after resection.
- **Congenital adrenal hyperplasia (CAH)**: give IV stress-dose hydrocortisone with glucose-containing fluid.

Spina bifida

- Confirm ventriculoperitoneal (VP) shunt function; postpone if malfunctioning. Scoliosis causes restrictive lungs.
- **Latex allergy** develops from repeated mucosal exposure and can cause anaphylaxis; †care should be latex-free from birth.
- **Autonomic dysreflexia** (lesion at or above **T6**): bladder distension or another noxious stimulus causes severe hypertension and bradycardia, risking stroke. †Stop the stimulus,

drain the bladder, sit the child up and give a rapid antihypertensive.

5 · Fluid and electrolyte management

Why this matters: hypotonic fluids can cause fatal hyponatraemia.

- **Holliday–Segar 4-2-1 rule**: **4 mL/kg/h** for the first 10 kg, plus **2 mL/kg/h** for the next 10 kg, plus **1 mL/kg/h** per kg above 20. For 22 kg: 40 + 20 + 2 = 62 mL/h.
- Large-volume saline causes hyperchloraemic acidosis. Add dextrose under 2–4 months; give no potassium if dialysed.

Maintenance fluid	Historical	Current
Tonicity	Hypotonic (0.45% saline)	Isotonic (0.9% saline or lactated Ringer) with dextrose
Potassium	10 mEq/L if < 10 kg; 20 mEq/L if > 10 kg	

KEY POINT

Why isotonic? Surgical stress, pain, nausea and hypoxia trigger non-osmotic antidiuretic hormone (ADH) release, which retains water through aquaporin-2. Hypotonic fluid adds free water and causes hyponatraemia (< 135 mmol/L). †A child's large brain-to-skull ratio leaves little room for cerebral oedema, so encephalopathy can be fatal.

6 · Blood loss and transfusion †

Why this matters: small losses are proportionally large.

- Estimated blood volume (EBV): preterm 90–100, term neonate 80–90, infant 75–80, child 70–75 mL/kg.
- Maximum allowable blood loss = EBV × (starting − target haematocrit) / starting haematocrit. Crossmatch when expected loss approaches it.
- Transfuse at haemoglobin **< 7 g/dL** in stable children (higher in neonates, cyanotic heart disease and active bleeding). 10–15 mL/kg raises haemoglobin by ≈ 2–3 g/dL.

7 · Temperature

Why this matters: infants cool quickly.

- Infants lose heat fast (large surface area, little brown fat), causing acidosis, coagulopathy and apnoea. Use a warm theatre, †forced-air warming, warmed fluids and core temperature monitoring. Laparoscopy adds cooling from the gas *(see Ch 39)*.

8 · Antibiotic prophylaxis and intraoperative crises

Why this matters: timing and redosing prevent wound infection.

- **Timing**: give within **60 min** of incision; vancomycin and fluoroquinolones within **120 min**.
- Doses: cefazolin **30 mg/kg** (redose at 4 h); ampicillin 50 mg/kg (redose at 2 h); gentamicin 2.5 mg/kg, vancomycin 15 mg/kg, metronidazole 15 mg/kg and ciprofloxacin 10 mg/kg (not redosed).

	Anaphylaxis	Malignant hyperthermia (MH)
Triggers	Neuromuscular blockers, β-lactams, latex	Succinylcholine, volatile agents (not nitrous oxide)

	Anaphylaxis	Malignant hyperthermia (MH)
First sign	Collapse, bronchospasm, hives minutes after induction	Unexplained hypercarbia, tachycardia, masseter spasm; late — acidosis, hyperkalaemia, rigidity
Treatment	**Adrenaline**, 100% oxygen, fluids; send tryptase	Stop triggers, **dantrolene**, clean circuit; admit 24–48 h (recrudescence)

9 · Analgesia

Why this matters: blocks and scheduled non-opioids spare opioids.

- Under 3 years, self-report is impossible, so use FLACC (Face, Legs, Activity, Cry, Consolability).
- **Regional blocks under general anaesthesia (GA)** are safe and opioid-sparing: penile, **caudal**, spinal and epidural (from 2 kg). An ultrasound-guided pudendal block for hypospadias gives less pain and opioid use than caudal. †Caudal 0.25% bupivacaine is 0.5 mL/kg (sacral) to 1 mL/kg (lower thoracic); maximum bupivacaine 2.5 mg/kg.
- Give **multimodal, scheduled** analgesia: acetaminophen 15 mg/kg 6-hourly and ibuprofen 10 mg/kg 8-hourly. Ketorolac 0.5 mg/kg IV 8-hourly is limited to ≤ 72–96 h (renal toxicity). No NSAIDs under 6 months. Adjuncts are IV lidocaine, ketamine and dexmedetomidine.
- Opioids are for breakthrough pain; patient-controlled analgesia (PCA) works from ≈ 4 years. **Codeine is contraindicated**: poor metabolisers get no analgesia, and ultrarapid metabolisers make lethal amounts of morphine. Methadone needs an ECG for the QT interval.
- Discharge opioids: 3–5 days' supply; teach safe disposal.

10 · Postoperative care, VTE and day-case discharge

Why this matters: airway events and delirium occur in recovery.

- Laryngospasm: positive pressure, then propofol, then reintubation. Bronchospasm: salbutamol, then adrenaline, steroids, magnesium.
- **Emergence delirium** occurs at 2–8 years, within 30 min of waking. The child does not recognise parents (a child in pain does). Treat with low-dose propofol, dexmedetomidine or fentanyl.
- **Venous thromboembolism (VTE)** †is rare before puberty. At-risk adolescents (immobility, obesity, oestrogen, central venous catheter, malignancy, pelvic surgery, prior VTE) need mobilisation and mechanical prophylaxis, with low-molecular-weight heparin (LMWH) if several factors are present.
- **Day-case** surgery suits †ASA I–II children outside the apnoea-risk ages having a short, low-blood-loss procedure. Discharge criteria: Aldrete or the Post-Anesthetic Discharge Scoring System.

11 · Landmark trials

Why this matters: one short anaesthetic appears safe.

Trial	Design	Result	Take-home
GAS	Randomised; infant hernia, awake regional vs sevoflurane < 1 h	No cognitive difference at 2 and 5 years	Single short anaesthetic safe
PANDA	Sibling-matched; single exposure < 3 years	No IQ difference at 8–15 years	Repeated exposure untested

Trial	Design	Result	Take-home
MASK	Cohort; exposure < 3 years	Negligible deficit after single exposure	Reassuring

39 Principles of Laparoscopic and Robotic Surgery in Children

1 · Evolution and indications

Why this matters: the robot made paediatric reconstruction feasible.

- P3 Discuss diagnostics and operative pediatric laparoscopy in urology — *evolution 1976, 1994, 1995, 2002; indications; age IAP limits; only contraindication; access; complications (right common iliac, inferior epigastric, port-site hernia 0.5–3%)*

Not in the paper answer:

- Robotic limits: no 5-mm ports, no haptics; single-port needs a 25-mm cannula.
- Surgeons doing > 12 cases a year have fewer complications.
- Same-day discharge is feasible after uncomplicated robotic renal surgery.
- The first paediatric urological laparoscopy was for non-palpable testis in **1976**. Intracorporeal suturing was the bottleneck; the wristed robot removed it, so **reconstruction** grew (pyeloplasty commonest; reimplantation, (hemi)nephrectomy, intra-abdominal testis, urachal/diverticulum excision; deep pelvis: utricle, urogenital sinus, bladder neck).
- The only strict contraindication is **inability to tolerate pneumoperitoneum**. Age and size are not contraindications; cardiopulmonary disease needs optimisation.

2 · Physiology of pneumoperitoneum

Why this matters: infants have least reserve against raised pressure and absorbed CO_2.

- P1 Pneumoperitoneum physiology; Laparoscopic access and gases — *pressure vs CO_2 effects on heart, lungs, kidney, ICP; vagal bradycardia; CO_2 choice; gas embolism signs and treatment (left lateral head-down, oxygen, aspirate)*

Not in the paper answer:

- Paediatric pressure ceilings rise with age (table below).
- Infant oliguria is not a guide to resuscitation, so do not bolus.
- Infants cool fast from cold, dry gas.

Effects depend on pressure and reverse on desufflation, so use the **lowest age-appropriate pressure**: intra-abdominal pressure (IAP) < 12 mm Hg; ≈ 7 mm Hg is feasible.

Age	0–1 month	2–12 months	1–2 years	2–8 years
Maximum IAP	**≤ 6**	**≤ 8**	**≤ 10**	**≤ 12 mm Hg**

- **Cardiovascular**: IAP > 10 mm Hg lowers venous return and cardiac index; catecholamines and renin–angiotensin–aldosterone (RAAS) activation raise vascular resistance. Insufflation can trigger a **vagal reflex**, so warn the anaesthetist, desufflate and reinsufflate slowly.
- **Respiratory**: infants have high chest wall compliance and low functional residual capacity (FRC), so tidal volume falls and peak pressure rises, worse with Trendelenburg. CO_2 absorption affects infants most.
- **Ventriculoperitoneal (VP) shunts**: raised intracranial pressure was feared, but minimally invasive surgery (MIS) is generally safe.
- **Renal**: oliguria is transient, most marked < 1 year and reversible, so give **no fluid boluses**.

- **Hypothermia** †: cold, dry gas cools infants; use warmed humidified gas, low flows and minimal leaks.
- **CO_2 embolism** †: falling end-tidal CO_2, desaturation, collapse, "mill-wheel" murmur.

3 · Access

Why this matters: most injuries happen at entry.

- Open (Hasson) and closed (Veress: aspirate, saline drop test, low opening pressure) entry do not differ in failure or vascular/visceral injury (Cochrane). Confirming intraperitoneal position protects, not the method.
- Place the Veress at the umbilicus; after scarring or failure, use **Palmer's point** (left midclavicular line ≈ 3 fingerbreadths below costal margin). A preperitoneal tip gives high opening pressure and subcutaneous emphysema.

4 · Ports, instruments and placement

Why this matters: small bodies limit port size and spacing.

- **3- and 5-mm** laparoscopic ports suit infants; robots need **8 mm**. Valveless trocars keep pressure constant and use less CO_2, but cause subcutaneous emphysema.
- Laparoscopic instruments have 4 degrees of freedom (DOF); robotic wristed ones 7 DOF.
- Energy: monopolar insulation failures cause remote burns; bipolar and ultrasonic devices seal vessels up to 7 mm.
- Triangulation is essential in laparoscopy; the robot tolerates linear placement. **HIdES** (hidden incision endoscopic surgery) puts ports below the Pfannenstiel line.
- **Positioning**: renal is modified flank at bed edge, 30–45°, padded, testing table rotation; pelvic is supine (frog-leg), steep Trendelenburg, strapped, face protected.

5 · Approaches

Why this matters: the approach is chosen for space, organ and dialysis status.

	Transperitoneal	Retroperitoneal	Vesicoscopic
Space	Largest; familiar	Limited	Limited; small bladder predicts complications
Strength	Complex reconstruction	Avoids peritoneum; preserves peritoneal dialysis	Intravesical access
Limitation	Colon mobilised or mesenteric window	Unfamiliar view; peritoneal tear collapses space	Learning curve; port-site closure
Uses	Testis, kidney, ureter, bladder	Pyeloplasty, (hemi)nephrectomy, lymph node dissection	Cross-trigonal reimplantation

- **Left pyeloplasty**: a transmesenteric window avoids colonic mobilisation.
- **Retroperitoneoscopy** †: open entry below the 12th rib tip, finger or balloon dissection, flank or prone. Nephrectomy outcomes equal transperitoneal; prefer it for peritoneal dialysis patients.
- **Vesicoscopy**: camera through the bladder dome under cystoscopic view, urethra occluded, CO_2 insufflation; the robotic version is limited by 8-mm ports.

6 · Robotics in children

Why this matters: small working space is the infant's main robotic problem.

- **Working space** is ≈ **1 L at 1 year vs 6 L in adults**, so arms collide. Aids: wide port spacing, linear placement, "burping" ports, minimal inhalational agent, enema and rectal tube.
- Infant robotic pyeloplasty is safe; compared with open, stay is shorter with less narcotic.
- **Limitations**: no 5-mm robotic ports; single-port needs a 25-mm cannula and ≈ **10 cm** to target, prohibitive in small children; no haptics; cost; training.
- **Benefits**: shorter stay (hours for pyeloplasty, days for Mitrofanoff/augmentation); cosmesis, as incisions < 15 mm contract and larger ones enlarge; pain benefit is least proven.

7 · Complications

Why this matters: complications cluster at entry and exit.

- Major vascular: **right common iliac** (behind umbilicus, Veress). Minor: **inferior epigastric** (insert ports under vision). Decompress bladder and stomach †; inspect on exit.
- Port-site hernia peaks around 1 year, so **close fascia** at every port.
- Experience is the strongest predictor (> 12 cases/year, fewer). Pyeloplasty complications are equal open, laparoscopic or robotic. Same-day discharge is feasible after uncomplicated robotic renal/ureteric surgery without drains.

> **KEY POINT**
> **Ergonomics**: laparoscopy strains hands, thumb and shoulders, so set the table at 70–80% elbow height and the monitor at eye level. Robot offloads the arm but fixes the neck; use the clutch to keep elbows on the armrest.

40 Clinical and Urodynamic Evaluation of Lower Urinary Tract Dysfunction in Children

1 · Principles and terminology

Most wetting is diagnosed without catheters; invasive tests are selective.

- History, examination and non-invasive tests first exclude an anatomical or neurological cause, then define **storage vs emptying** dysfunction (International Children's Continence Society, ICCS 2014).
- **Expected bladder capacity (EBC)**: infant ≤1 y = **weight (kg) × 7 mL**; older child = **(age in years + 2) × 30 mL**. Small is <65% EBC; large is >150% (some use >130%).
- Functional capacity (largest diary void, or void plus post-void residual [PVR]) is normally ≈70–115% EBC.

2 · History

History, diary and bowel habit find most causes.

- **Secondary enuresis** is wetting after **≥6 months** dry, so look for stressors.
- In girls, a poor **toilet posture** (legs together, perching, dangling feet) activates the pelvic floor., causing high voiding pressure, incomplete emptying and urinary tract infection (UTI); fix with a footstool or sitting backwards.
- **Holding manoeuvres** (potty dance, squatting, perineal grabbing) suppress detrusor overactivity (DO) via the sacral reflex arc, the same mechanism as neuromodulation.
- **Bowel** and bladder share S2–S4 nerves; bowel dysfunction may be the *sole* cause of urinary symptoms. Target a painless **Bristol 3–4 stool daily**.
- **Rome IV functional constipation** (developmental age ≥4 y, ≥1 month): **≥2 of**: ≤2 stools/week; ≥1 soiling/week; retentive posturing; painful or hard stools; large rectal mass; stools blocking the toilet.
- Bagged urine counts only if negative; febrile culture-proven UTI suggests stasis. **Neurological red flags** are perineal numbness, or gait or limb change with new lower urinary tract symptoms (LUTS).

ALGORITHM

Bowel and bladder dysfunction (BBD)

1. Apply Rome IV; obtain a 7-day diary, urinalysis and pelvic ultrasound (US).
2. **Treat the bowel first.**
3. If LUTS persist, then do uroflow with electromyography (EMG) and start urotherapy.
4. If that fails, add an anticholinergic, then a β3-agonist; then botulinum toxin, biofeedback or neuromodulation.

- The **7-day bowel–bladder diary** (voids, stools, Bristol type, wetting) removes recall bias.
- The **48-h frequency–volume chart** (2 × 24 h, need not be consecutive) shows capacity vs EBC and polyuria; it is **mandatory** when first-line therapy fails.
- Scores: Dysfunctional Voiding Symptom Score (**DVSS**) best tracks treatment; **DVISS** (with incontinence) is best for diagnosis; PinQ (quality of life) measures burden, not response.

- Psychological comorbidity: highest with **faecal incontinence**, then daytime wetting, then enuresis.

3 · Physical examination

Look for a hidden neurological or anatomical cause.

- Height, weight and blood pressure (BP) at **every visit**.
- **Neurological**: include perineal sensation and anal tone. Hand function decides self-catheterisation. Functional level and ambulation **do not correlate**.
- **Sacral reflexes** (anal wink, bulbocavernosus) make the external anal sphincter contract; absence means the S2–S4 arc is interrupted.
- **Spine**: a simple sacral dimple (<2.5 cm from anus) is low risk and gets spinal US up to 6 months. High-risk signs (hair tuft, lipoma, vascular malformation, multiple stigmata, VACTERL association) need **magnetic resonance imaging (MRI)**. US works only until the arches ossify (about 10–12 months).
- **Genitalia**: meatal stenosis matters only with stream deviation; labial fusion responds to topical oestrogen. **Continuous incontinence in a girl suggests an ectopic ureteric orifice.**

4 · Laboratory tests

One urinalysis suffices for most children.

- Do a **urinalysis for every child with LUTS**; a negative result reassures. **Culture is not routine.**
- On clean intermittent catheterisation (CIC), culture only with fever, pain, malaise, new or worse incontinence, or spasm. Bloods only if renal disease is suspected.

5 · Non-invasive urodynamics

Flow, EMG, US and residual answer most questions if valid.

- P3 P1 Uroflowmetry; EMG — *bell/tower/staccato/interrupted/plateau patterns and meaning; Qmax peak ≥2 s; patch EMG at 3 and 9 o'clock; lag time <2 s DO, 2–6 s normal, >6 s bladder-neck dysfunction*

Not in the paper answer:

- In children the adult ≥150 mL rule is replaced by voided volume ≥50% EBC, with more than one curve.
- A plateau can also mean an underactive detrusor; needle EMG adds pain, not accuracy.
- Uroflow is valid only if toilet-trained, voiding **≥50% EBC** (a pattern needs about 50–100 mL), with >1 curve; it shows emptying only.
- A staccato curve never reaches zero; an interrupted one does. The pelvic floor should be **silent during voiding**.

> **KEY POINT**
> The curve reflects detrusor against outlet. Too much detrusor gives a tower; too much outlet gives a plateau; intermittent outlet activity gives staccato; intermittent detrusor effort gives an interrupted curve. Patch EMG tells which side is at fault.

- **Pelvic US**: bladder wall is abnormal if **>3 mm full / >5 mm empty**. A rectal diameter **>30 mm** without urge means faecal loading.

- **PVR** varies, so repeat it. It is abnormal at 4–6 y if **>30 mL or >21%** of capacity and at 7–12 y if **>20 mL or >15%**; repeated >20 mL is abnormal.
- PVR is valid only **<5 min** after voiding with the bladder at 50–115% EBC; a late or over-distended value is invalid, not abnormal.

6 · Invasive urodynamics

Filling pressure, not wetting, decides renal risk.

Indications

- **Neurogenic**: spinal dysraphism, tethered cord, sacral agenesis, sacrococcygeal teratoma, transverse myelitis, spinal cord injury (SCI), or suspected neurogenic bladder with new LUTS.
- **Severe obstruction**: posterior urethral valves (PUV), urethral atresia, large ectopic ureterocele.
- Non-neurogenic LUTS: study only after failed 2nd/3rd-line therapy, and only if it will change management.

Technique

- **Avoid sedation or general anaesthesia**, which alter the detrusor; if unavoidable, use ketamine rather than midazolam.
- Routine culture or prophylaxis is not needed, because post-study UTI is about **1.4%**.
- Use a **6 Fr dual-lumen** urethral and a rectal catheter with perianal patches. **Fill at 5–10% of EBC per minute** with warm fluid.
- **Stop filling** at strong urge, discomfort, voiding, leak greater than fill rate, detrusor pressure (Pdet) persistently **>40 cmH_2O**, or volume **>150% EBC**.
- Report the first fill, which gives the highest Pdet (kidney-protective). Natural fill gives smaller capacity, higher voiding pressure.
- **Videourodynamics** adds trabeculation, diverticula, vesicoureteric reflux (VUR) with the volume and pressure at its onset, and bladder-neck shape.

Storage phase

- **Pdet = Pves − Pabd** (vesical minus abdominal, zeroed at the symphysis); normal filling is flat. **DO** means phasic involuntary rises (exclude cough or strain). An **underactive** bladder is large and acontractile and empties by straining.
- **Compliance = ΔV/ΔPdet**, calculated every study: **>20 normal; 10–20 moderately; <10 severely impaired**. In small contracted bladders absolute Pdet (<30–40) is the better guide; an **early** rise means most of the day at high pressure.
- P1 Leak point pressures (VLPP, ALPP, DLPP) in SUI: clinical implications — *DLPP = outlet resistance, high = upper-tract risk; ALPP = sphincter competence, low = bad; McGuire 1981*

Not in the paper answer:

- No leak at high filling pressure is as dangerous as a high detrusor leak point pressure (DLPP).
- The adult abdominal leak point pressure (ALPP) cut-off (<60 cmH_2O) does not apply in children.
- A DLPP **>40 cmH_2O** (or no leak at high pressure) signals danger; ALPP has no paediatric norms.

KEY POINT

McGuire (1981), myelodysplasia: with leak pressure >40 cmH_2O, hydronephrosis or secondary VUR developed in **85%**, because pressure is transmitted up the ureters. Aim for the lowest achievable pressure.

Voiding phase

- Voiding data are valid if ≥50–100 mL and ≥50% functional capacity are voided. **High Pdet with low maximum flow (Qmax) means obstruction.**
- Anatomical obstruction gives a plateau with a **quiet EMG**; functional obstruction (detrusor–sphincter dyssynergia [DSD] or dysfunctional voiding) gives staccato or interrupted flow with an **active EMG**.
- Normal Pdet at Qmax is about 66 cmH_2O in boys and 57 in girls; infant boys reach 118, which is physiological MCQ.

KEY POINT

Upper-tract threats: compliance <20 (especially <10); DLPP >40 cmH_2O; sustained high filling Pdet; DSD with poor emptying; high voiding pressure with poor flow. Interobserver agreement is poor.

- **UMPIRE** (Urologic Management to Preserve Initial Renal Function, spina bifida) studies at about **3 months**, then at 1, 2 and 3 years, then yearly to 5 y if VUR or a hostile bladder.
- Hydronephrosis is late and insensitive; re-study on any change and before reconstruction.

7 · Evaluation in specific conditions

Typical pattern and timing by condition.

Condition	Typical findings	When to study
Non-neurogenic LUT dysfunction / enuresis	DO, dysfunctional voiding	Only if refractory
Spinal dysraphism	UMPIRE risk by DLPP/end-fill Pdet: **<25 low; 25–39 or DO intermediate; ≥40 high**	Baseline, serial, pre-surgery
Sacral agenesis	Areflexia with weak sphincter, or DO with DSD; level does not predict; ⅓ diagnosed late (flat buttocks, short cleft)	If voiding dysfunction
Anorectal malformation	Tethered cord common	Baseline ≈1 y after repair; on change
PUV	DO, then poor compliance, then myogenic failure; polyuria worsens it ("valve bladder" ≈15%)	Serial
Cerebral palsy	Upper motor neuron; DO, large or small capacity, DSD rare	Only if conservative therapy fails
SCI	Spinal shock causes retention; higher lesions have more DSD	Video-urodynamics 3–6 months after injury
Transverse myelitis	DO > poor compliance > DSD > DLPP >40 (by frequency); motor recovery does not predict bladder	Early, with CIC

KEY POINT

UMPIRE dropped DSD from high risk because reviewers disagreed on who had it, not because DSD is benign. Neurogenic bladder goals: normal capacity, low storage pressure, complete emptying, upper-tract protection.

8 · Landmark trials

Trial	Design	Result	Take-home
McGuire 1981	Myelodysplasia urodynamic cohort	DLPP >40 cmH_2O led to upper-tract damage in 85%	40 cmH_2O = safety threshold

42 Bladder Anomalies in Children

1 · Classification and embryology

Embryology explains each anomaly.

- P3 Embryological development of urinary bladder and management of Hutch Diverticulum? — *cloacal division weeks 4–6, UGS fates, trigone, urachus months 4–5; anomaly table; Hutch primary vs secondary, VCUG, treat outlet first, diverticulectomy + reimplant*

Not in the paper answer:

- Syndromic diverticula can exceed bladder size and need urodynamics after resection.
- **Structural** (exstrophy, duplication, agenesis, urachal, diverticula) or **functional** (usually secondary) anomalies; prune-belly syndrome (PBS) is both.
- Every urachal anomaly is failed obliteration of the allantoic duct (normally the median umbilical ligament) at a different level.

2 · Fetal bladder on ultrasound

A bladder not seen once is not yet absent.

- The wall is ≤3 mm, and the bladder is seen in nearly all fetuses by 13 weeks. It empties every **15–20 min**, so if unseen, **repeat in the same sitting**. Amniotic fluid is mainly fetal urine after **16 weeks**.
- A non-visualised bladder is absent, malformed or not filling; the kidneys and amniotic fluid discriminate. In exstrophy there is no filling with **normal amniotic fluid**.

	Agenesis	Hypoplasia	Exstrophy
US	Absent after repeat scan	Small, never reaches capacity	No filling; bladder template on abdominal wall
Core defect	Cranial urogenital sinus (UGS) atrophy or no trigone	Inadequate filling or storage	Open bladder plate
Outcome	Survives only with ectopic ureteric drainage	May grow; augmentation often	Reconstruction (*see Ch 43*)

- **Agenesis** (~1:600,000, mostly female): ureters drain into müllerian structures in girls or the rectum in boys. Renal dysplasia or agenesis and absent genitalia coexist. Treat by ureterosigmoidostomy or stoma.
- **Hypoplasia** occurs with severe epispadias, UGS anomalies, bilateral renal dysplasia or agenesis, and ureteric ectopia.

3 · Megacystis

Size, sex and karyotype predict the cause.

- In classic congenital megacystis, **massive vesicoureteral reflux (VUR) causes the dilation** because urine recycles; it is not bladder-neck obstruction. Today the term means any prenatally dilated bladder.

Gestation	Normal	Abnormal
10–14 weeks	Length ≤6 mm; <10% of crown–rump length (CRL)	**≥8 mm**; ≥13% of CRL
Later (functional)	Empties within 45 min	No emptying in 45 min

Gestation	Normal	Abnormal
15–40 weeks	Length ≈ gestational age (GA) − 5 mm	> GA + 12 mm = megacystis

- **Causes**: lower urinary tract obstruction (LUTO), with posterior urethral valves (PUV) ~50%, urethral atresia or stenosis, anterior valves. Non-obstructive: PBS, megacystis–microcolon–intestinal hypoperistalsis syndrome (MMIHS), cloacal anomalies. About 1/3 are undiagnosed; male:female 8:1.
- **Aneuploidy is ~1/5** (trisomy 13/18), so karyotype; outcome is poor.
- In euploid fetuses, 7–15 mm mostly resolves, while **>15 mm** never does (all obstructive). Resolution before 23 weeks means no postnatal surgery; later, most need intervention. **Persistent megacystis with oligohydramnios means LUTO.**

MMIHS

- Visceral myopathy (**ACTG2**); ~70% female; prenatally megacystis with normal or increased fluid. Exclude Hirschsprung disease (ganglion cells). Total parenteral nutrition (TPN) complications are the leading cause of death.
- Start clean intermittent catheterisation (CIC) or vesicostomy; urodynamics shows an acontractile bladder; most need **lifelong CIC**. VUR is uncommon.

> **KEY POINT**
> A female fetus with megacystis cannot have PUV, so think of MMIHS, cloacal anomaly or urethral atresia.

4 · Bladder duplication

Sphincter function drives management.

- **Complete sagittal is commonest**. External genital duplication occurs in up to 90% and lower gastrointestinal (GI) duplication in up to 42%, with VUR and renal ectopia or dysplasia. A bladder without a urethra causes ipsilateral dysplasia. Incomplete duplication (one urethra) may need nothing.
- > **ALGORITHM**
  > 1. Early: relieve obstruction and prevent infection.
  > 2. Later, if both sphincters are competent, then join the distal urethras.
  > 3. If one is incompetent, then close that bladder neck and excise its urethra.
  > 4. Join duplicated vaginas, with vulvoplasty.
- **Sphincter competence, not anatomy, decides reconstruction.**

5 · Bladder diverticula

Primary or secondary decides treatment.

- A primary paraureteral (Hutch) diverticulum is observed if small and excised if VUR; dextranomer/hyaluronic acid injection is still possible. For a secondary one, **treat the outlet first**.
- **Syndromic** diverticula (**Ehlers–Danlos, Williams, Menkes**) are multiple, posterolateral and giant. Observe them and resect for urinary tract infection (UTI) or stones; recurrence is common, so vesicostomy is an option.

6 · Urachal anomalies

The level of patency sets the picture.

	Patent urachus	Sinus	Cyst	Vesicourachal diverticulum
Open	Whole canal	Umbilical end	Isolated segment	Bladder apex
Presents	Urine from the umbilicus in a neonate	Draining umbilicus	Infection (*S. aureus*), painful mass; commoner in adults	Incidental
Imaging	US; fistulogram or voiding cystourethrogram (VCUG)	Sinugram	US; CT/MRI if extensive	Incidental
Treat	Excise with cuff	Complete excision	Drain, then excise	Only if a narrow neck causes stones or UTI

- Differential: patent **omphalomesenteric duct**. Urachal adenocarcinoma is unreported in children. For an incidental remnant, the choice is "excise all" or "only if symptomatic" (increasingly adopted).

ALGORITHM

1. If infected, drain and give antibiotics first.
2. Infraumbilical incision; circumscribe the tract at the umbilicus, preserving it.
3. Dissect extraperitoneally to the dome and excise with a **bladder cuff**.
4. Two-layer closure; catheter overnight.

- Laparoscopic or robotic excision is feasible but intraperitoneal, risking spill and urine leak.

7 · Acquired lesions

Several mimic tumour.

Lesion	Key features	Treatment
Haemangioma	Klippel–Trénaunay, Sturge–Weber; haematuria	Fulguration or Nd:YAG laser; partial cystectomy if large
Fibroepithelial polyp	2nd commonest paediatric bladder tumour after rhabdomyosarcoma	Endoscopic resection
Nephrogenic adenoma	Female 5:1; after infection, stones, surgery, augmentation	Transurethral resection; **80% recur**, so surveillance cystoscopy
Cystitis cystica	Almost all female; mimics tumour	Long-term antibiotic prophylaxis
Eosinophilic cystitis	Male	**Biopsy mandatory**; resection with steroids and antihistamines
Haemorrhagic cystitis	Cyclophosphamide or ifosfamide; BK virus, cytomegalovirus, adenovirus; radiotherapy	Mesna prevents; pentosan polysulfate

43 Exstrophy–Epispadias Complex

1 · Spectrum, epidemiology and embryology

One defect, rupture of the cloacal membrane, explains the whole spectrum.

- P3 P1 Bladder and ureter development, exstrophy–epispadias complex and its management; Describe embryology of bladder… theories for EEC… skeletal abnormalities — *spectrum, CBE 1 in 46,000, M:F 2.3:1, offspring 1 in 70; 22q11.2, ISL1, p63; IVF/progesterone/smoking/folate; failed mesodermal reinforcement of cloacal membrane, Marshall–Muecke and other theories; pelvis 12°/18° rotation, anterior pubis 30% short, diastasis 4.8 cm; MSRE stages, osteotomy indications, CPRE outline*

Not in the paper answer:

- Spica cast fails in 66% without osteotomy; diastasis >6 cm needs staged closure.
- Capacity <30 mL at 1–2 years may preclude continence surgery; not dry by 2 years after BNR is failure.
- Expert series: dehiscence or prolapse ~3%, femoral palsy ~2%.
- The complex has five entities: cloacal exstrophy, male and female classic bladder exstrophy (CBE), and male and female epispadias. Family recurrence is ~1 in 100, and offspring risk is ~500-fold raised.
- Rupture timing explains three phenotypes: early, extensive rupture opens bladder **and** hindgut (cloacal); rupture after urorectal septation opens the bladder only; rupture at the genital tubercle only gives epispadias.

2 · Anatomy of classic bladder exstrophy

Each defect explains a step of the repair.

- **Pelvis**: acetabula are retroverted; the posterior segment is normal. Consequences (short pendular penis, waddling gait, out-toeing) improve with growth. Significant spinal dysraphism is uncommon (unlike cloacal).
- **Pelvic floor**: the levator ani lies posterior and flattened. Continent patients have the shortest intersymphyseal distance and deepest bladder neck, hence **deep placement, osteotomy and pelvic fixation**.
- The **intrasymphyseal band** (divergent urogenital diaphragm) tethers the posterior vesicourethral unit to the pubic rami. It contains **no striated sphincter**, so radical incision down to the levator hiatus is essential.
- The umbilicus is low; omphalocele is rare (common in cloacal). **Indirect inguinal hernia** is common, often bilateral in males, so repair it at closure and explore the other side; osteotomy lowers its incidence.
- **Anorectum**: the anus is anterior. **Rectal prolapse before closure** resolves with closure. Prolapse **after closure** suggests outlet obstruction and needs urgent cystoscopy.
- **Male**: the **anterior corpora are ~50% shorter** and ~30% wider,, so "lengthening" is only apparent. The prostate is never circumferential, and prostate-specific antigen (PSA) is below age norms. Testes are normal and mostly retractile. Cavernous nerves lie laterally, so potency is retained; retrograde ejaculation may follow closure or bladder neck reconstruction (BNR).
- **Female**: the vagina is short (~6 cm) with a stenotic, anterior orifice; the **cervix sits in the anterior vaginal wall**; the clitoris is bifid. A levator hiatus twice normal width causes

uterine prolapse, even in nulliparous women; adult diastasis predicts it and **osteotomy does not prevent it**.

- **Bladder**: it is **immature, not neuropathic**, with normal muscarinic receptors; raised type III collagen normalises after closure. The exposed plate develops polyps, squamous metaplasia and cystitis glandularis, which carry an **adenocarcinoma risk**. Delayed closure lowers uroplakin and increases metaplasia. Compliance and stability fall after BNR, so follow-up is lifelong.
- Upper tracts are usually normal (duplex is the commonest anomaly). The ureters enter laterally without obliquity, so **vesicoureteral reflux (VUR) occurs in 100%** of closed bladders and is corrected by reimplantation at BNR.

3 · Variants

Except superior vesical fissure, variants are managed like CBE.

Variant	Feature	Management
Skin-covered (split symphysis) — commonest	Musculoskeletal defect; urinary tract intact	Formal closure at birth
Duplicate exstrophy	Everted mucosal patch plus a closed pelvic bladder	Excise patch, close wall
Superior vesical fissure	Rupture only at the top; sphincter intact	Toilet-trains normally; no BNR
Pseudoexstrophy	Musculoskeletal defect only; low umbilicus	—
Epispadias with bladder prolapse	Prolapse may show only on Valsalva	Repair, with or without osteotomy

KEY POINT
Superior vesical fissure inflated the belief that variants do better. All others need formal closure and follow-up like CBE.

4 · Prenatal diagnosis

Diagnosis allows delivery at an exstrophy centre.

- Only ~25% are diagnosed prenatally; many are mislabelled omphalocele or gastroschisis.
- Signs are **absent bladder filling**, a low umbilicus, wide pubic rami, small genitalia and an **enlarging lower abdominal mass**. The cord insertion–genital tubercle distance is <5th centile and diastasis >10 mm.
- Fetal magnetic resonance imaging (MRI) helps if CBE vs cloacal exstrophy is unclear. Plan **delivery at an exstrophy centre**.

5 · Newborn care and selection

A good first closure best predicts continence, so a poor plate is closed later, not badly.

ALGORITHM

1. Tie the cord with 2-0 silk (no clamp, which traumatises mucosa).
2. Cover the plate with non-adherent plastic wrap; irrigate with saline at each nappy change.
3. Assess heart and lungs; obtain a renal ultrasound (US).
4. Transfer to an exstrophy centre for multidisciplinary team (MDT) counselling.

- **Assess** the template (≥**3 × 3 cm** is suitable), the diastasis and pelvic malleability under anaesthesia, the urethral plate and penile length. A plate that **bulges on crying or indents under anaesthesia** is adequate.
- A plate <3 cm, polyp-covered or fibrotic is **inadequate**. Penoscrotal duplication, hypoplastic bladder and bilateral hydronephrosis preclude primary closure; ectopic bowel in the plate is relative.
- A small plate is closed **later, at 6–9 months, with osteotomy**, because a bad closure ruins continence. If it has not grown by 6–12 months, excise it and use a colon conduit or ureterosigmoidostomy.
- After delayed closure, continence comes later and a **catheterisable channel is needed far more often**; capacity loss is steepest beyond 9 months.
- Early closure gives more differentiated urothelium (uroplakin II); once surface trauma stops, a small bladder with modest outlet resistance enlarges.

6 · Surgical philosophies

Staged vs single-stage: most still need an outlet procedure.

Repair	Stages / feature	Weakness
Modern staged repair (MSRE)	Closure at birth; epispadias repair at 6–12 months; BNR with reimplant at 4–7 years. A balanced outlet grows the bladder	Multiple operations
Complete primary repair (CPRE, Mitchell)	Newborn closure with penile disassembly and neck tailoring	Hypospadias; glans loss; early reimplant
Kelly	Closure, then radical soft-tissue mobilisation at 3–6 months (pudendal bundles freed, muscles wrapped round the urethra), then hypospadias repair ~3 years; avoids osteotomy	Fistula, stenosis
Erlangen	Total repair ~8 weeks; no osteotomy (obturator-foramen coaptation); epidural 5 days	—
Mainz	Closure, then sigma pouch ~2 years; bladder remnant serves as seminal receptacle; oral alkalinisation	Colonic malignancy

- **Common principles**: radical posterior mobilisation; tension-free closure (osteotomy if needed); strict selection for newborn closure.

KEY POINT

Erlangen, Kelly and CPRE claim to avoid BNR, yet almost all still need an outlet procedure. Compressing the reconstruction adds ischaemic and outlet risk without removing that need.

7 · Primary closure (MSRE stage 1)

Closure creates an incontinent epispadias with enough outlet for bladder growth.

ALGORITHM

1. General anaesthesia with **epidural** (catheter to T10–T12).
2. Outline a 2-cm mucosal strip from the trigone to mid-shaft (boy) or vaginal orifice (girl); incise the bladder–skin junction.
3. Free the bladder from the rectus and strip peritoneum off the dome.
4. **Divide the urogenital diaphragm fibres sharply down to the levator hiatus**; free the corpora from the rami.
5. Close bladder, neck and posterior urethra; **taper over a 12-Fr sound**.
6. Place a suprapubic Malecot (4 weeks) and ureteric stents; leave the **urethra unstented**.
7. Close the pubis with a **No. 2 nylon horizontal mattress** suture, knot away from the neourethra; fashion a neoumbilicus.

- **Tapering** gives enough resistance for growth and against prolapse, but not enough to harm the kidneys.
- **Ureteric stents** stay 10–14 days, because oedema can obstruct the ureters. A urethral stent causes necrosis.
- The **commonest error** is inadequate urogenital diaphragm dissection; the unit is pulled forward when the pubes close and prolapses.

8 · Osteotomy and immobilisation

Osteotomy and rigid immobilisation cut closure failure several-fold.

- Aims: close the ring, relieve tension, place the unit deep, reduce hernias.
- In a **newborn <72 h** whose malleable pubes meet on trochanteric rotation, osteotomy is not needed. **If in doubt, osteotomise.**
- Osteotomise if closure is **>72 h** after birth, diastasis is **>4 cm**, the pelvis is stiff, the template is large, after **failed closure**, and in **cloacal exstrophy**. Diastasis >6 cm needs staged closure with a fixator.

Type	Position	Features
Combined bilateral anterior innominate and vertical iliac (preferred)	Supine	Best malleability; least recurrence
Bilateral posterior iliac (original)	Prone	Worse approximation, more recurrence
Anterior innominate alone	Supine	Less mobility

- Technique: Gigli-saw transverse innominate cut between the anterior superior and inferior spines, plus a vertical closing-wedge iliac cut lateral to the sacroiliac joint (posterior cortex hinge). Then pins, repair, pubic suture and **external fixator**.
- Aftercare: radiograph at 7–10 days and adjust the fixator; **modified Buck traction ~4 weeks with the fixator for 4–6 weeks**. Without osteotomy, use **modified Bryant traction** (hips 90°) for 4 weeks. Pubic repair is weaker than intact symphysis, so immobilise rigidly.

Immobilisation (CBE)	Failure overall	With osteotomy	Without osteotomy
Spica cast	47%	30%	**66%**
External fixation with traction	**7.5%**	8%	— (few)

- Immobilisation matters on its own: failure falls with Buck traction plus fixator for **>4 weeks** and rises with a non-paediatric orthopaedic surgeon. Osteotomy type and epidural make no difference.
- Complications are pin-site inflammation (near-universal), transient femoral nerve palsy (recovers by 12 weeks) and delayed union. Partial diastasis recurrence (pin loosening, ischiopubic undergrowth) occurs, but the net effect is beneficial.

9 · After primary closure

The months after closure decide bladder growth.

- Keys: **bladder drainage, ureteric drainage, immobilisation, pain control**. At 4 weeks calibrate the outlet, scan, and clamp for residual. Give **continuous antibiotic prophylaxis**, because all have VUR.
- Scan at 3 months, then every 6–12 months; perform **yearly cystoscopy and cystogram under anaesthesia** to measure capacity.
- If capacity is **<30 mL at 1–2 years**, a continence procedure may not be feasible.
- If the **outlet is too tight**, escalate from dilation to incision, early reimplant and revision. Recurrent urinary tract infection (UTI) with a distended bladder suggests **intrapubic stitch erosion**. Keep residuals <50 mL.

10 · Epispadias repair (stage 2)

Urethroplasty raises outlet resistance, so the bladder grows.

- Done at 6–12 months, before BNR, because urethroplasty **increases capacity**. Give **testosterone enanthate 2 mg/kg intramuscularly (IM) at 5 and 2 weeks** before every urethroplasty.
- **Modified Cantwell–Ransley** (preferred): the plate is freed but its **distal 1 cm stays on the glans**; it is tubularised over 8 Fr and the **corpora are rotated over the neourethra**, which then lies ventral. Stent for 10–12 days. Always excise remnant suspensory ligaments and scar and free the corpora from the rami.
- **Chordee** is corrected by dorsomedial corporotomy with cavernocavernostomy or, in adolescents, a dorsal dermal graft. **Skin** cover uses the ventral prepuce as lateral or buttonholed flaps, with a Z-plasty at the base. In-situ dorsal tubularisation, free grafts and transverse island flaps are abandoned. Adolescent males often value appearance over continence.
- Results: a downward penis and catheterisable urethra. Early fistula ~1 in 6 (many close by 3 months), at the base; **coronal fistula is rare**. Scarce tissue: tunica vaginalis flap.
- **Mitchell–Bägli disassembly** risks plate ischaemia, and most end **hypospadiac**. Glans or corporal loss is linked to closure **without osteotomy**, so if ischaemia is seen, osteotomise immediately.
- Control bladder spasm (epidural and oxybutynin), because spasm causes extravasation and fistula.
- **CPRE**: ~half need reimplant in year 1; girls closed after 72 h risk retention; complications are as frequent as MSRE but more serious. Voided continence without outlet surgery is ~1 in 5; 65–80% need an outlet procedure, and most end on clean intermittent catheterisation (CIC).

- **Mitchell**: osteotomy if >3 days old or wide diastasis; in girls neck, urethra and vagina move as one unit; a short urethra gives a hypospadiac meatus.
- **Kelly** stage 2: plate beneath corpora, pubic "flake", pudendal bundles freed in the ischiorectal fossa. Unaided voided continence <1 in 5, half partially continent; glans loss reported.
- **Combined closure and epispadias repair** suits failed or delayed closures and boys >6 months; ~two-thirds are dry after Young–Dees–Leadbetter (YDL) neck repair, but ischaemic penile loss is a risk (experts only).
- **Female closure**: osteotomy less often (>4 cm or stiff); bladder, urethra and vagina move deep as one unit, with **monsplasty, joining of the clitoral halves** and Y–V labioplasty.

11 · Bladder neck reconstruction (stage 3)

Capacity at BNR best predicts dryness.

- **Prerequisites**: capacity **≥100 mL** (under anaesthesia), stable urodynamics, a motivated child (5–7 years), a voiding programme for ≥6 months and a psychology review. Reimplant earlier for pyelonephritis, worsening hydronephrosis or falling function.

ALGORITHM

Modified YDL

1. Open the bladder neck transversely.
2. Perform **Cohen cross-trigonal reimplantation**.
3. Outline a posterior mucosal strip **15 × 30 mm**; denude the lateral muscle triangles (no cephalad transverse cut).
4. Tubularise the strip and **double-breast the muscle flaps** over it.
5. Suspend to the rectus fascia; suprapubic tube for 3 weeks, then clamp trials.

- After MSRE, ~**70–75%** are dry by day (intervals >3 h); CPRE with YDL is lowest (~56%).
- **Predictors** are template size, **successful primary closure** and **capacity at BNR**. Capacity is the most important: >100 mL halves the time to dryness.
- Closure <72 h or with osteotomy makes continence likely; delayed closure without osteotomy gives ~10%. After two closures, fewer than 1 in 5 void continently.
- High "continence" in large cohorts mostly reflects **bladder-neck closure with a catheterisable stoma**.
- Day dryness comes within ~12 months; **not dry by 2 years means failure**; night dryness takes 2–3 years.

Category (after BNR)	Definition
Continent	Dry day and night, interval ≥3 h
Socially continent	Dry 3 h by day, occasional wet nights
Wet	Dry <3 h by day, wet at night

12 · Failures and complications

Each failed closure shrinks the bladder.

- **Failed closure**: dehiscence, **bladder prolapse**, outlet stricture, soft-tissue loss or vesicocutaneous fistula. Causes: incomplete pelvic-diaphragm mobilisation, inadequate immobilisation, infection, distension and tube malfunction.

- Expert series: major dehiscence or prolapse ~3%; minor urological ~11%; major orthopaedic ~6% of osteotomised; femoral palsy ~2%.
- **Evaluate** template, perivesical scarring, diastasis and prior osteotomy, ≥4 months after failure. A vesicocutaneous fistula (lower midline tract, widened diastasis) usually signals failed closure.
- **Reclosure**: wait ≥4–6 months; use **osteotomy** (repeat osteotomy is safe), rigid immobilisation and radical urogenital diaphragm division. Epispadias repair may be combined after testosterone. After tissue loss from CPRE, grafts or tissue expansion come first.
- Stricture (mostly within 60 days) presents with UTI, enlarging bladder, stones, long dry intervals or rectal prolapse; escalate from dilation or incision to urethroplasty, then diversion. Obstruction >6 months usually means augmentation.
- After secondary closure bladders are smaller; voided continence is ~1 in 6; bulking disappoints. Fellowship-trained paediatric urologists reclose better.
- **Failed BNR** means no 3-h dry interval by 2 years. A near-miss (dry >2 h) may respond to bulking, but there is no benefit if dry <1 h.
- **Redo YDL** only if the neck is patulous, capacity >100 mL and urodynamics stable. Otherwise do **bladder neck transection, augmentation and a Mitrofanoff channel**; stones follow in ~30%. No pubertal "prostatic" continence.
- **Genital**: urethrocutaneous fistula is commonest. Glans or corporal loss is managed with tissue expansion, grafts or **radial forearm phalloplasty**.

13 · Diversion and augmentation

Diversion gives dryness and protects the kidneys.

- **Ureterosigmoidostomy** needs anal continence. It risks pyelonephritis, hyperchloraemic acidosis and **colonic adenocarcinoma about three decades later**, so **yearly colonoscopy** is required. A Mainz sigma pouch lowers pressure; with severe renal impairment use a colon conduit.
- If the bladder is too small, use a **non-refluxing colon conduit**. After failed BNR, do **augmentation with an appendiceal channel, with or without neck closure**; stones are the commonest complication. Augmentation does not harm long-term glomerular filtration rate (GFR).
- **Continent diversion** is indicated for failed closures not growing after reclosure, newborn closures not growing, inadequate capacity or wetness after BNR, and upper-tract decompensation. Move early if adequate BNR capacity is unlikely.

14 · Long-term issues

Care is lifelong.

- **Malignancy**: augmented exstrophy bladders develop multifocal adenocarcinoma; survey endoscopically from 5 years after reconstruction.
- **Male sexual function**: libido, erection and orgasm are preserved; concerns are length, appearance and chordee (adolescent dorsal graft, ligament release).
- **Male fertility** is impaired by verumontanum injury, retrograde ejaculation and oligo- or azoospermia; fewer BNRs mean better counts. About 1/4 achieve paternity, half via intracytoplasmic sperm injection (ICSI).

- **Female**: a stenotic introitus needs **perineal flap vaginoplasty**, because a cutback worsens prolapse. A split mons is corrected by monsplasty (rhomboid flaps or expanders, after topical oestrogen). **Uterine prolapse** is treated by sacral suspension; consider prophylactic suspension at adolescent reconstruction.
- Males need routine PSA and digital rectal examination (DRE) like other men. **Pregnancy** ends in a **planned caesarean with a urologist present**.
- Psychosocial: acting-out, sexual anxiety and suicidal ideation in older males need lifelong support and transition care.

15 · Epispadias

Better than exstrophy, unless diastasis is wide.

- P3 Development of the urethra, congenital urethral anomalies and management of epispadias — *incidence, male/Davis classification, VUR rates, >3 cm means exstrophy pathway, Cantwell–Ransley then YDL at ~100 mL, female repair, continence males 47–82% / females 70–87%*

Not in the paper answer:

- Glanular epispadias is never incontinent.
- Penile disassembly in isolated epispadias causes no ischaemia but often leaves hypospadias; old plication and suspension operations failed.
- Female continence after BNR (~87%) equals that of girls with CBE.
- Fistula is usually basal; coronal fistula is rare.

	Male	Female
Incidence	1 in 117,000	1 in 484,000
Classification	Glanular, penile, **penopubic** (~70%, incontinent)	Davis: vesicular, subsymphyseal, retrosymphyseal (incontinent), in rising severity
VUR	30–40%	30–75%
Presentation	At birth	May present only with incontinence

ALGORITHM

Female repair (6–12 months)

1. Incise vertically through the urethral wall at 9 and 3 o'clock.
2. Excise the urethral roof (dorsal wedge taper) and close over a 10-Fr Foley.
3. Unite the clitoral halves and labia minora; bring mons fat over the suture line.
4. Catheter 5–7 days; BNR at 4–5 years.

- Treat as exstrophy (osteotomy and closure) if diastasis is >3 cm, the intrasymphyseal bar is poor, sphincter is scant or capacity is small.
- In **males** with a small bladder, do urethroplasty first, then BNR at ~100 mL (4–5 years). With good capacity, repair in one stage; osteotomy is usually not needed.
- Female continence after BNR is ~**87%**, as good as in CBE girls. Epispadias outperforms exstrophy because the bladder was never exposed.

16 · Cloacal exstrophy

A multisystem defect; neurological level sets disability.

- Incidence is 1 in 200,000–400,000. **OEIS** stands for Omphalocele, Exstrophy, Imperforate anus and Spinal defects. **Two hemibladders flank exstrophied ileocaecal bowel**, with a

rudimentary hindgut, separated hemiphalluses and a diastasis about twice that of CBE.

Feature	CBE	Cloacal
Spinal defects	Uncommon	**85–100%** (lipomeningocele commonest)
Omphalocele	Rare	Near-universal
Upper-tract anomalies	Uncommon	41–60%; solitary kidney commonest
Osteotomy	Selective	Always
Continence	Mostly voiding	Mostly augmentation with CIC

- **Neurospinal**: dorsal mesenchyme is disrupted and only ~1 in 5 has a sacral skin sign, so all need MRI. Medial autonomic nerves are at risk; watch for tethering lifelong.
- Müllerian duplication in most girls; clubfoot; short gut (~1/4). A 46,XY child may have an **intravesical phallus**, so biopsy before calling it aphallia.
- **Prenatal** diagnosis (~15%) shows a non-visualised bladder, omphalocele and myelomeningocele; the **"elephant trunk" sign** is prolapsed ileum.
- **Gender** follows **karyotype**; historic female assignment of 46,XY is abandoned.
- **Bowel**: never discard hindgut; tubularise the caecal plate with end colostomy and keep it for vaginoplasty or augmentation.

ALGORITHM

1. Stage 1 (newborn): close the omphalocele, separate the caecal plate, join the hemibladders and form a colostomy.
2. Stage 2: bladder closure with **combined osteotomy** and fixation for 6–8 weeks.
3. At 5–7 years: YDL only if capacity >100 mL (rare); usually augmentation with a continent stoma.

- Combined osteotomy gives ~90% success and ~sixfold odds of closure, but no continence gain. If diastasis is >10 cm, a fixator is cranked for 1–2 weeks first.
- Staged closure has ~4-fold odds vs single-stage. Predictors of success: staging, osteotomy, immobilisation >4 weeks and a paediatric urologist.
- Primary closure fails in ~31–43% (prolapse). Continence relies on augmentation with CIC, with a median of two procedures. The **neurological level** determines disability. Female cervical prolapse is worse than in CBE, so deliver by caesarean.

44 Prune-Belly Syndrome

- P1 P3 Prune Belly Syndrome. Antenatal diagnosis, pathophysiology and postnatal management; Prune belly syndrome — *triad, 1 in 29,000–44,000, 95% male; four theories, MYOCD/FLNA; pathology by organ (dysplasia ~50%, VUR ~75%, failed pump); extra-GU rates; Woodard I–III; neonatal work-up; vesicostomy, reconstruction after 3–6 months; orchidopexy ~6 months; Randolph/Ehrlich/Monfort; creatinine <0.7 mg/dL, 30% CRF*

Not in the paper answer:

- MAG3/DTPA renograms are unreliable with massive dilation (section 5).
- Normal infant germ cells justify early orchidopexy; the inguinal route fails (sections 2, 8).
- Renal failure drivers: pyelonephritis, dysplasia, polyuria; transplant outcomes equal (section 10).
- Female PBS ~5% with normal gonads (section 4).

1 · Definition and aetiology

Kidneys and lungs, not the belly, decide survival.

- The **triad** (Eagle–Barrett) is deficient abdominal wall muscle, bilateral intra-abdominal testes and a dilated urinary tract. Incidence is ~**1 in 29,000–44,000** (similar to exstrophy) and **95% male**. Commoner in twins, African ancestry and young mothers. **Osler** gave the name MCQ.
- **Survival follows the kidneys (and lungs), not the belly**; the wall is least prognostic. A wrinkled, flaccid neonatal abdomen is diagnostic on sight.
- Heredity: 1 in 23 children with prune-belly syndrome (PBS) is a twin, but most twins are discordant; chromosomal associations (Turner, trisomy 13/18, monosomy 16, Beckwith–Wiedemann) are exceptional. No inheritance pattern; multiplex families show smooth-muscle genes **MYOCD**, **FLNA** MCQ.
- Theories: early urethral obstruction, a lateral plate mesoderm defect, an intrinsic urinary tract defect and a yolk sac defect.

> **KEY POINT**
> Weakness of the obstruction theory: most have **no obstruction after birth**. The tract is **dilated, collagen-rich and largely non-obstructed**; this drives management.

2 · Genitourinary pathology

A weak pump, not a blocked pipe: stasis and infection are the enemies.

- **Kidneys**: dysplasia in ~**50%**, often asymmetric, worst with outlet obstruction **without urachal decompression**. Dilation **does not correlate** with dysplasia; **infection, not obstruction, is the main threat**.
- **Ureters**: distal is worse, so the **proximal ureter is preserved** in reconstruction. Vesicoureteric reflux (VUR) occurs in ~**75%**; pelviureteric or vesicoureteric junction (PUJ/VUJ) obstruction is uncommon. The ureter is a **failed pump, not a blocked pipe**, causing stasis and infection.
- **Bladder**: massive, **thick but smooth** (no trabeculation), with a urachal pseudodiverticulum; the urachus is patent at birth in **25–30%**. Splayed trigone, lateral orifices, wide neck.
- It has **normal compliance, large capacity and poor sensation** ("unbalanced voiding"); ~half void normally but may deteriorate, so reassess.

- **Prostatic hypoplasia** dilates the posterior urethra and causes ejaculatory failure; distal obstruction occurs in ~20%. Vasa and seminal vesicles are often atretic; retrograde ejaculation occurs.
- **Urethral atresia** is lethal unless the urachus is patent. **Megalourethra** is commoner in PBS than any syndrome: scaphoid (spongiosum only, ventral ballooning) is milder; fusiform (cavernosa and spongiosum, whole phallus dilates) is **more severe**.
- **Testes** lie over the iliac vessels. Infant germ cells are normal, so the **abdominal environment** destroys later potential, hence early orchidopexy.

3 · Extragenitourinary anomalies

Lungs often decide early survival.

About **75%** have them; nearly **half premature**.

- **Abdominal wall**: worst medially and inferiorly; flared costal margin, weak cough; heals well.
- **Pulmonary** (>50% of survivors): hypoplasia (newborn death), pneumothorax and recurrent pneumonia. **Cardiac** (~10%): patent ductus arteriosus (PDA), atrial or ventricular septal defect (ASD/VSD) and Fallot.
- **Gastrointestinal** (≥30%): malrotation, volvulus, atresia, splenic torsion, constipation.
- **Orthopaedic** (up to 65%): talipes (commonest), hip dysplasia, scoliosis, **lateral knee dimples** and limb hypoplasia from iliac compression.

> **KEY POINT**
> Two mechanisms: **oligohydramnios** (lung hypoplasia, limb deformity, knee dimples) and **absent abdominal wall pressure** (weak cough, atelectasis, constipation).

4 · Presentation and spectrum

The Woodard category guides how aggressive to be.

- **Prenatal**: mimics posterior urethral valves (PUV) and megacystis–megaureter: massive bladder, hydroureteronephrosis, irregular abdominal circumference and urachal diverticulum. Dilation does **not** predict function. A dilated prostatic urethra with anhydramnios suggests urethral atresia.

Woodard category	Kidneys / lungs	Course	Urological management
I	Dysplasia; **pulmonary hypoplasia**; oligohydramnios, Potter sequence, urethral atresia	Stillbirth or early death (~20% perinatal mortality overall)	Supportive; simple bladder drainage
II	Full triad; minimal or unilateral dysplasia; no lung hypoplasia	Risk of progressive renal insufficiency	Individualised — **controversial group**
III	Incomplete or mild; no dysplasia; stable	Majority; good	Surveillance; surgery for urinary tract infection (UTI) or deterioration; orchidopexy in year 1

- The abdominal wall is **not** a sorting criterion. **RUBACE** scores severity in survivors; renal weighs most MCQ.
- **Pseudo-prune**: partial flaccidity, unilateral undescended testis (UDT); late diagnosis brings recurrent UTI and **higher late renal failure**.

- **Female PBS** (~5%): normal gonads; outlet obstruction and anorectal anomalies frequent; mortality as in males.

5 · Evaluation

Stabilise the chest first; image selectively.

ALGORITHM

1. Team; **heart and lungs first** (chest X-ray).
2. Creatinine and blood urea nitrogen (BUN) trend; day-1 value is maternal.
3. Renal–bladder ultrasound (US).
4. Prophylaxis and circumcision; then selective imaging.

- Voiding cystourethrogram (**VCUG**) only for renal insufficiency or suspected outlet obstruction, with antibiotic cover and strict asepsis; VUR up to 70%.
- Dimercaptosuccinic acid (**DMSA**) scan at **4–6 weeks**. MAG3 or DTPA renograms are unreliable with massive dilation; MR urography (MRU) shows anatomy. Outlet obstruction is drained by a **percutaneous suprapubic tube**.

6 · Management philosophy

Watch a non-obstructed tract, or rebuild it before infection strikes?

	Conservative	Early reconstruction
Premise	Dilation is non-obstructive and may improve	Stasis, reflux and infection drive renal loss
Operate when	Proven obstruction, intractable infection	Progressive dilation, high-grade VUR, recurrent upper tract infection, declining function
Best for	Category III	Selected category II

- Delay major reconstruction to **≥3–6 months** (lungs). Single-stage comprehensive surgery (reconstruction, orchidopexy, abdominoplasty, circumcision) adds no morbidity. Correct high-grade VUR.

7 · Urinary tract surgery

Drain widely, keep the proximal ureter, improve emptying.

- **Blocksom vesicostomy** (sepsis, renal failure, atresia): excise the urachal diverticulum; make the stoma **larger than usual**, as low intra-abdominal pressure (IAP) favours stenosis. **Avoid proximal ureterostomy**.
- **Ureteroplasty**: excise the distal ureter, taper the upper few cm and reimplant transtrigonally. A pelviureteric kink needs ureterolysis; true obstruction needs pyeloureteral anastomosis.
- **Reduction cystoplasty**: capacity recurs, so use it only for a large urachal diverticulum or within wider reconstruction. Clean intermittent catheterisation (**CIC**) empties better.
- **Internal urethrotomy** (Otis 24–30 Fr) helps short-term only; abandoned in infants. Atresia: progressive dilation. Megalourethra: deglove, excise or infold.
- **Circumcision** reduces UTI.

8 · Orchidopexy

Approach and timing decide success.

- A broad mesorchium defeats the inguinal approach. Do a **bilateral transabdominal orchidopexy at ~6 months**, while the vessels still mobilise, often with other surgery.
- Later, use **Fowler–Stephens** (vessels divided; testis lives on vasal collaterals); the **staged** version (collaterals enlarge) beats single-stage or microvascular autotransplant. At laparoscopy, the lax wall needs careful port insertion.

9 · Abdominoplasty

Cosmetic gain is certain; functional gain likely.

- Mild laxity may improve; a corset helps. Functional gains (cough, defecation, emptying) are plausible but confounded.
- Operate from ~6 months with other surgery; infants may need postoperative ventilation.

Technique	Incision / umbilicus	Key step / note
Randolph	Transverse, 12th rib to 12th rib via the pubis	Full-thickness excision of the lower wall (worst on electromyography [EMG]); lateral bulge persists
Ehrlich	Vertical; umbilicus kept on an inferior epigastric pedicle	Vest-over-pants advancement; good long-term
Monfort	Vertical ellipse; umbilicus left in situ	Central fascial plate with lateral flaps; **transperitoneal**, so ideal with reconstruction or orchidopexy
Dénes	Fusiform; umbilicus buttonholed	Modified Ehrlich, double-breasted
Furness	Midline, limited	Plication; **peritoneum not opened**

ALGORITHM

Monfort

1. Skin ellipse, umbilicus spared; incise the central plate at the lateral rectus borders.
2. Do the genitourinary surgery.
3. Score peritoneum over lateral muscle; suture plate edges to the scored line.
4. Overlap lateral flaps over drains.

10 · Long-term outcome

Early creatinine predicts the renal future.

- Creatinine **<0.7 mg/dL at 1 year** predicts stable function unless pyelonephritis or obstruction intervenes. Up to **30%** develop chronic renal failure (CRF), driven by **pyelonephritis, dysplasia and polyuria**.
- **Transplant** outcomes are equal; ensure bladder emptying first.
- **Bladder**: timed and double voiding, Credé; if these fail, CIC. Constipation affects over half; asthma occurs. Musculoskeletal correction improves quality of life (QoL) most.
- **Fertility**: normal puberty and erections, but **azoospermia means no spontaneous paternity**; use sperm retrieval with intracytoplasmic sperm injection (ICSI). Females can carry pregnancies.

KEY POINT

Long-term care is a contest against **stasis**: lowering residual urine and infection protects the kidneys, while dysplasia and polyuria set a ceiling that surgery cannot raise.

45 Posterior Urethral Valves

1 · Definition and classification

Why this matters: valve damage starts before birth.

- **Posterior urethral valves (PUV)** are the commonest cause of bladder outlet obstruction in infant boys. Ablation relieves obstruction but **not the embryological injury** to bladder and kidney, so care is lifelong (urology and nephrology).
- Lower urinary tract obstruction (LUTO): ~2.2/10,000 births; PUV is commonest (~1.4/10,000), then urethral atresia, prune-belly. Commoner in Black and minority ethnic groups.
- P1 Embryology of Posterior Urethral Valves. — *Young 1919; types I–III and origin; COPUM (one membrane torn by catheter); genetics: chr 11 duplications, CNVs (worse renal outcome), TBX5, PTK7*

Not in the paper answer:

- Randall performed the first endoscopic resection.
- Type III resembles a congenital stricture; contentious but accepted.

Young type	Anatomy	Status
I (95%)	Leaflets from verumontanum fuse anteriorly just proximal to external sphincter	Obstructing; the lesion treated
II	Folds from verumontanum to bladder neck	Non-obstructing; probably not real
III (5–10%)	Annular membrane, central aperture (persistent urogenital membrane)	Obstructing

- Copy number variant (CNV) burden predicts higher nadir creatinine and chronic kidney disease (CKD) stage 5: it links genotype to renal prognosis, not to valve presence.

2 · Pathophysiology

Why this matters: the bladder, not the valve, drives later renal damage.

- Obstruction causes detrusor hypertrophy; then **fibrosis replaces smooth muscle**, reducing contractility. The bladder is the **driver**: renal dysfunction, vesicoureteral reflux (VUR) and voiding dysfunction act through it.
- Voiding cystourethrogram (VCUG) triad: **dilated posterior urethra, hypertrophied bladder neck, flattened verumontanum**; it regresses after ablation.
- Ureters dilate from **pressure transmission, VUR (up to 70%) and polyuria**. Chronic dilation loses peristalsis and coaptation, causing stasis and infection.
- Renal injury is **obstructive uropathy** (partly reversible) plus **dysplasia** (fixed at birth; sets the ceiling).
- Tubular injury and reduced medullary flow cause **nephrogenic polyuria** and post-obstructive diuresis.
- Renin rises before creatinine. Angiotensin II induces transforming growth factor-β1 (TGF-β1), causing fibrosis. Angiotensin-converting enzyme inhibitors (ACEIs) lower TGF-β1 and tumour necrosis factor-α (TNF-α), so may help selected children; both fall after ablation (candidate biomarkers).

- In fetal sheep, dysplasia is irreversible by term, so the renal rescue window closes early.

Organ / lesion	Effect	Natural history
Lung — hypoplasia	Respiratory failure	Commonest cause of neonatal death
Kidney — obstructive uropathy	Reversible insufficiency	Improves; may recur with bladder dysfunction
Kidney — dysplasia	Irreversible insufficiency	Progressive failure, hypertension
Kidney — tubular injury	Salt and water wasting	Nephrogenic diabetes insipidus (DI); worsens with age
Bladder	Poor sensation, hypercontractility, low compliance; later myogenic failure	Lifelong, changes with age
Ureters	Poor coaptation, stasis	Most retain hydronephrosis — **not** a sign of residual obstruction

"Pop-off" mechanisms

- **VURD** (VUR and dysplasia): unilateral high-grade VUR into a non-functioning kidney, thought to be a pressure reservoir. This is **refuted**: most with normal creatinine at 2 years lose it by 8–10 years; glomerular filtration rate (GFR) is abnormal early; urodynamics are worse, with more clean intermittent catheterisation (CIC); the contralateral kidney is often scarred.
- Whether **urinoma**, **urinary ascites** and large **bladder diverticula** protect or harm is debated; management is the same (relieve obstruction).

> **KEY POINT**
> Every pop-off is false reassurance: normal creatinine in a growing infant hides an abnormal GFR. Follow VURD as closely as any PUV child.

3 · Diagnosis

Why this matters: every suspected case needs a VCUG before discharge.

Antenatal

- Ultrasound (US) shows a **thick distended bladder, bilateral hydroureteronephrosis, "keyhole sign" and oligohydramnios**; an echogenic or cystic cortex means dysplasia. Sensitivity ~95% and specificity ~80% for LUTO, but only ~50% accurate for the cause.
- Fetal magnetic resonance imaging (MRI) is an adjunct. ~Half are diagnosed antenatally, but this has **not** improved renal outcome. Postnatally, half present as neonates and half with urinary tract infection (UTI).
- A male fetus with thick bladder and bilateral dilation needs postnatal US and **VCUG before discharge**.

Postnatal

- **US** first: the wall is thick if **>3 mm distended / >5 mm empty**; perirenal fluid means urinoma.
- **VCUG** (definitive) shows an elevated bladder neck, dilated elongated posterior urethra and **abrupt funnelling** at the valve, with trabeculation and diverticula; ~half have high-grade VUR. Use a 5–8 Fr feeding tube and **withdraw the catheter during voiding**. Contrast-enhanced voiding urosonography (ceVUS) is a radiation-free alternative.

- Mercaptoacetyltriglycine (**MAG3**) renogram at **2–3 months**; slow washout from a dilated system mimics ureterovesical junction (UVJ) obstruction.
- First-48 h creatinine is maternal. Follow the trend and **nadir** (1 month, 6 months, 1 year); the plateau may come late in year 1.

4 · Fetal intervention

Why this matters: it may save lungs, not kidneys, at high fetal risk.

- Decisions hinge on **oligohydramnios, hydronephrosis degree and laterality, and gestational age**. Favourable: oligo- (not an-)hydramnios, severe hydronephrosis, normal karyotype, no cortical cysts, favourable urine.

Fetal urine (after 20 weeks)	Favourable
Sodium / chloride	**<100 / <90 mEq/L**
Osmolality	**<200 mOsm/L**
β2-microglobulin	**<6 mg/L** (most sensitive)

- Low solute means the tubules still reabsorb. **Triple serial sampling** 48–72 h apart beats one sample; rising values are ominous. Prediction was poor on systematic review. The Society for Fetal Urology (SFU, 2016) standardised LUTO severity classification.
- A **vesicoamniotic shunt** may prevent **pulmonary hypoplasia** but gives **no renal benefit**. Nitinol (Somatex) dislodges less than double-pigtail (Harrison).
- **Fetal cystoscopic laser ablation**: no randomised controlled trial (RCT); no better than shunting.
- Both carry **40–50%** pregnancy loss. Deliver at a neonatal surgical centre. The paediatric urologist counsels on diagnosis, alternatives and outcomes, starting lifelong care.

> **KEY POINT**
> Fetal intervention can rescue the lung, not the kidney: fluid deficiency is still in progress, but dysplasia is already established.

5 · Neonatal presentation and initial management

Why this matters: lungs and fluids kill early; stabilise and drain first.

- Signs: Potter facies, club feet, weak stream, palpable bladder. **Pulmonary hypoplasia** is the commonest early death and usual cause of delayed surgery: oligohydramnios and huge kidneys and bladder compress the chest, so lungs lack the fluid volume and pressure for alveolar growth.

> **ALGORITHM**
> 1. Stabilise the airway.
> 2. Pass a **5–7 Fr feeding tube**. If it curls: coudé, curved stylet, or rectal finger.
> 3. Confirm on US; minimal balloon fill.
> 4. Creatinine, electrolytes, fluids.
> 5. US, then VCUG; antibiotic prophylaxis.
> 6. Ablate when stable.

- **Post-obstructive diuresis** causes large salt and water losses. †Monitor output, weight and electrolytes; replace IV.

- **Hyponatraemia** (often with UTI) is transient pseudohypoaldosteronism; give sodium supplements.
- **Urinoma or ascites** (3–10%) comes from forniceal rupture, transudation or bladder rupture. It resolves with bladder drainage; tap only for respiratory distress or tense distension.
- **Delayed presentation** is common (normal prenatal US): UTI, lower urinary tract symptoms (LUTS), overflow incontinence, haematuria, renal failure, rarely ejaculatory dysfunction. >40% of series present after 6 months; late does not mean milder. Do US, then VCUG.

6 · Surgery

Why this matters: incise precisely; over-treatment causes strictures.

Primary valve ablation (preferred)

- Ablation restores flow and **normal bladder cycling**, superior to passive diversion.
- Use a **7.5–9 Fr cystoscope** with holmium laser or Bugbee. A 9.5 Fr resectoscope with cold (Collins) knife, if the urethra allows, gives a better view but risks meatal injury. Whitaker hook or Fogarty work without an endoscope.
- Incise at **5 and 7 o'clock, with or without 12 o'clock**. Valves are thin and avascular, so **incise, don't resect**.
- Small urethra: vesicostomy or **PADUA** (progressive augmentation by dilation of the urethra anterior), up-sizing the tube from 5 to 8–10 Fr over 3–5 days.
- Catheter ≥24 h. **Repeat VCUG within 1 month**.

Complication	Cause / prevention	Management
Urethral stricture	Over-resection, hot loop, oversized instrument	†Dilation / urethrotomy; urethroplasty if dense
Meatal injury	Resectoscope through small meatus	Smaller scope / PADUA
Residual valve	Incomplete incision	†Persistent funnel on VCUG: re-ablate
†Sphincter injury	Ablating distal to valves	Incontinence — precise incision

- **Bladder neck incision** gives no benefit and risks retrograde ejaculation; bladder-neck botulinum fails. Neither is recommended.

Vesicostomy

- For a very small or premature infant, failed catheter or ablation decompression, or persistent deterioration. It **does not defunctionalise** the bladder, matches ablation in final outcome, and keeps the bladder usable for transplant.
- P4 Vesicostomy. — *Blocksom: 2 cm transverse incision midway, dome (not anterior wall) to skin to stop prolapse; indications, complications, closure*

Not in the paper answer:

- The urachus is excised with a small dome cuff; using the dome keeps the posterior wall taut.
- Same final outcome as primary ablation.

Upper tract diversion

- It gives **no renal protection**; biopsies are all dysplastic and undiversion is complex. Use only for worsening function or dilation, with or without sepsis, **despite full lower tract drainage**, after nephrology input.

Circumcision and nephrectomy

- UTI risk is 50–60% in PUV vs ~1%; circumcision cuts it >80%. Offer it to all, always before reimplantation.
- VURD **nephroureterectomy** is rare (polyuria beats anuria); **keep the ureter** for augmentation.

7 · VUR in PUV

Why this matters: reflux here signals bladder pressure.

- VUR occurs in 50–80% and is **secondary**. It resolves after ablation or vesicostomy in 25–40% (more if low grade, functioning kidney) and does not predict renal outcome.
- Treat the **bladder**. Reimplant only for recurrent febrile UTI despite circumcision and optimal bladder care; an unrehabilitated bladder gives stricture and persistent VUR. Endoscopic injection is safer, less successful.

> **KEY POINT**
> VUR in valves is a pressure gauge, not a lesion: fix the pressure, not the reading.

8 · Bladder dysfunction and valve bladder syndrome

Why this matters: the bladder keeps harming kidneys after ablation.

- P2 Valve bladder: management — *Mitchell definition; polyuria, poor sensation, residual; age pattern; continence ~5.5 y; ladder: behavioural, then oxybutynin 0.1 mg/kg bd / α-blocker, then CIC, then overnight drainage 7–10 h, then Mitrofanoff, then ureterocystoplasty*

Not in the paper answer:

- Do not force toilet training.
- Stop oxybutynin if residual and capacity rise: myogenic failure can follow.
- Mirabegron is a β3-agonist.

> **ALGORITHM**
> 1. High-pressure voiding causes fibrosis and weaker contraction.
> 2. Post-void residual (PVR) rises; nephrogenic polyuria adds volume.
> 3. Upper tracts store urine and refill the bladder, so it is never empty.
> 4. Pressure is back-transmitted; renal function declines, causing more polyuria.

- Continence comes at ~5.5 vs 2.9 years. History underestimates dysfunction, so do **uroflow and PVR** in every toilet-trained boy. New incontinence after dryness is a red flag.
- It is a continuum, not inevitable; effective drainage alters the course.
- Drug logic: α-blocker for early functional bladder-neck obstruction; antimuscarinic for the thick overactive wall; both harm once hypocontractile. **Mirabegron** improves compliance; early safety data.
- Do **videourodynamics** serially, especially when the picture changes.
- After Mitrofanoff, ~a third still deteriorate (dysplasia, polyuria). Augmentation is now rare.

9 · Prognosis for renal function

Why this matters: year-1 nadir creatinine guides counselling.

- Lifetime end-stage renal disease (ESRD) is **20–50%**. Risks: age at diagnosis, dysplasia with or without VUR, **year-1 nadir creatinine**, recurrent UTI and bladder dysfunction.

Nadir creatinine	ESRD risk
1 year <0.8 mg/dL	Minimal (best single predictor)
1 year >1.2 mg/dL	High
<0.4 / 0.4–0.69 / 0.7–0.99 / >1.0 mg/dL	**0% / 2% / 27% / 100%**

- **0.7 mg/dL is the watershed**: risk leaps. The 1-year value beats the 1-month value; 30-day creatinine velocity also predicts.
- On US, echogenicity, cysts, lost corticomedullary differentiation and smaller **renal parenchymal area** predict worse outcome.
- Higher ESRD in the post-US era and in those presenting <1 year is **survival bias**. Late presenters may be more azotaemic.
- Emerging: VCUG posterior urethral ratios, deep-learning US.

10 · CKD, ESRD and transplantation *see Ch 61*

Why this matters: grafts do well only if the bladder is managed.

- Obstructive uropathy is a leading cause of paediatric transplantation.
- Do **videourodynamics in every candidate**; optimise with CIC or overnight drainage. Native nephrectomy only for proteinuria or severe polyuria.
- There is more **ureteric obstruction**, but graft and patient survival are equal if the bladder is managed; late graft loss follows bladder dysfunction. Initial treatment does not change outcome.
- Augmentation is no longer "always before"; afterwards does as well. It raises UTI risk, but the graft is safe with good CIC. A vesicostomy-drained bladder is a safe bridge.

11 · Long-term follow-up, sexual function and fertility

Why this matters: PUV is lifelong.

- Transition to adult urologists who know valves. Attention-deficit/hyperactivity disorder (ADHD) and neurodevelopmental disorders are commoner; assess early.
- Adult LUTS are 2–3× the population rate. Erectile dysfunction (ED) and infertility are similar overall, but worse with incontinence or renal insufficiency.
- †Lifelong checks: blood pressure (BP), creatinine, urine protein, culture when febrile, US, and uroflow with PVR.

12 · Related urethral lesions

Why this matters: these mimic PUV.

- P2 Anterior urethral valve — *25–30× rarer than PUV; diverticulum; penile bulge; vesicostomy if premature, else endoscopic ablation; open repair of large diverticulum; 80% bladder dysfunction, 78% normal renal function; 25-fold risk with azotaemia+VUR+UTI*
- **Urethral atresia** is a membrane at the distal prostatic urethra; survival needs decompression (urachus, shunt). Treat with vesicostomy, then PADUA.
- **Urethral duplication** (Effmann I blind, II complete, III caudal duplication; "Y"-type to perineum): the **ventral** channel has the sphincter and verumontanum, so move it to the glans. Blind accessory: observe or Bugbee. Two glanular openings: septum incision and dorsal

urethroplasty. "Y": staged flaps. Meatus in anal sphincter: anterior sagittal trans-ano-rectal approach (ASTRA).

- **Urethrorrhagia**: peripubertal boys with blood spots after clear urine. Incomplete sphincter relaxation or meatal stenosis causes turbulence and mucosal engorgement. Up to 92% resolve, so treat voiding dysfunction (history, US, uroflow, PVR). Cystoscopy only if atypical: stricture eventually appears in 14–28%, and instrumentation may provoke it.
- Anorectal malformation: rectourinary fistula is common; do US and VCUG. At posterior sagittal anorectoplasty (PSARP), catheterise first, excise the fistula flush, catheter ≥1 week.

13 · Landmark trials

Why this matters: they shape counselling.

Trial	Design	Result	Take-home
PLUTO	RCT: vesicoamniotic shunt vs conservative in LUTO	Trend to better 28-day survival; poor long-term survival and renal function in both	Shunt may help lungs, not kidneys
Circumcision RCT	Circumcision and prophylaxis vs prophylaxis	Febrile UTI 3% vs 49% over 2 years	Circumcise boys with PUV
Oxybutynin RCT	Oxybutynin to toilet training vs observation post-ablation	Better hydronephrosis/VUR; no change in creatinine, UTI, continence	Selective, monitored use

46 Neuromuscular Dysfunction of the Lower Urinary Tract in Children

1 · Definition and causes

Why this matters: kidneys decide late outcome.

- **Neurogenic lower urinary tract (LUT) dysfunction** is bladder and/or sphincter dysfunction from aberrant innervation: congenital or acquired, central or peripheral, transient or permanent.
- The commonest cause is **neural tube defect (NTD)**, ≈ spina bifida (SB). Survival is ≈90% to 10 years, so **renal failure** is a leading late morbidity; **10%** of SB deaths are urological.
- Pivotal advance: **clean intermittent catheterisation (CIC)**, Lapides 1971.

Congenital	Acquired
Myelomeningocele; occult NTD (lipomeningocele, tethered cord); sacral agenesis; anorectal malformation (ARM)	Brain: cerebral palsy (CP), tumours, infarcts. Cord: trauma, transverse myelitis. Pelvis: extensive surgery

2 · Myelodysplasia (NTD)

Why this matters: spinal level cannot predict the bladder.

- P1 Neurogenic bladder in myelodysplasia — *lesion levels; detrusor × outlet patterns; EAU work-up; McGuire risk; early CIC outcomes*

Not in the paper answer:

- Antimuscarinic side effects: dry mouth, constipation, flushing, blurred vision. Trospium (quaternary amine) has few central effects.
- Botulinum toxin causes no fibrosis.

Classification and prevention

- **Myelomeningocele** is open, commonest and most severe; >90% lumbar, lumbosacral or sacral. **Meningocele** is meninges only. **SB occulta** is closed, in up to 15% of people, usually benign.
- The neural tube closes by **day 35**. Prenatal detection: maternal α-fetoprotein (AFP) and ultrasound (US).
- **Folic acid** cuts NTDs 50–70%: all women of child-bearing age take **400 µg/day**, from ≥4 weeks before conception through the first month; women with SB take **4–5 mg/day**.
- Recurrence is **5%** after one affected child; offspring risk is 3.7% if one parent has SB (≈15% if both).
- Almost all open NTDs have **Chiari II**; most have hydrocephalus (treated by ventriculoperitoneal (VP) shunt or endoscopic third ventriculostomy (ETV)).
- **Bony level differs from neurological level by 1–3 segments; LUT function is not predictable from bony level or legs**, so urodynamics defines every bladder.

KEY POINT

Management of Myelomeningocele Study (**MOMS**): prenatal repair (<26 weeks) reduced death or shunt need (relative risk (RR) 0.70) and improved motor and cognitive scores at 30 months. Costs: prematurity (34 vs 37 weeks), uterine-scar thinning or dehiscence (⅓), more re-tethering. Rationale: a "two-hit" in-utero injury.

Pathophysiology

- In **detrusor–sphincter dyssynergia (DSD)**, the sphincter fails to relax during detrusor contraction. It presumably drives remodelling and renal risk; diagnose it by electromyography (EMG), not fluoroscopy.

KEY POINT

Every pattern is detrusor (over/under) × outlet (competent/incompetent/dyssynergic). Dangerous: **overactive or poorly compliant detrusor against a closed or dyssynergic outlet**: storage pressure rises, causing vesicoureteric reflux (VUR) and hydroureteronephrosis. An incompetent outlet "pops off": the child is wet but the kidneys are safe.

Newborn evaluation

ALGORITHM

1. Before discharge: renal–bladder US and post-void residual (PVR) (need for CIC?).
2. After week 1: serum creatinine.
3. Once stable: urodynamics, **safe from ≈3 months** (within 3 months per ICCS, SBA, EAU).
4. Cystography: ideally videourodynamics, otherwise voiding cystourethrogram (VCUG).
5. Optional baseline dimercaptosuccinic acid (DMSA) scan (EAU).
6. US at 6, 12, 18 and 24 months, then yearly.

- Guidelines: International Children's Continence Society (ICCS 2012); British Association of Paediatric Urologists (BAPU 2016: most selective, "no consensus" on early CIC); Spina Bifida Association (SBA 2018); European Association of Urology (EAU 2022).
- UMPIRE (CDC newborn SB protocol) baseline: US normal 56%, mild hydronephrosis 40%, moderate–severe 4%; dilating VUR **7%**; DMSA defects ≈8%.
- **Cystatin C** beats creatinine, as low muscle mass hides a falling glomerular filtration rate (GFR). Gold standard: nuclear clearance.
- High-grade VUR acts as a pop-off, so urodynamic pressures underestimate risk.

Urodynamic risk factors

Category	Definition
Hostile / high	Detrusor leak point pressure (DLPP) or end-filling pressure **≥40 cmH_2O**; neurogenic detrusor overactivity (NDO) with DSD (DSD removed in UMPIRE revision because of poor agreement)
Intermediate	NDO, reduced compliance, 25–39 cmH_2O
Abnormal but safe	<25 cmH_2O
Normal	Normal capacity, compliance >15 mL/cmH_2O, no NDO/DSD, minimal residual

- Poor compliance independently predicts renal injury.

Tethered cord

- The conus lies below L1–L2; growth stretches the cord, so gait, sensation and LUT deteriorate. Secondary tethering after repair is common (more after fetal repair).
- † Suspect it with new wetting, rising pressure or falling compliance, new urinary tract infections (UTIs), worsening scoliosis, or back and leg pain.
- Do urodynamics **before untethering and 3–4 months after**. Needle EMG helps when magnetic resonance imaging (MRI) is equivocal.

3 · Proactive (early) management

Why this matters: early pressure control prevents renal failure.

Priorities in order: **kidneys, then continence, then UTI avoidance, then sexual function and fertility**.

- **Early CIC and an antimuscarinic** (Bauer) treat hostile dynamics (DSD, poor compliance, high outlet pressure). All paediatric societies endorse this; there is **no benefit if the child empties spontaneously**.
- CIC bypasses DSD and is better accepted from infancy. If hydronephrosis persists, add overnight indwelling drainage.
- Give **no** antibiotic prophylaxis on CIC; treat bacteriuria only if symptomatic.

Outcome	Expectant	Early CIC + antimuscarinic
Hydroureteronephrosis	18–80%	0–15%
Renal scarring	27–50%	4–12%
End-stage renal disease (ESRD)	≈30%	0–1.6%
Augmentation	—	18–24% fewer

> **KEY POINT**
> Proactive won because it treats the cause (pressure, failure to empty) before the consequence (reflux, dilatation, scarring, renal failure).

4 · Management ladder

Why this matters: surgery comes last.

> **ALGORITHM**
> 1. CIC and antimuscarinic, with or without a β3-agonist.
> 2. If that fails: intradetrusor botulinum toxin.
> 3. If CIC is impossible: vesicostomy as a bridge.
> 4. Then augmentation, with or without a catheterisable channel and an outlet procedure *see Ch 48*.

Pharmacotherapy

- **Antimuscarinics** are first-line (M2 receptors are most numerous; M3 mediates contraction). They raise capacity and reduce leaks, with no cognitive effect in a randomised controlled trial (RCT).
- US Food and Drug Administration (FDA)-licensed: **oxybutynin** (≥5 y; 0.1–0.6 mg/kg/day, max 5 mg/dose; oral, patch, or **intravesical**, which avoids first-pass), **solifenacin** (≥2 y), **fesoterodine** (≥6 y, >25 kg) and **mirabegron** (≥3 y; may cause hypertension).

- **α-Sympathomimetics** (ephedrine, pseudoephedrine) treat low-outlet incontinence: unlicensed; a minority respond.

Botulinum toxin A

- It blocks release of acetylcholine (and ATP, substance P), acting on motor and sensory arms.
- Under general anaesthesia, by rigid cystoscopy, inject **≈10 U/kg (max 360 U)** into ≈30 sites, **sparing the trigone**; minimum age usually 2 y.
- Continence is 65–87%, lasting 6–10 months; repeats stay effective (no tachyphylaxis). Poor pre-existing compliance predicts poor response.

Surgery

- **Vesicostomy**: temporary bridge when CIC is unmanageable; reverses hydronephrosis and VUR, fewer UTIs.
- **Augmentation** (ileum preferred) treats the refractory bladder. Add a channel if urethral CIC is hard, and a sling, bladder-neck reconstruction or artificial urinary sphincter (AUS) for a weak sphincter *see Ch 48*. After ileum, check B12 yearly from 2 y and do cystoscopy from 10–15 y.

VUR in the neurogenic bladder

- VUR is secondary; treating the bladder resolves 30–55%. Grade I–III emptying fully: prophylaxis. IV–V or poor emptying: CIC. Poor compliance: add antimuscarinic.
- **Avoid Credé** with a reactive sphincter, because it raises voiding pressure and worsens VUR.
- Antireflux surgery: breakthrough febrile UTI or persistent dilation despite low pressure; with CIC, ≈95% success (Jeffs). At augmentation, **reimplant grade ≥III** (else it persists in 20–47%).

5 · Bowel management †

Why this matters: a loaded bowel worsens the bladder.

- † Neurogenic bowel causes constipation and faecal incontinence. Escalate: diet and timed toileting; laxatives with or without suppositories; transanal irrigation; then **Malone antegrade continence enema (ACE)**, often at augmentation. Treat constipation around reconstruction.

6 · Latex allergy †

Why this matters: anaphylaxis in theatre is preventable.

- † Repeated mucosal exposure brings high sensitisation risk and intraoperative anaphylaxis. Latex-free **from birth**; latex-free condoms.

7 · Sexual function and transition

Why this matters: adolescents need counselling and handover.

- Function is **inversely proportional to neurological level**.
- **Central precocious puberty** affects ≈50% of girls and 10–30% of boys.
- **Males**: erectile dysfunction (ED) ≈75%; ejaculation 40–75% (some retrograde); a lesion at S1 or below usually allows normal reproduction. **Sildenafil** works in ≈80%.
- **Females**: 70–80% conceive; caesarean and late incontinence are common. Avoid the combined pill (thrombosis) and intrauterine device (IUD; pelvic infection).

- Offer SB-specific sex education and † a structured **transition**.

8 · Other causes

Why this matters: lesion level predicts pattern and risk.

Sacral agenesis

- † Linked to maternal diabetes. The child may walk normally; look for **flat buttocks, a short gluteal cleft** and absent sacrum on plain radiograph.
- Level does not predict the bladder; ≈⅓ present late. Expect an areflexic bladder with weak sphincter, or NDO with DSD.

Spinal cord injury (SCI)

- **Spinal shock**: CIC. After ≈3–4 months, a **suprasacral lesion gives NDO with DSD**; cauda equina recovers little. Do urodynamics after shock.
- **Autonomic dysreflexia** (lesion **≥T6**): bladder filling triggers a sympathetic surge, with severe hypertension, stroke or seizures. Monitor blood pressure; keep α-blockers ready during urodynamics.

Transverse myelitis

- Acute retention in >95%; treat with steroids, plasma exchange, intravenous immunoglobulin (IVIG). Areflexia (≈70%) evolves into NDO 59–90%, poor compliance ≈47% and DSD 17–80%; 57–73% need CIC.
- The clinical picture does not predict risk; **late CIC leads to poorer compliance**, so do early urodynamics and CIC.

Cerebral palsy

- CP is a coordinated upper motor neuron (UMN) lesion: late continence, urgency; NDO 30–61%, DSD ≈12%, and ⅓ are normal. Upper-tract risk is **<5%**, so do urodynamics only if conservative care fails, or for DSD symptoms or recurrent UTI.

ARM and pelvic surgery

- ARM: cord anomaly in 18–50% (more if high), so do **spinal US before 3 months**. Neuropathy comes from the **cord, not the posterior sagittal anorectoplasty (PSARP)**; urodynamics before and after pull-through. Currarino: tethered cord >80%.
- Pelvic splanchnic injury gives an underactive bladder; hypogastric injury, poor compliance and an open bladder neck. Sacrococcygeal teratoma (35–45% neuropathic): urodynamics before and after; pelvic rhabdomyosarcoma survivors need surveillance.

> **KEY POINT**
> Lesion level predicts pattern. **Suprapontine** (CP) gives coordinated overactivity and low risk. **Suprasacral cord** (SCI, myelitis, tethering) gives NDO with DSD and high risk. **Sacral/peripheral** gives underactivity. Myelodysplasia is mixed and unpredictable.

9 · Landmark trials

Why this matters: they set the 40 cmH_2O line.

Trial	Design	Result	Take-home
MOMS	RCT, prenatal vs postnatal repair	Less shunting, better motor score; more prematurity, uterine dehiscence	Fetal repair in selected centres

Trial	Design	Result	Take-home
McGuire 1981	Myelodysplasia cohort	DLPP >40 cmH_2O led to upper-tract damage	40 cmH_2O = safety line
Sildenafil in SB	Double-blind placebo RCT	≈80% improved erections	First-line for ED

47 Functional Disorders of the Bladder and Bowel in Children

1 · Normal bladder and bowel function

Why this matters: dysfunction is judged against normal.

- P1 Describe neurophysiology of lower urinary tract and various types of voiding dysfunction in children *— innervation, PAG–PMC control, maturation, ICCS functional types, flow patterns, management principles*

Not in the paper answer:

- The International Children's Continence Society now uses expected bladder capacity = (age + 1) × 30 mL; the paper's (age + 2) is the older formula†.
- Infant voiding pressures are higher in boys than girls.
- The rectum, like the bladder, is a cloacal derivative and compliant store.
- The mature lower urinary tract (LUT) has **storage** (normal compliance, no overactivity, closed outlet) and **voluntary coordinated emptying**.
- In **infancy**, voiding pressures are often **>100 cm H_2O** and interrupted voiding is normal; both are gone by **~2 yr**. Infants arouse before voiding. Dyscoordination later is **dysfunction, not immaturity**.
- **Rectal** distension triggers the recto-anal inhibitory reflex (**RAIR**): the internal sphincter relaxes to "sample" contents while the external sphincter contracts.
- Toilet training proceeds from bowel to day bladder to night bladder. Methods are equivalent; **late initiation** is linked to incontinence, enuresis, urinary tract infection (UTI) and withholding.

> **KEY POINT**
> Bladder and bowel are one unit: they share pelvic space ("zero-sum" volume) and converging dorsal-root afferents (cross-organ sensitisation). A loaded rectum causes overactivity and poor emptying, so treating constipation improves lower urinary tract symptoms (LUTS). Anticholinergics for the bladder can recreate the constipation that drove the symptoms.

2 · Definitions — International Children's Continence Society (ICCS)

Why this matters: precise labels lead to the right treatment.

Bladder and bowel dysfunction (BBD) is LUT dysfunction plus bowel dysfunction (constipation and/or faecal incontinence) in a neurologically normal child.

Group	Symptom	ICCS definition
Storage	Increased frequency	**≥8 voids/day**
	Decreased frequency	**≤3 voids/day**
	Urgency	Sudden immediate need to void
	Nocturia	Waking at night to void

Group	Symptom	ICCS definition
Voiding	Hesitancy	Difficulty initiating when ready
	Straining	Valsalva to initiate and maintain
	Intermittency	Non-continuous stream
	Retention	Cannot void despite effort; distended bladder
Other	Holding manoeuvres; incomplete emptying; recurrent UTI	Incomplete-emptying sensation counts even with minimal post-void residual (PVR); recurrent UTI is itself a BBD symptom

- **Daytime urinary incontinence (DUI)** is leakage while awake at age **≥5 yr**. **Enuresis** is intermittent leakage while asleep at ≥5 yr: **monosymptomatic (MNE)** has no other LUTS, **non-monosymptomatic (NMNE)** has any LUTS.
- **Faecal incontinence** is either **retentive** (the majority; overflow around a faecal mass) or **functional non-retentive (FNRFI)** (~1 in 5; no constipation; up to half are also wet).
- † **Expected bladder capacity (EBC)** = (age in years + 1) × 30 mL (ICCS, to ~12 yr).

Daytime LUT conditions

Condition	Definition / mechanism	Key features
Overactive bladder (OAB)	Urgency, with or without frequency, nocturia, incontinence; no UTI or other pathology	Urgency incontinence, holding manoeuvres
Dysfunctional voiding	Habitual sphincter/pelvic-floor contraction **during voiding**, neurologically intact; staccato flow	Learned; high PVR, UTI, secondary vesicoureteral reflux (VUR)
Underactive bladder	Raises abdominal pressure to initiate, maintain or complete voiding	Infrequent, interrupted flow, PVR; overlaps dysfunctional voiding
Voiding postponement	† Habitual deferral with holding until urgency incontinence	† Low frequency; oppositional defiant disorder
Giggle incontinence	Extensive emptying during or after laughing; LUT otherwise normal	Distinguish from laugh-provoked OAB leak, vaginal reflux, postponement
Vaginal reflux	Vagina fills retrograde during voiding	Prepubertal girl; small leak just after voiding
Extraordinary daytime frequency	† ≥ hourly small voids, no incontinence or nocturia	† Benign, self-limiting; exclude UTI, diabetes
Hinman syndrome (non-neurogenic neurogenic bladder)	End stage of dysfunctional voiding	Trabeculation, VUR, hydronephrosis, renal damage, hypertension

KEY POINT

Trap: dysfunctional voiding (neurologically intact) is not detrusor–sphincter dyssynergia (DSD), which needs a neurological lesion. Say "LUT dysfunction", not "voiding dysfunction". Urgency incontinence is not always OAB: a voiding postponer also leaks with urge, and anticholinergics help briefly, then worsen emptying.

3 · Clinical presentations and associations

Why this matters: BBD hides behind other complaints.

- **Acute retention** (peak 2–4 yr) follows genital inflammation, instrumentation or **faecal impaction**; exclude tumour and neurological causes. If comfortable, try **analgesia, sitz bath and/or enema before a catheter**: retention is a reflex to a pelvic noxious stimulus.
- **Functional constipation** affects ~14% worldwide, peaking at 2–3 yr. In the **withholding cycle**, painful stool leads to withholding, harder stool and more pain. Organic causes: anorectal malformation (ARM), Hirschsprung, coeliac disease, cystic fibrosis, hypothyroidism, hypercalcaemia, hypokalaemia, spinal causes, drugs.
- **Abdominal pain**: constipation causes ~half of acute cases. Alarm features (weight loss, gastrointestinal (GI) bleeding, fever, severe diarrhoea, vomiting) need gastroenterology.
- **Genital pain** is often referred from the bladder (boys grab the penile tip).
- **Attention-deficit/hyperactivity disorder (ADHD), autism spectrum disorder (ASD) and intellectual disability (ID)** have the highest BBD rates and respond worse. Treating ADHD improves enuresis (not constipation); risperidone causes incontinence. Adapt urotherapy.
- Anxiety, depression and stress are associated. Quality of life (QoL) improves even with education. Childhood BBD predicts adult OAB, incontinence and interstitial cystitis/bladder pain syndrome (IC/BPS).

4 · Evaluation

*Why this matters: evaluation is mainly to **exclude organic disease**; children are over-investigated.*

Warning signs

Finding	Points to
Blood in stool without fissure, fever, vomiting, weight loss, severe distension	Organic GI/systemic disease
Failure to thrive, raised blood pressure	Chronic kidney disease (CKD)
Lumbosacral dimple, hair tuft, vascular lesion, asymmetric cleft	Occult dysraphism: spinal magnetic resonance imaging (MRI)
Weak/numb legs, abnormal gait; absent anal or cremasteric reflex	Neurogenic bladder/bowel
Abnormal anal position or small anus	ARM
Low specific gravity (SG), glucosuria, proteinuria	Diabetes insipidus (DI), diabetes mellitus, CKD
Girls: constant dribble despite normal voiding	Ectopic ureter

- Ask **concretely** (stools this week, Bristol pictures); ask about bowels in **every** child with LUTS. Use a **3-day bladder diary** and **bowel diary**. Review prior treatment, school toilets, neurological history.
- **Bristol** scale: types 1–2 are hard (constipation), 3–4 normal, and 5–7 range from soft to watery.
- Examine growth, gait, back, neurology, abdomen, anogenital area. A **gush on separating the labia** means vaginal reflux. Routine digital rectal examination is **not** needed.

Investigations (stepwise, non-invasive)

- Scores: Dysfunctional Voiding Symptom Score (DVSS) and Vancouver. **ICCS response**: complete 100%, partial 50–99%, none <50%.
- **Urinalysis**: SG >1.005 makes DI unlikely. Persistent protein ≥1+ needs a first-morning protein:creatinine ratio (>**0.2 mg/mg** is abnormal).
- Measure **PVR** within 5 min of voiding, with the bladder at 50–115% EBC. Renal ultrasound (US) is not routine; DVSS ≥12 with UTI predicts an abnormal US.
- **Uroflow** (3 flows, >50% EBC): bell normal, tower OAB, staccato dysfunctional voiding, interrupted underactive bladder, plateau obstruction.
- **Pelvic floor electromyography (EMG)** active during voiding means dysfunctional voiding. An **EMG lag time <2 s** means detrusor overactivity.
- **Video urodynamics** is only for treatment failure or suspected obstruction or upper-tract risk.

Bowel — Rome IV functional constipation (child ≥4 yr)

ALGORITHM

1. ≥2 of, ≥1/week for ≥1 month, irritable bowel syndrome (IBS) criteria not met:
2. ≤2 defecations in toilet/week; ≥1 faecal incontinence/week; retentive posturing/withholding; painful or hard stools; large rectal faecal mass; large-diameter stools that may block the toilet.

- Pain is **not** a criterion; if pain persists after constipation resolves, think IBS.
- Do no routine labs or **plain radiograph** (only if impaction is suspected and exam impossible, or if refractory). A rectal diameter **≥30 mm** on US suggests constipation (European Association of Urology/European Society for Paediatric Urology, EAU/ESPU), but gastroenterologists advise against routine US.
- If refractory: transit study, contrast enema, **anorectal manometry** (absent RAIR needs rectal biopsy for Hirschsprung), and colonic manometry.

5 · Management of daytime incontinence

Why this matters: most children improve with bowel care and urotherapy alone.

ALGORITHM

1. History, exam, urinalysis, PVR, with or without uroflow.
2. Exclude neurogenic causes, diabetes, CKD and anatomical causes.
3. Optimise neurodevelopmental disorders; **treat constipation first**.
4. If wetting is only at night, treat as MNE.
5. Otherwise, **standard urotherapy** (≥4 wk); re-evaluate.
6. Then **specific urotherapy** by diagnosis.
7. If refractory: botulinum toxin or neuromodulation.

Standard urotherapy

- A structured cognitive-behavioural programme: information, instruction, lifestyle (fluids, diet, regular emptying), diaries and follow-up.
- Posture: hips and knees ~90°, feet supported, diaphragmatic breathing. **Timed voiding** 6–7 times a day, anchored to events, with a timer watch.
- **Reward adherence, not dryness**. Urotherapy cures DUI in more than half.

Dysfunctional voiding

- Learned functional obstruction causes detrusor hypertrophy and overactivity, then poor compliance and decompensation, with VUR and hydronephrosis (Hinman). Diagnose by **staccato flow with active pelvic floor EMG**.

Renal risk	Management
High (hydronephrosis, VUR, rising creatinine)	Video urodynamics; lower storage pressure; clean intermittent catheterisation (CIC) past the sphincter
Low	Urotherapy, double voiding, treat constipation, short-course prophylaxis for recurrent UTI

- **Biofeedback is the first-line specific therapy** (fewer UTIs and better flow than urotherapy). **α-blockers** (doxazosin) lower PVR. If refractory, give **intrasphincteric onabotulinumtoxinA** 50–300 units.

Overactive bladder

Use drugs only after urotherapy and bowel management.

Agent	Dose / point
Oxybutynin (non-selective)	Immediate release **0.3–0.6 mg/kg/day** in 2–3 doses (max 15 mg); extended release (ER) 5–20 mg/day (do not crush); patch 3.9 mg/day. Only US-approved drug for paediatric non-neurogenic OAB. Dry mouth, flushing, constipation, cognition.
Tolterodine	1–2 mg twice daily or ER 2–4 mg; ER randomised controlled trial (RCT) negative (under-dosing >35 kg)
Solifenacin (M3-selective)	2.5–10 mg daily; long half-life; positive phase III RCTs (LION, LEOPARD)
Trospium (quaternary amine)	Does not cross the blood–brain barrier; take on an empty stomach
Mirabegron (β3-agonist)	25–50 mg if >35 kg; approved ≥3 yr for **neurogenic** detrusor overactivity only; similar to solifenacin in OAB; safe add-on

- Check PVR and stools before and during antimuscarinics.
- **Intradetrusor onabotulinumtoxinA** (refractory): **5 units/kg, max 100 units**; it lasts ~12 months, and retention needing CIC is rare.

Neuromodulation	Technique	Verdict
Parasacral transcutaneous electrical nerve stimulation (TENS)	Surface electrodes over S2–S3	Beat oxybutynin; true-sham RCT negative (placebo effect?)
Transcutaneous/percutaneous tibial nerve stimulation (TTNS/PTNS)	34-gauge needle ~5 cm above medial malleolus; great-toe flexion confirms	Weaker than parasacral TENS
Sacral nerve stimulation (SNS)	Implanted lead	Off-label; helps bladder and bowel; migration, infection, revision

Other conditions

- **Underactive bladder**: timed and double voiding, biofeedback, interferential stimulation; CIC if PVR is high (poorly tolerated).

- **Postponement**: timed voiding and treat comorbidity; † avoid anticholinergics.
- **Giggle**: **methylphenidate** (relapse off drug) or **biofeedback** (more durable).
- **Vaginal reflux**: void with legs abducted (sit backwards on the toilet) and cough at the end.
- **Extraordinary frequency**: † reassure.

6 · Nocturnal enuresis

Why this matters: common, treatable; the alarm cures durably.

- P1 P3 Nocturnal enuresis / Enuresis — *ICCS definitions, primary vs secondary (6 months dry), triad, sleep apnoea, evaluation, alarm vs desmopressin, second line*

Not in the paper answer:

- In NMNE, minor daytime symptoms need not delay enuresis treatment.
- Desmopressin works best with normal daytime volumes; special nights include camp.
- Imipramine is fatal in overdose; keep it away from siblings.
- Prevalence is **5–10% at 7 yr**; ~15% resolve each year, and ~2% persist into adulthood. Loci are on chromosomes 6 and 13.
- The **triad** is nocturnal polyuria (relative antidiuretic hormone (ADH) deficiency), reduced nocturnal capacity or overactivity, and failure to arouse.
- In **NMNE**, treat bothersome daytime LUTS and constipation first. In **MNE**, treat from **~6 yr**.
- Ineffective: night waking, fluid restriction alone, rewards, punishment, pants.

	Enuresis alarm	Desmopressin
Use	Nightly; the parent wakes the child; continue until **14 consecutive dry nights**	**0.2–0.4 mg** tablet or 120–240 µg melt ~1 h before bed; stop if no effect after ~1 week
Response	50–70%; slow	Immediate; rule of thirds (complete/partial/none); best with nocturnal polyuria
Durability	**Durable**; overlearning lowers relapse	**Not durable**; drug holidays or special nights
Safety	Nil	Hyponatraemia: † no fluids from 1 h before dose

ALGORITHM

1. MNE at ≥6 yr: start the alarm or desmopressin.
2. If it fails, check use, then switch or **combine**.
3. Re-evaluate: hidden daytime LUTS, constipation, sleep-disordered breathing, neurodevelopmental disorder; uroflow and PVR.
4. Then **add an anticholinergic** to desmopressin.
5. If refractory: imipramine, a β3-agonist or specialist drugs.

- Sleep-disordered breathing: adenotonsillectomy helps about half. If an anticholinergic works and then relapses, suspect constipation.
- **Imipramine** 25–50 mg is **cardiotoxic** (overdose, long QT), so do an electrocardiogram (ECG) if there is syncope, palpitations or a family history of sudden death.

7 · Functional constipation — treatment

Why this matters: bowel first; LUTS often settle once the rectum is empty.

- P4 Management of Defecation Disorders resulting into LUTS — *why bowel causes LUTS, Rome IV, bowel first, disimpaction/maintenance PEG, biofeedback, irrigation, ACE, toxin, neuromodulation*

Not in the paper answer:

- Daily enemas for 3–6 days are equivalent if PEG is unavailable; monitor electrolytes in renal impairment.
- "Dependence" is really relapse after abrupt stopping; laxatives do not cure dysmotility.

ALGORITHM

1. If impacted, **disimpact** with oral polyethylene glycol (PEG) or enemas.
2. Give education, a diary, toilet sitting and **maintenance PEG**.
3. Review at 2 wk; if failing, check adherence, dose, impaction, mental health.
4. Continue for **≥2 months**. Wean when ≥1 month symptom-free, with ≥3 stools/week and Rome IV not met.
5. Relapse: reinstate, or refer to gastroenterology.

Laxative	Dose
PEG 3350 — disimpaction	**1–1.5 g/kg/day × 3–6 days** (>3 yr)
PEG 3350 — maintenance	**0.2–0.8 g/kg/day**, start 0.4 (>6 months)
Lactulose — maintenance	1–2 g/kg once or twice daily (safe in infants)

- Electrolyte-free PEG is as effective with fewer adverse events; caution in renal impairment. **Titrate**: if no stool in 24 h, increase; if watery accidents occur, add a **stimulant** (senna, bisacodyl).
- There is **no dependence**; wean gradually. Keep fluid and fibre normal (~10 g rising to ~20 g/day). Extra water, fibre and probiotics do not help and must not delay laxatives.
- If refractory: pelvic floor biofeedback for dyssynergic defecation, transanal irrigation (¼–⅓ stop), anal sphincter toxin, or SNS. For an antegrade continence enema (**ACE**), do **colonic manometry first**, because diffuse dysmotility may need resection or diversion *see Ch 48*.

8 · Recurrent UTI and BBD

Why this matters: BBD outranks reflux as a UTI risk.

- In RIVUR/CUTIE, recurrence was **BBD + VUR 51% > BBD alone 35% > VUR alone 20%**. A wider caecal diameter predicted more recurrence.
- Dysfunctional voiding, postponement and underactive bladder carry more UTI than OAB; secondary VUR often resolves with treatment.
- **EAU/ESPU**: exclude BBD in every toilet-trained child with febrile or recurrent UTI, and give prophylaxis during BBD treatment. The American Urological Association (**AUA**) advises prophylaxis for BBD with VUR. ICCS: continue until symptoms improve.

9 · Landmark trials

Why this matters: they guide prophylaxis and drug choice.

Trial	Design	Result	Take-home
RIVUR + CUTIE (BBD analysis)	Prophylaxis vs placebo, <6 yr after febrile UTI, with or without VUR	BBD + VUR highest recurrence; BBD alone > VUR alone	Treat BBD; prophylaxis for BBD + VUR
LION / LEOPARD	Phase III placebo RCTs, solifenacin	Improved all efficacy measures; mild–moderate side effects	Best paediatric antimuscarinic evidence

48 Lower Urinary Tract Reconstruction in Children

1 · Goals and principles

Why this matters: kidneys come first, dryness last.

- The priorities, in order, are: **protect the kidneys, then prevent urinary tract infection (UTI), then achieve continence**.
- The indication is a hostile bladder (poor compliance, high pressure, poor emptying) despite medical care, mostly in spina bifida; also exstrophy, posterior urethral valves and bilateral ectopic ureters.
- **Clean intermittent catheterisation (CIC)** (Lapides) is the key enabler, because spontaneous voiding after reconstruction is a hope, not a requirement. The main determinant of outcome is **patient and family commitment**.

> **KEY POINT**
> Every reconstructive step trades storage for emptying. Raising capacity or outlet resistance turns a leaking, self-venting system into one wholly dependent on emptying. A continent reservoir that is not catheterised causes hydronephrosis, sepsis and perforation, so incontinence can be "protective". CIC and motivation are prerequisites.

2 · Patient selection and preoperative evaluation

Why this matters: the work-up decides what one operation must include.

- **Storage**: a chronic filling pressure **>40 cm H_2O** causes hydroureteronephrosis, a falling glomerular filtration rate (GFR) and acquired vesicoureteral reflux (VUR). Start anticholinergics and CIC; persistent poor compliance needs **augmentation**.
- **Outlet**: resistance must exceed bladder pressure during all activity; bladder neck or external sphincter alone suffices. A deficient outlet needs a procedure.
- **Emptying**: nothing predicts voiding after augmentation, so every patient practises CIC beforehand.

Urodynamic finding	What it decides
Capacity and fluoroscopic bladder shape	Length of bowel segment
Detrusor leak point pressure (DLPP), end-filling pressure, open bladder neck	Need for a bladder neck procedure
VUR with a history of UTI	Reimplantation
Sphincter electromyography (EMG; needle is more accurate)	Dyssynergia

- Fill at **≤10% of predicted capacity per minute**, because fast filling falsely lowers compliance; use body-temperature fluid. The UMPIRE protocol (Urologic Management to Preserve Initial Renal function) standardises care.
- Every child has ultrasound (US). Hydronephrosis needs a renogram **with a bladder catheter** (to exclude primary obstruction) and voiding cystography or videourodynamics.
- Check electrolytes, blood urea nitrogen (BUN) and creatinine. If raised, involve nephrology and measure 24-h urine **volume**: the reservoir must hold **3–4 h** of output.
- In **renal insufficiency**, acidosis **always worsens** with ileum or colon, so consider stomach. In **liver failure**, ammonia reabsorption is dangerous.

- **Reflux**: low-grade secondary reflux usually resolves after augmentation; correct high-grade reflux at reconstruction.
- **Bowel**: note short gut (cloacal exstrophy, resection, irradiation); diarrhoea or faecal incontinence means sparing the ileocaecal valve. Add an antegrade continence enema (ACE) if needed.
- **Route**: a sensate urethra, wheelchair use or prior bladder neck repair favour an abdominal stoma, which patients prefer. † Assess hand function and body habitus.
- **Timing**: early if kidneys are threatened, otherwise when incontinence is socially unacceptable; all steps in **one operation** where possible.
- **Sterile urine** is mandatory. Bowel preparation is optional; enhanced recovery after surgery (ERAS; carbohydrate up to 2 h) reduces 90-day complications.
- **Ventriculoperitoneal (VP) shunt**: infection is rare with sterile urine, but shunt failure increases; remove drains early.

3 · Managing the ureters

Why this matters: reflux or obstruction defeats the reconstruction.

- Reimplant into the native bladder, with or without a **psoas hitch**: 3–5 non-absorbable sutures go through muscle into the psoas tendon, avoiding the genitofemoral and femoral nerves, and allow an 8–10 cm tunnel. A short ureter needs transureteroureterostomy.
- Into bowel: **colon** under a taenia; **stomach** submucosally; **ileum** needs a **nipple valve** (hardest).

4 · Bladder augmentation (enterocystoplasty)

Why this matters: augmentation makes a high-pressure bladder safe.

- P1 Instestinal segments in urology — *segments compared with electrolyte disturbance; NH_4Cl absorption mechanism; other metabolic, stone, cancer complications; renal function for continent diversion*

Not in the paper answer:

- Base deficit is the most sensitive marker; height can fall even without frank acidosis.
- A composite stomach-and-bowel augment is near neutral.
- Mucus: colon more than ileum more than stomach; stomach forms uric acid stones.

Principles

- Do cystoscopy first. **Bivalve the native bladder widely** (sagittal or stellate); a narrow anastomosis makes a diverticulum.
- **Always detubularise** and **reconfigure into a sphere**. Intact bowel contracts at **60–100 cm H_2O**, and a sphere gives the most volume per surface area.
- Length: ileum **20–40 cm**, colon **~20 cm**; err on the large side.
- **No permanent sutures or staples**: they seed stones. Place a suprapubic tube through native bladder.

Segments

- **Ileocystoplasty**: take ileum at least **15–20 cm proximal to ileocaecal valve** (B12, †bile salts); open antimesenterically into a U, S or W fold or cup patch.
- **Ileocaecocystoplasty** uses caecum and 15–30 cm ileum. Valve loss causes diarrhoea in neurogenic bowel.

- **Sigmoid**: 15–20 cm. Strong contractions demand detubularisation (U-fold beats Mitchell).
- **Gastrocystoplasty**: † a gastric wedge on a gastroepiploic pedicle. For **renal insufficiency with acidosis, short gut and irradiated bowel**.
- **Jejunum** is avoided (salt-losing acidosis).

Segment	Advantages	Disadvantages
Ileum (default)	Least pressure activity; plentiful, mobile mesentery	Hyperchloraemic acidosis; B12 loss; mucus; hard to make antireflux
Ileocaecal	Taenial reimplant; appendix; imbricated valve	Diarrhoea in neurogenic bowel; more mucus
Sigmoid	Taenial reimplant; close to bladder	Strongest contractions; most mucus; more reaugmentation
Stomach	No acidosis; least mucus; fewer struvite stones; usable in renal failure	Hypokalaemic hypochloraemic alkalosis; haematuria–dysuria syndrome; tumours
Ureter	No mucus, no metabolic effect	Needs a massively dilated ureter of a non-functioning kidney; limited volume
Autoaugmentation (detrusorectomy)	Extraperitoneal; no bowel	Seldom adds capacity; compliance may not last

- **Ureterocystoplasty**: after nephrectomy the ureter, on its blood supply, is opened opposite it and sewn into the bivalved bladder. **Autoaugmentation**: the detrusor is excised off intact dome mucosa and its edges hitched to the psoas.
- Seromuscular enterocystoplasty and tissue-engineered bladder are experimental (no benefit in a multicentre trial). Technique matters more than segment, so use ileum unless there is a reason not to. Robotic outcomes mirror open.

> **KEY POINT**
> Augmentation is needed less now, thanks to early anticholinergics, CIC, botulinum toxin and neuromodulation (immediate vs expectant treatment: 17% vs 41% augmented). Not trial-proven, but the rationale for treating at diagnosis.

Incontinent alternatives

- **Vesicostomy** is an infant bridge for worsening hydronephrosis, febrile UTI with high-grade VUR, or high pressure. Problems are stenosis and prolapse.
- **Ileovesicostomy** joins 10–15 cm of ileum from the dome to a rosebud stoma. It is catheter-free, for those who cannot catheterise.

5 · Continent catheterisable channels

Why this matters: a channel allows CIC when the urethra cannot be used.

- P2 Discuss principles of continence in continent urinary diversion. Application of Mitrofanoff principle in continent urinary and GI reconstruction. — *flap vs nipple valve, conduit options (appendix, Monti, tapered ileum, ureter, Casale), appendicovesicostomy steps, MACE, results, stenosis*

Not in the paper answer:

- Stent 10–14 days; the surgeon does the first catheterisation.
- Stenosis is treated with an L-stent and steroid cream; stomal leakage needs urodynamics.
- Kock needs ~40 cm ileum, so keep a useful native bladder.

- In a **flap valve**, a supple narrow tube tunnelled into a **low-pressure** reservoir is **compressed** as pressure rises; the appendix needs only **~2 cm** of tunnel. Nipple valves (Kock, Mainz, Hanna) are **effaced** by the same pressure (about 85% vs more than 90–95% continent). A flap valve has **no pop-off**.

ALGORITHM

1. Detach the appendix with a caecal cuff (less stenosis; a caecal flap lengthens it) and preserve the mesoappendix.
2. Tunnel the **distal end** submucosally.
3. Bring the caecal end to skin with no tension or twist, and sew a **skin flap** into the stoma.
4. Keep the channel **short and straight**; fix the bladder to the abdominal wall.
5. Test catheterisation after each step.
6. Stent with 6–12 French (Fr) for 10–14 days; the surgeon does the first catheterisation.
7. Then CIC at least 4-hourly by day.

- Stoma: umbilicus or right lower quadrant; † mark it sitting, where the child can reach it. The appendix can be **split** for both an ACE and a channel.
- **Yang–Monti**: a 1–2 cm ileal ring is retubularised **transversely** over 12 Fr. Obese: double or spiral Monti. About twice the appendix revision rate.
- **Indiana** (imbricated ileocaecal valve) is simple but causes diarrhoea. The **Casale** bladder flap spares bowel but uses capacity and has high stenosis. Ureter and fallopian tube have more stenosis.
- **Stomal stenosis** is commonest (umbilical appendix more than ileum) and early. Difficult catheterisation (redundancy, kinking) needs subfascial revision.
- Stomal incontinence (rare) means a short tunnel or high reservoir pressure. Also: polyps or granulation, caecal-extension necrosis and reservoir stones.
- Pure continent diversion uses ~40 cm ileum vs ~20 cm for augmentation. Orthotopic neobladder is rare.
- **Ureterosigmoidostomy and Mainz II** (acidosis, pyelonephritis, cancer) are largely abandoned; they need a competent anal sphincter (oatmeal enema test). **Bladder buttons** give drainage only.

6 · Bladder outlet procedures

Why this matters: a tighter outlet can make the bladder hostile.

Outlet procedures sacrifice voiding and pop-off, so neurogenic repairs need **concurrent augmentation** and surveillance for **≥5 yr** (many need augmentation or toxin within 2–5 yr).

Procedure	Mechanism	Voiding	Main problem / place
Fascial sling	Coaptation and elevation	Rarely adequate, so CIC	Better in girls; only predictor of success is augmentation; AUS still possible later
Artificial urinary sphincter (AUS)	Inflatable bladder-neck cuff	**Preserved**	Infection, erosion, malfunction; best if voiding
Bulking injection	Submucosal	Preserved	Poor durability
Young–Dees–Leadbetter	Tubularised trigone	Possible	Exstrophy (*see Ch 43*); poor in neurogenic

Procedure	Mechanism	Voiding	Main problem / place
Kropp	Anterior bladder tube under trigone (flap valve)	Abolished	Highest continence; hard urethral CIC (boys); new VUR
Pippi Salle	Anterior flap onlaid on trigonal strip	Abolished	Easier CIC; lower night continence; fistula
Bladder neck closure	Complete separation and continent stoma	Abolished	After failed repairs; no pop-off

- **Sling** (rectus fascia 1 cm, with or without a circumferential wrap): incise endopelvic fascia, find the bladder neck by balloon traction, open the plane to vagina or rectum, fix to rectus sheath. Suits a marginal bladder neck or deficient sphincter in an augmented child on CIC.
- **AUS**: bladder-neck cuff in girls and prepubertal boys, not outgrown; left **deactivated ~6 weeks** so a pseudocapsule forms. Most neurogenic children still need CIC. Infection: remove. New hydronephrosis means unmasked hostility (anticholinergic or augment). Simultaneous and staged augmentation do equally well.
- **Kropp**: a 6 × 2 cm anterior flap is tubularised and passed through a trigonal tunnel. **Salle**: a 1 × 4–5 cm flap is closed in two layers with a wide base. Both often need later augmentation.
- **Closure**: wide mobilisation, posterior lip rolled forward, layered closure and **interposed omentum or rectus** keep fistula low; reliable catheterisers only.

> **KEY POINT**
> Choosing the outlet: a child who voids gets an AUS (the only option preserving voiding). A child on CIC who is being augmented gets a sling (girls) or flap-valve lengthening with a channel. After failed repairs, do bladder neck closure with a Mitrofanoff. Always re-study the bladder afterwards.

7 · Malone antegrade continence enema (MACE)

Why this matters: faecal continence matters as much as urinary.

- † An appendicocaecostomy with an antireflux tunnel gives scheduled antegrade washouts and faecal continence in spina bifida, anorectal malformation and refractory constipation. Alternatives are a Monti, a caecostomy button or a left-sided ACE.
- Complications: leakage, stenosis, granulation. Non-neurogenic: manometry guides selection (*see Ch 47*). Plan with urinary reconstruction.

8 · Postoperative care

Why this matters: mucus and overdistension are early dangers.

- Irrigate the suprapubic tube **≥3 times daily**. Do a **cystogram at ~3 weeks**, then clamp the tube and start CIC 2–3-hourly by day and 1–2 times at night, **never beyond 4–5 h** by day.

9 · Complications of augmentation

*Why this matters: further surgery is needed in up to ⅓ (about half at 10 years), because **bowel is not bladder**.*

Segment	Derangement	Management
Ileum, colon	**Hyperchloraemic metabolic acidosis** (overt mainly with renal impairment)	Bicarbonate; **base deficit** is most sensitive; chlorpromazine or nicotinic acid

Segment	Derangement	Management
Jejunum	Hyponatraemic, hypochloraemic, hyperkalaemic acidosis	Avoid
Stomach	**Hypokalaemic hypochloraemic alkalosis** with vomiting; **haematuria–dysuria**	Fluids, H_2 blockers or proton pump inhibitor
Stomach and bowel	Near neutral	Complex patients

- Acid load demineralises bone and lowers height percentile, so treat base deficit early. Check B12 long term. Obstruction is uncommon; diarrhoea follows valve loss. † Drugs such as methotrexate and phenytoin can be reabsorbed.
- **Mucus** (colon more than ileum more than stomach) needs **daily irrigation**.
- **Bacteriuria** is expected with CIC. Treat symptomatic infection (pain, new incontinence, fever) and **urea-splitters**; optionally haematuria, foul urine, more mucus.
- **Stones** are mostly **struvite** (bacteriuria, mucus, poor emptying; worst with abdominal stoma). Prevent them with **daily irrigation ≥240 mL** saline. Stomach augments form uric acid stones.

Delayed spontaneous perforation

- Counsel a **~9%** risk, mostly in myelodysplasia. **Outlet procedures double the risk**, as do missed CIC and substance abuse. Overdistension makes the wall ischaemic; recurrence is common.
- Traumatic catheterisation and adhesions contribute. Signs: pain, distension, fever, sepsis, reduced output and shoulder-tip pain. Poor sensation means **late presentation** and deaths; gastric augments present early.
- Do a **cystogram** with full filling and a post-drain film, and a computed tomography (CT) cystogram if negative. Treat by **immediate laparotomy and repair**; manage conservatively only if stable, sterile and contained.

Malignancy

- P1 Cancer in augmented bladder — *adenocarcinoma at anastomosis, earliest 4 yr, N-nitroso pathogenesis, gastrocystoplasty/exstrophy/transplant risk, SB 2–4%, nephrogenic adenoma, surveillance debate, cystectomy*
- After ureterosigmoidostomy, adenocarcinoma arises at the anastomosis with a latency of **~26 yr**, because urine and faeces form N-nitroso compounds.
- After augmentation it is reported from **4 yr**, is highest after **gastrocystoplasty**, and is aggressive and often metastatic. Bladder cancer in spina bifida (SB) is ~2–4%, mostly stage III–IV.
- Nephrogenic adenoma (benign) is the commonest tumour. There is no consensus on annual cystoscopy; consider it after gastrocystoplasty, in exstrophy and after transplant. † Investigate any haematuria.

Pregnancy

- The pedicle deflects laterally. Hydronephrosis (nephrostomy over stent) and UTI occur. Vaginal delivery is possible; caesarean after major bladder neck repair or AUS, with a urologist present.

10 · Follow-up and transition

Why this matters: late kidney damage comes from missed CIC.

When	Tests
3, 6, 12 months	US; electrolytes, BUN, creatinine, urine culture (with or without urodynamics by 6 months)
Yearly, lifelong	US, chemistry (with or without blood gas, B12), CIC and irrigation compliance; urodynamics if new hydronephrosis, incontinence or VUR
Symptomatic	† Haematuria needs cystoscopy; pain or sepsis needs perforation excluded

- Early high-pressure contractions after augmentation need repeat urodynamics; true reaugmentation is rare.
- After augmentation with a bladder neck procedure, many develop scarring and chronic kidney disease (CKD) stage ≥3, driven by **CIC non-compliance**. Adult urology must police CIC, irrigation and the upper tracts.

11 · Landmark trials

Trial	Design	Result	Take-home
Early vs expectant neurogenic bladder care	Cohort, treated at diagnosis vs expectantly	Augmentation 17% vs 41%	Treat hostile bladder early
Daily bladder irrigation	Augmented cohort, ≥240 mL saline daily	Marked fall in stones	Routine daily irrigation
Tissue-engineered bladder	Industry multicentre trial	No gain in compliance or capacity	Bowel remains standard

★ Putting it together: common paediatric pathways

A · Antenatal urinary tract dilation (UTD)

Why this matters: most dilation is transient.

ALGORITHM

1. **Prenatal:** A1 before 28 weeks: ultrasound (US) after 32 weeks. A2–3: US 4-weekly. Male, thick bladder, bilateral: suspect posterior urethral valves (PUV).
2. **Postnatal:** first US **after 48 hours**, except suspected PUV or bilateral severe dilation (before discharge).
3. If normal (anteroposterior diameter <10 mm), discharge; if mild, repeat US; if moderate–severe or dilated ureter, voiding cystourethrogram (VCUG), MAG3, consider continuous antibiotic prophylaxis (CAP) (→ *Ch 34*).

B · First febrile urinary tract infection (UTI)

Why this matters: prompt treatment prevents scarring.

ALGORITHM

1. **Diagnose:** catheter or suprapubic sample if not toilet-trained; bag only rules out.
2. **Treat** within 48 h (>72 h doubles new scarring); oral if well.
3. **Image:** US all; VCUG if US abnormal, atypical or recurrent (American Academy of Pediatrics 2–24 months); DMSA at 4–6 months if scarring matters.
4. **Prevent:** treat bladder and bowel dysfunction (BBD) and constipation; CAP if high risk; circumcision lowers infant UTI (→ *Ch 37, 41*).

C · Newborn with PUV

Why this matters: early drainage protects the kidneys.

ALGORITHM

1. Stabilise breathing → 5–7 Fr feeding tube (minimal or no balloon) → creatinine, electrolytes, fluids.
2. US, then VCUG; prophylaxis.
3. When stable, **endoscopic ablation**; if the urethra is small, vesicostomy.
4. Prognosis: first-year nadir creatinine (<0.8 mg/dL good). Lifelong: valve bladder, chronic kidney disease (CKD), blood pressure; transplant, Ch 61 (book 1) (→ *Ch 45*).

D · Neurogenic bladder in spina bifida

Why this matters: a hostile bladder damages kidneys silently.

Step	What	Why
Birth	US and post-void residual (PVR); creatinine after week 1; latex-free care	Detect retention; baseline
≈3 months	Urodynamics (with or without video)	Hostile: leak point or end-filling ≥40 cmH_2O, poor compliance, dyssynergia
Hostile bladder	Early clean intermittent catheterisation (CIC) and antimuscarinic	Kidney protection

Step	What	Why
Failing	Botulinum toxin, then vesicostomy (infant), then augmentation with or without channel or outlet	Safe storage, emptying (*see Ch 46, 48*)
Always	Bowel programme; watch for re-tethering	Safety

E · Daytime wetting and bedwetting

Why this matters: exclude serious causes before urotherapy.

ALGORITHM

1. Urinalysis, PVR with or without uroflow; exclude neurogenic, anatomical, medical causes.
2. **Constipation first**, then urotherapy ≥4 weeks.
3. Specific: antimuscarinic for overactive bladder, biofeedback for dysfunctional voiding. If refractory: toxin, neuromodulation.
4. **Monosymptomatic nocturnal enuresis:** alarm or desmopressin (no fluids 1 h before to morning); combine if partial (→ *Ch 47*).

F · Augmentation segment

Why this matters: the segment sets lifelong metabolic effects.

Segment	Advantages	Disadvantages
Ileum	Default; compliant, reliable supply	Mucus, hyperchloraemic acidosis, B12 and bile-salt loss (terminal ileum)
Ileocaecum / colon	Appendix for channel; taenial reimplant	Valve loss causes diarrhoea in neurogenic bowel; acidosis
Stomach	Secretes acid, so suits renal failure; little mucus	Haematuria–dysuria, alkalosis
Ureter	Urothelium, so no mucus or metabolic effect	Needs a massive ureter of a non-functioning kidney

KEY POINT

Reconstruction trades storage for emptying. An unreliably catheterised continent reservoir causes hydronephrosis, sepsis and perforation, so CIC commitment comes first. Perforation means pain and sepsis; it is urgent (*see Ch 48*).